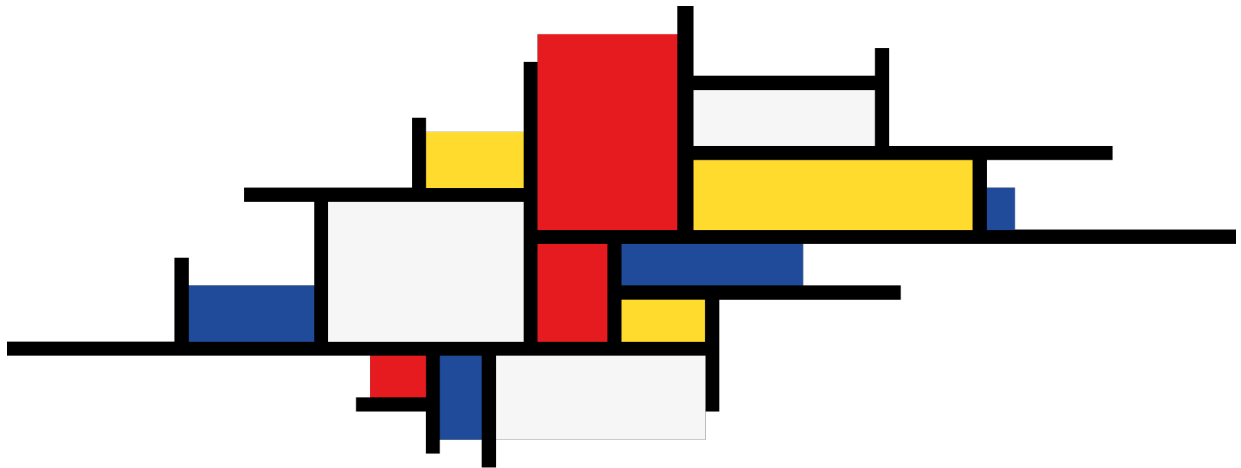
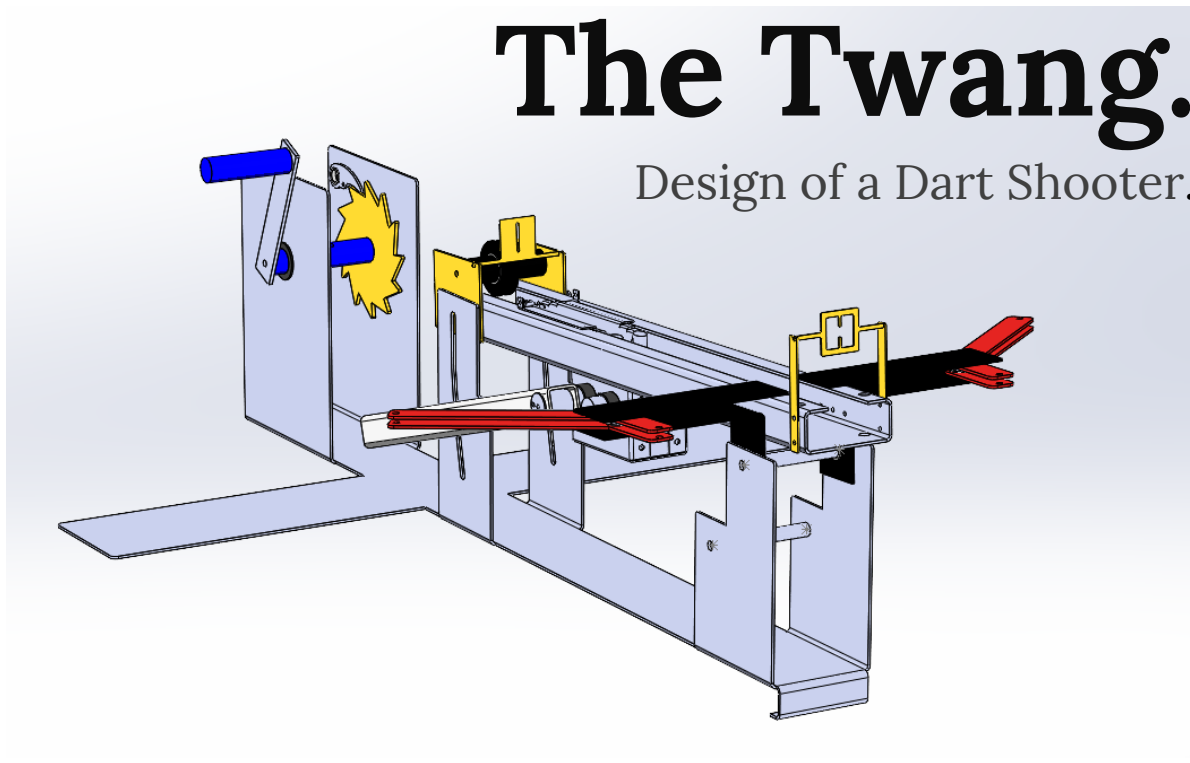




The Twang.

Design of a Dart Shooter.



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Executive Summary

The challenge of shooting plastic darts over a distance of 4 to 8 meters, using fully mechanical methods, no electrical components and a deadline that is always just a bit too close. From challenge to working prototype to further improvements, this report documents the development and prototyping process of a mechanical dart shooter. The project began with a comprehensive analysis, understanding user needs, given requirements, and the general scope of the project. From there a list of requirements and a morphological chart paved the way for a series of concepts, out of which a final concept was derived. This final concept was then designed in detail and made ready to be manufactured into a functioning full-scale prototype of a fully mechanically operating dart shooter.

The final result came not without challenges. The dart shooter's design was adjusted and modified throughout the design phase to address unforeseen issues, including the strength of the springs and the stability of certain parts like the main T-shaped frame and the frame of the cart holding the dart. Manufacturing the prototype required a variety of different techniques including welding, milling, turning, and laser cutting with the materials and machines available in the University of Twente workshop. Parts were carefully measured and assembled with the help of a work preparation sheet to ensure accuracy and reduce errors. Once the prototype was completed, testing revealed aspects of the prototype that required further improvements. Adjustments were made to enhance the functionality and durability of the dart shooter, such as reinforcing parts and improving mechanisms. The final product was evaluated based on whether it met the initial requirements. While the dart shooter was largely successful, several areas still require improvement, including the spring power and safety features.

This report concludes with a series of recommendations to further refine the design, ensuring the dart shooter meets all functional requirements and is ready for broader use.

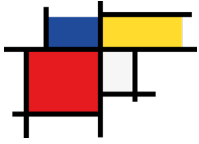


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Introduction

A normal game of darts is in essence quite simple. You stand in front of a dart board with three darts at hand and a total score of 501 points. One after the other you throw the darts onto the board; every hit reduces your point total. The first player without any points wins the game. [5]

Can we do better?

In this project the goal was to design and manufacture the prototype of a fully functional mechanical dart shooter. The dart shooter should be able to shoot the three given darts over a variable distance between 4 and 8 meters onto a dart board. This should ideally be done as accurate and reliable as possible.

In this technical report the design process for the dart shooter will be presented. Starting at the analysis of the design process, past the different concepts and concept selection, using 3D CAD models, technical drawings, tolerances, and material and production choices, with work preparation sheets to produce the prototype all the way through to the evaluation of the functional prototype.

This is the report of the design process of Group 12 - The White Lightning in the Project Prototyping in Module 2 of the first year of the Industrial Design Study at the University of Twente. in 2024-2025. This is the report describing how The Twang came to be.



Chapter One - Problem Analysis

The primary objective of this project is to develop a reliable mechanism capable of shooting the provided darts over the required distance. Additional goals include user-friendliness and portability. The design must prioritize functionality, safety, and ease of operation, aimed to use in a competitive environment.

The dart shooter must be operable by a single individual, easily transported, and easy to install on a table. It will be designed to fire three darts at a target located at a distance ranging from 4 to 8 meters away, set at a height of 1500 mm, and with a diameter of 600 mm. The target consists of a 2 mm-thick PET plate with a 1-10 point scoring system in concentric rings. The dart shooter itself will rest on a 900 mm high table with a surface area of 500 x 1000 mm. The operational requirements dictate the ability to shoot three darts within two minutes and be ready to shoot within one minute of setup. [1] The final mechanism must require no more than 115N of force to operate (Appendix One - List of Requirements).

The dimensions of the darts provided for this project are; 170 mm in length and width, and a height of 42mm, and a grip diameter of 20 mm. The distance from the back of the fin to the back of the grip is 85mm. [1] These specifications directly influence the design of the dart shooter, as the device must accommodate these measurements.

To achieve the project goals, the dart shooter must balance several factors:

1. The system must operate within the maximum manual operating force of 115N while maintaining sufficient shooting power. Hence, the dart shooter will incorporate a lever or winch mechanism, to make optimal use of the mechanical advantage in order to release the dart with more force than put in.
2. Accuracy and consistency will be prioritized through stable construction and an adjustable shooting mechanism. These features will enable users to adapt to varying target distances, to reach a consistent performance across the entire range.
3. While portability is a secondary goal, the design will focus on making the design manageable in terms of size and mass. The dart shooter will fit within the dimensional constraints of the designated table: 500x1000mm [1], ensuring suitability for various settings. Efforts will be made to maintain a balance between the structure and ease of operation, without compromising the performance.
4. Achieving consistent performance across the 4–8 meter range also requires a calculated balance between the force exerted by the springs and the angle under which the darts is shot. Further research will be conducted to address the relevant physical considerations and optimal geometrical features to accommodate the necessary performance.

The material and production choices should align with the resources available at the University of Twente's Hangar, adhering to specific requirements:

1. The materials and production processes are limited to the resources at the Hangar or within an additional budget of 25 euro. Only a maximum of two parts will be 3D-printed with maximum dimensions of 60x60x60mm. [1]

2. All parts are manufactured within four workshop sessions. [1]
3. Laser cutting will be prioritized for precision and time efficiency. The choice of materials and joining methods will be considered carefully to enhance durability, ensuring the device can withstand repeated use.

The device should create a safe and reliable experience for all users. To ensure the safety:

- The device will incorporate rounded edges and enclosed moving parts.
- The total mass will remain under 23kg, ensuring manageable handling, and require less than 115N of force operation. (Appendix One - List of Requirements)
- The mechanisms will be designed for single-person operation, with a maximum of two simultaneous and easily reachable manual interaction points, hence reducing complexity and improper handling of operation. (Appendix One - List of Requirements)

Testing will take place within the facilities of the university, with the target group for the testing phase consisting of university students and/or staff members. Success criteria include the ease of operation, trajectory consistency, and overall user experience.

The List of Requirement (LoR), serves as a comprehensive guide for the project, outlining constraints and specifications derived from various phases of development. While some requirements are based on the problem analysis outlined in this chapter, others were established in different aspects of the project.

For a detailed overview of the complete LoR, refer to Appendix One - List of Requirements.



Chapter Two - Conceptualisation

Ideation and Concept Generation

[4] After having analysed the problem at hand and having drawn up a list of requirements for the design, the second phase of the project was initiated: Concept generation. With the use of a broadened morphological chart (Appendix Two - Morphological Chart) concepts were generated on an individual basis. These concepts were turned into concept drawings (Figure 1, Figure 2, and Figure 3) and presented to the rest of the group.

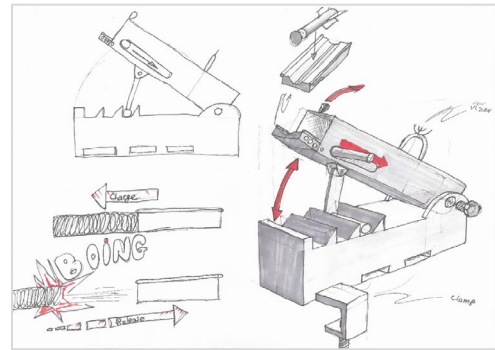


Figure 1: Concept drawing of 'The Boing'.

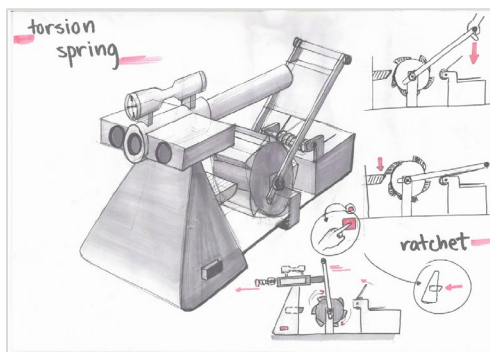


Figure 2: Concept drawing of 'The Woosh'.

Together the concepts were evaluated, considering aspects such as feasibility, required knowledge, and overall preference. Three were selected for further development:

These concepts were not yet fully defined, with certain

mechanisms undecided and without formal names. Names for the concepts were chosen based on prominent sounds produced by the device. After detailing the concepts more, concept selection took place.

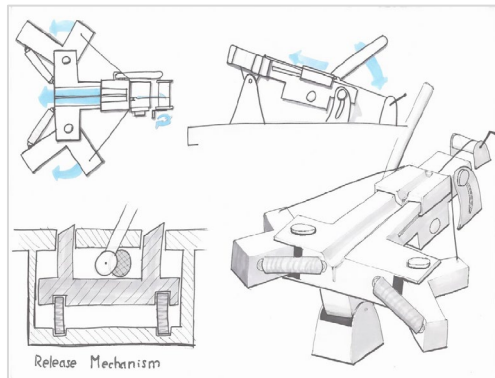


Figure 3: Concept drawing of 'The Twang'.

Concept Selection

The MoSCoW method was chosen, because of prior experience with this selection method. The MoSCoW selection method uses the List of Requirements (Appendix One - List of Requirements) as the foundation of the concept selection. The requirements are turned into criteria, each with its own performance scale. Each of the criteria are then given a weighing factor based on importance. After this, every concept is graded on all criteria.

A selection table (Three A - Selection Table) was created listing each concept, requirement, and corresponding weight factors. Scores were defined in the performance scale (Three B - Grading Scales), describing to what extent the concept needed to fulfil the requirement to obtain a certain score. Requirements were categorized as Must (M), Should (S), Could (C), or Want (W), correlating to the weight factors.

Despite close results, "The Twang" emerged as the highest scoring concept. Not only the score was considered when the final selection was made, but also how much room for improvement was left based on what this concept lacked when compared to the other two concepts. Based on the group's estimations, the Twang had the highest potential.

Evaluating the concept selection, it was noted that 'The Woosh' did not meet all the Musts and should have therefore been eliminated preliminary. The other concept scored full points on all Musts.

While mechanisms, proportions, and sizes were considered, materials for the device had not yet been looked at, assuming this would fall under the "Detailed design" phase. The feedback resulted in the cart being marked as a point of attention, the initial design would create a substantial amount of friction.

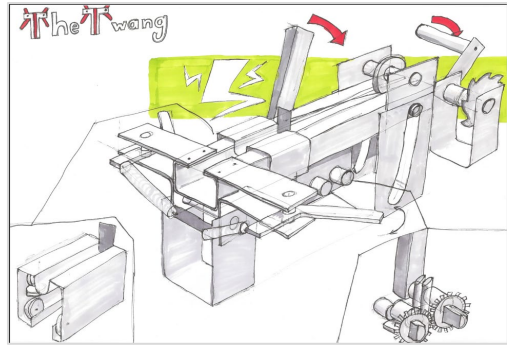


Figure 4: Final concept drawing 'The Twang', 3D view.

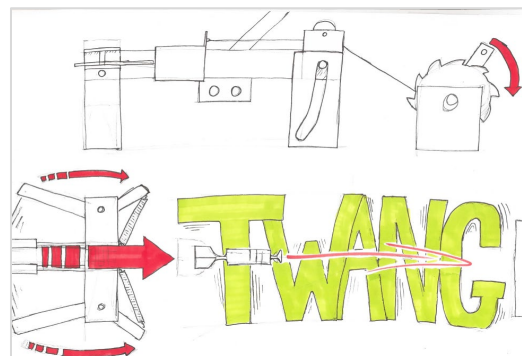


Figure 5: final concept drawing 'The Twang', orthogonal view.



Chapter Three - Detailed Design

Working with the feedback on the concept by people more experienced in this field, the concept was taken to a detailed design using 3D models and technical drawings.

After finishing the final concept, it was analysed out of which parts it would consist. All parts were sorted and grouped by looking at how every part was related to the other parts. This was visualised in a Tree of Parts (Appendix Four - Tree of Parts). Herein all parts were ordered in subassemblies which in turn formed the total assembly.

While making the 3D models, the possible (and optimal) production methods were considered, as well as the joining methods within the (sub)assemblies. To ensure all parts were feasible to make and feasible to join together.

For some of the design requirements and details in the final design, calculations were necessary. Requirements regarding the physics of the design, the velocity of the darts, the angle under which to shoot them and the subsequent geometry of the device, were all determined using the physical processes described below.

Impact

The first of these to determine was the impact; On what range of (end) velocities does the dart stick to the board? And, on what range of angles does the dart stick to the board?

The easiest way to evaluate the required end velocities was to release the dart from an ever-increasing height and measure when it would stick properly. To do this a particular setup was required. First a perspex sheet -the same material as the surface of the board- was laid on the ground, with a ladder next to it. Then a person would hold one of the darts by the tail, using a measuring tape to measure the distance from the suction cup to the plate on the ground, and, when the dart was on the proper height, release the dart. The dart now had the distance to the ground to accelerate over. After every drop the dart was picked up from the board, feeling how securely it was connected to it. Every height was tested thrice, with increments of 0.15 m ranging from 0.15 m to 1.65 m. The results can be seen in Table 1.

Table 1: Results required end velocity test.

Distance from plate (m)	Test #1	Test #2	Test #3	Notes	End velocity dart (m/s)
0.15	Fail	Fail	Fail		1.72
0.30	Succes	Succes	Fail	Weak suction strength.	2.43
0.45	Succes	Fail	Succes	Weak suction strength.	2.97
0.60	Succes	Succes	Succes	Stronger; not yet strong enough.	3.43
0.75	Succes	Succes	Succes	Quite firm suction strength.	3.84
0.90	Succes	Succes	Succes	If angle not perpendicular to plate, dart does not stick.	4.20
1.05	Succes	Succes	Succes	With small angles, dart sticks.	4.54
1.20	Succes	Succes	Succes	Trustable suction strength.	4.85
1.35	Succes	Succes	Succes		5.15

1.50	Succes	Succes	Succes	Trustable suction strength for 3 min. on board.	5.42
1.65	Succes	Succes	Succes		5.69

After these measurements were taken, the corresponding end velocities per height were determined using the formulas $F_{res} = m \cdot a$ and $F_z = m \cdot g$ to get $v = g \cdot t + v_0$ -where $v_0 = 0$, and the formula $h = \frac{1}{2}gt^2$. Neglecting air resistance, rewriting, and substituting these formulas into one another results in the formula $v = \sqrt{2hg}$. Using a gravitational constant of $g = 9.81 \text{ m/s}^2$ and the heights listed in Table 1, another column is added to this table to show these velocities.

Out of this table, the velocity upon impact desired was chosen to be (at least) $v_{impact} = 5.15 \text{ m/s}$. From this experiment it was concluded to be useful to also evaluate how well the dart sticks when dropped under an angle, since this seemed to affect the data. It was then determined to only use the height of $h = 1.35 \text{ m}$, since the corresponding velocity was going to be used.

The dart was then dropped, while being held at an increasing angle (orthogonal to the sheet), with increments of 10 degrees. Testing when it would stick properly. The results are shown in Table 2.

Table 2: Results required maximum angle.

Angle	Test #1	Test #2	Test #3	Notes
10°	Succes	Succes	Succes	No noticeable difference in suction strength.
20°	Fail	Fail	Fail	Slight decrease in suction strength.
30°	Fail	Fail	Fail	
40°	Succes	Fail	Succes	
50°	Succes	Succes	Fail	Severe decrease in suction strength.
60°	Fail	Succes	Fail	
70°	Succes	Fail	Fail	Unnecessary to continue with steeper angles.

From this the following design requirement was formulated: "The product shall 'shoot' the darts in such a manner that the angle under which the dart hits the board shall lie within a 15° range relative to the ground."

Trajectory

Next to these experiments, an analytical model was made to determine the theoretical optimal starting velocities and angles per distance. The model itself can be found in Appendix Eight - Trajectory Model.

In this model, the velocity of the dart in both x and y direction was analysed, alongside the corresponding position, neglecting air resistance for simplification. These standard formulas were then used to separate two branches of answers; The first one used the given that at the top of the dart-described parabola the

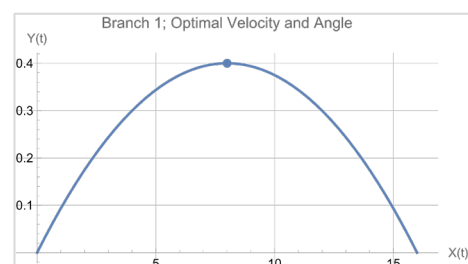


Figure 6: Branch 1; trajectory graph of dart over 8 m.

velocity in the y-direction is zero. In this branch of solutions, the difference in height between the initial point and the peak, and the distance between them could be plugged in, and the optimal combination of starting angle and velocity would be computed.

Table 3: Branch 1; optimal velocity and angle for different distances.

Distance (m)	Optimal Velocity (m/s)	Optimal Angle (°)
4.0	14.29	11.31
4.5	16.01	10.08
5.0	17.73	9.09
5.5	19.46	8.28
6.0	21.20	7.59
6.5	22.93	7.02
7.0	24.67	6.52
7.5	26.41	6.09
8.0	28.15	5.71

The second branch does not have the requirement of reaching the input height and distance combination at the peak of the parabola, it just needs to reach those at the same time. This parameter is swapped for the possibility to use the start velocity as input and has the corresponding optimal angle as output. To check if the angle of impact would stay within the determined range, an extra part was added to the model, with which the angle of the dart at the point of impact could be computed.

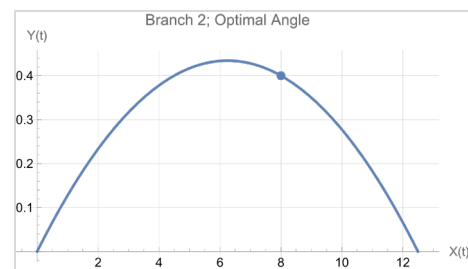


Figure 7: Branch 2; trajectory graph of dart over 8 m.

It was chosen to have a difference in height of 0.4 m, this results in a design requirement; "The product shall shoot the darts from a height of 0.2m above the table.". This was done to simplify the computations by changing an unknown value into a known constant.

A combination of both branches is used to compute the ideal starting velocity. The first branch is used as a start, for every half meter on the interval 4.0 m to 8.0 m, to find the optimal range of starting velocities. These results can be seen in Table 3. The range of optimal starting velocities starts at 14.29 m/s and goes on until 28.15 m/s. It was deemed next to impossible to design for a continuous range of starting velocities corresponding to this range, therefore it was decided to try to split it up into multiple different intervals. The second branch is used to compute the shooting angle corresponding to the start velocity for all values on the intervals. First the entire range was split up into three intervals: from 4.0 m to 5.3 m, from 5.3 to 6.7, and from 6.7 to 8.0. For all these ranges the average distance was used to compute a starting velocity, with which a range of angles would be computed. Next to this the angle of impact relative to the ground for the upper bound of the interval was determined, to check if this did not exceed the previously determined range. The results are shown in Table 4.

Table 4: Computed starting velocities, ranges, and impact angles for three sections.

Interval (m)	Starting velocity (m/s)	Range of angles (°)	Maximum deviation in angle of impact (°)
4.0 - 5.3	16.58	9.7 - 9.8	1.5
5.3 - 6.7	21.20	7.6 - 7.7	1.0
6.7 - 8.0	25.83	6.2 - 6.3	0.6

Since the ranges in angles is very small, while the maximum angle of impact stayed well within range, splitting the entire interval into two sections was tried next in a similar fashion. The results are shown in Table 5.

Table 5: Computed starting velocities, ranges, and impact angles for two sections.

Interval (m)	Starting velocity (m/s)	Range of angles (°)	Maximum deviation in angle of impact (°)
4.0 - 6.0	17.73	9.1 - 9.3	2.1
6.0 - 8.0	24.67	6.5 - 6.6	1.0

Since this, again, resulted in very small changes in shooting angles and impact angles within range, it was determined to work with one singular starting velocity and a continuous range of starting angles. The values are shown in Table 6.

Table 6: Computed starting velocities, ranges, and impact angles for one section.

Interval (m)	Starting velocity (m/s)	Range of angles (°)	Maximum deviation in angle of impact (°)
4.0 - 8.0	21.20	7.6 - 8.2	3.2

Therefore, the design requirements "The product shall ensure that a velocity of 21.20 m/s can be achieved by the darts at the moment when they leave the product." and "The angle under which the product shoots the darts shall be adjustable within the range of 7.5° till 15.2°." were determined. Where the lower bound angle is slightly lowered with 0.1 degree, and the upper bound is heightened with 7.0 degrees, since the neglected friction will play a role in the actual device.

Forces

When looking at the forces on the dart, and how to apply them, it is best to start by simplification. Approximating the velocity with which the dart would be released from the prototype is done by first assuming all the forces are acting linearly on the dart, parallel to the direction of the trajectory. Next to that, the friction forces will be ignored entirely, which simplifies the calculations enormously.

Starting with equating the work done by an n number of springs to the kinetic energy of the dart when released, gives $nW_{spring} = E_{kinetic}$. Since $W_{spring} = \int_{x_1}^{x_2} F_{spring}(x)dx$ (where x_1 is the minimal stretching of the spring and x_2 the maximum stretching), $F_{spring}(x) = kx$ (where k is the spring constant and x the length stretched) and $E_{kinetic} = \frac{1}{2}mv^2$, this can be rewritten into where m is the mass of everything that is accelerated.

$$v_{release} = \sqrt{\frac{2nk(x_2^2 - x_1^2)}{m}}$$

Since the chosen concept did not have the forces of the springs parallel to the path of travel, some restriction had to be made to the design of the prototype to allow for a better approximation formula. The force of the springs can be split over its components, and then by approximation only the force on the string will be taken into account, to do this the work of the springs is multiplied with the cosine of the angle between the string and the travel path. This resulted in the following formula where α is that angle. Since this angle is not constant the formula only works as a rough approximation.

$$v_{release} = \sqrt{\frac{nk(x_2^2 - x_1^2)}{m} \cdot \cos \alpha}$$

This approximation would only yield usable results, if other geometric angles were to be restricted. To ensure the best results the angle between the string and the L-shaped parts (when the springs are in tension), the angle between the springs and the T-shaped (at all times), and the angle between the springs and the L-shaped parts should all be (as close to) 90° to ensure a proper transmission of forces.

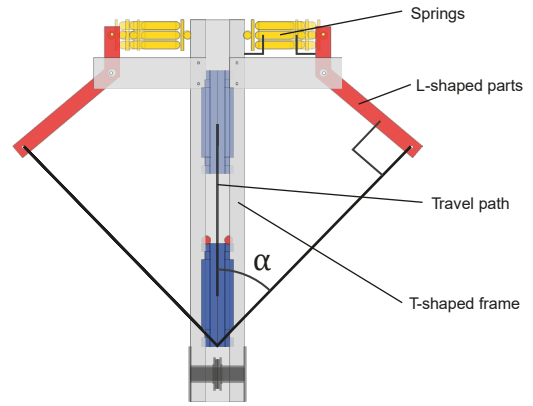


Figure 8: Image of top view of dart shooter including specified angles.

Geometry

Even though there is technically no minimum length the runway should have when friction forces are neglected, there were some things to consider while designing for the geometry of the device. Therefore, a simplified geometric model of the concept was put together, where all the geometric features were described relative to the others. This model can be found in Appendix Nine - Geometry Model.

Some dimensions are derived from fixed points. Mainly, the dimensions of the strongest springs available in the Hangar [2][3]. These dimensions (plus a little extra to allow for the connection between parts) determined the length of the 'small' part of the T-shaped frame.

While keeping all the dimensions within reasonable bounds, it was found to be quite complex to get a long runway of the dart, which was thought to be necessary, since inertia played a role in the acceleration of the dart. Therefore, the model was mostly used to figure this out.

To maximize the runway, the small part of the L-shapes needed to be minimized, and the big part of the L-shapes needed to be maximized. The angle between these two also played an important role, it was found that an angle close to 130° generally provided the longest runway. Using this and the dimensions of the springs the dimensions for the L-shaped parts (for the 'small' part of the L-shape 50 mm, for the big part 150 mm) were

determined, as well as the length of the runway (226 mm). Subsequently a large unknown for the length of the T-shaped frame was now known.

FMEA and Safety Analysis

To make sure the final design would not become a lethal weapon, instead of a fun dart shooter, and to make sure the design aligned with the list of requirements, an FMEA and safety analysis was conducted.

After defining the scope and goals of the investigation, the analysis was limited to the parts of the device and not necessarily taking the potential environments into account in order to find major flaws and safety issues in our design. Using the given templates, failure modes for each part were defined as well as their corresponding functions from the structure and functional analysis. The functional model (Eleven B - Functional Model) from the functional analysis was used to gain insight to the flow of energy and which parts would be subjected to most force. This gave a useful overview following the risk analysis, from which the risk priority numbers for all relevant failure modes were obtained.

FMEA Results and Implementation

From the FMEA (Eleven A - FMEA) and safety analysis it was concluded that the highest risk in the design lies mostly within the strength of the material, failure of joints and human error. The measurements which could influence the success of the design functionality wise, and the risk factor are not necessarily part of the safety analysis in the same way as other parts. The measurements must be altered in order to assemble the device, so the measurements will correspond with each other in the end, otherwise the device could not even function. Faulty measurements as a result of misuse of Hangar machinery will be taken into account when evaluating the device during the prototyping. The results of the FMEA and safety analysis did not disappoint in the slightest, a lot of useful information gathered from the FMEA and safety analysis were used to improve the design. More appropriate techniques and ways to machine the parts of the device will be used. Certain materials will also be replaced, such as swapping out steel bars for sheet metal.

The L-shapes on the sides, attached to the rope and the springs, were redesigned in this way: instead of using steel bars and machining them, the L-shapes will be laser cut to reduce mass and the time used to machine them. This way of designing the L-shapes will result in more accurate results and a slightly safer experience for the user.

The “Big-T”, the guide along which the cart will glide, was redesigned along with the cart to accommodate a necessary change: The design of the cart switched from using rolling wheels to sliding planes. This is done because there would be close to no time for the wheels to accelerate, since they would need a lot of friction with the T-frame, this would result in a lot of energy loss which would reduce the power of the device and the accuracy of our calculations. Now that the cart glides along the guide, it is easier to accelerate, and the risk of the cart malfunctioning is also reduced.

In the original design for the winch system, the rope, with which the cart would be pulled, was supposed to be attached to the axle of the winch with a laser-cut part, which would be welded on. However, since this part would be under immense force and not confidently strong enough, it might have bent easily or broken apart entirely. Therefore, the design of the axle was altered to support a hole through the axle in radial direction through which the rope could be fastened.

To increase the chance that the users do not harm themselves whilst using the device, the decision was made to indicate which way the lever should be moved to shoot the dart. We will be painting this indication in the form of an arrow on the side of the “Big-T” in a bright colour that will stand out to the user.

Finally, the materials of some parts were changed to be laser cut to minimize human error during machining, improve accuracy and simplify the assembly process.

The results of the FMEA and safety analysis were positively used, improving both the safety for the user and the functionality of the device. The most important changes that came from the FMEA results were implemented and a new and improved version of The Twang was now under construction.

After initiating the production process of the design, some additional changes were made during the workshop moments:

1. The cart that carries the dart was subjected to many changes during the detailed design phase. The first iteration of the cart was a cart that was connected both above and below the Big-T and slid over the surface (metal on metal), this was changed to a cart with 6 wheels, 3 on both sides on all contact surfaces (top, bottom and on the side) to reduce friction. However, during the FMEA and discussions with people from the Hangar the discovery was made that it was not possible to make the geometry designed for the cart, as the number of smart bends integrated into the design could not be bent. After it was redesigned, it turned out to be way too heavy, so the cart was changed from a perfect fit to a V shape. The cart was redesigned to make it as light as possible. Taking these restrictions into account, the decision was made to make the cart from one plate of 1 mm sheet metal instead of 3mm sheet metal.
2. The springs in the front were also changed, in the concept two springs were used (one on each side). From the calculations discussed earlier in this chapter, the conclusion was drawn that instead of two springs, a minimum of 14 springs were required to launch the dart with the desired velocity. This was changed to 18 springs, to account for the neglected friction forces and simplify the connection between these springs. These springs were to be bound on both sides of the T-frame between two square plates which in turn would be connected to the T-frame and the L-shapes.

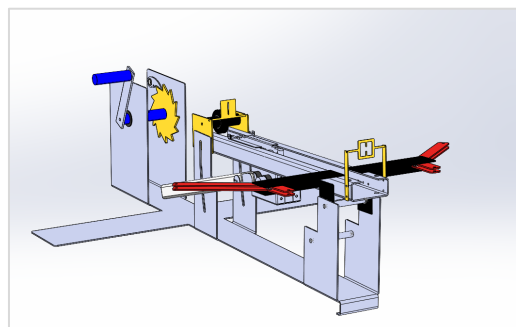


Figure 9: Total assembly of CAD models of dart shooter.

Even after the detailed design phase and the FMEA there were some defects in the design which only came to light during the making of the prototype. Some materials were not available in the Hangar, some parts were too complex for us to manufacture or worked better with different materials. Next to that human errors in the design of the separate parts also played a role in some defects, luckily quite some of them could already be filtered out using SolidWorks assemblies, shown in Figure 9.

Design Justifications

Subassy; Baseplate

Part; Baseplate frame

The shape of the baseplate resembles a rectangular ground plane, with two plates to the sides to secure it onto the table, these will be secured with clamps. On both sides of this rectangle multiple upward flanges are present to connect the different subassemblies. The front of the ground plane includes a straight bent edge for easy alignment with the edge of the table.

Part; Pawl

The pawl works together with the ratchet wheel to function properly. Since this mechanism does not rely on high accuracy the exact dimensions are not that relevant. Since the shape is quite complex, it will be laser cut.

Part; T-hinge/Baseplate spacer

This part provides added structural support for the baseplate flanges as well as a pivot axle for the T-frame to revolve around. They are made from solid bars with a diameter of 10 mm. On both sides there are threaded holes for M4 screws.

Subassy; Winch

Part; Lever

To create mechanical advantages, a lever is implemented in the winch, this lever must be as large as possible without interfering with other parts. A flat and rectangular lever is chosen, since it was the easiest to produce.

Part; Handle

For the handle, a bar with a diameter of 20 mm is selected, because the handle must be comfortable in the user's hand. A 20 mm solid bar is chosen, because this is one of the standard diameters in the workshop. The handle has a threaded hole for a M5 screw on one side to connect it to the lever.

Part; Ratchet

In combination with the pawl the ratchet only works in one direction which ensures that the design is safe. There is opted for a relatively large ratchet, because it greatly

reduces the number of rotations needed to clamp the cart and the user does not have to turn the lever a lot of times.

Part; Axle

The axle is an important part in the design, not only because the rope is attached to this bar, but mainly because it determines exactly how many rotations you need, the thicker the bar the more rope goes around it per revolution and the faster the cart goes backwards. A diameter of 20 mm is used, since this was standard size bar. A hole is made with a diameter of 4 mm straight through the middle of the tube to connect the rope to. The axle connects to both the lever and the ratchet with an M5 screw. On both sides of the axle is a hole for such an M5 screw.

Part; Cart hook

The cart hook is secured to one end of the rope. it is made to be able to easily hook onto the cart without slipping over.

Subassy; Frame

Subassy; Cart

Part; Cart frame

The frame of the cart went through many different versions before it ended up looking the way it does now. In the first concept it had ridges which would connect into a slot in the sides of the T-frame. Then it wrapped around to the bottom side of the T-frame and included wheels, to reduce friction. But since the wheels would only work with enough friction, this was changed back to ridges, but now on all sides. Then it was found out to be too difficult to manufacture the shape for the dart containment part. This was changed, but then the mass had to be reduced as well. Therefore, the cart is now as light as possible. The place for the dart is a 50 degree bend to simplify the production method, it is made from 1 mm sheet metal to reduce the mass. For alignment purposes the cart has flanges to go on top of the T-frame, just below the surface and at the same level of the surface.

Part; Backplate

The backplate went through multiple different iterations as well. At first it would have a precise cut out for the dart, to ensure it would not fly off, but this was reduced to an as small part as possible. Therefore, it is now just a small plate, bent to connect to the cart frame.

Subassy; L-Shape

Part; L-shape

The dimensions of the L-shapes are already in detail discussed in Geometry.

Subassy; T-shape

Part; Big T-shape

The cross section of the Big T-shape was determined before the cart was changed to its final form. The outside dimensions would have correlated with the necessary size for the cart to still have wheels without interfering with the Release Mechanism. The inside gap is to allow the suction cup of the dart to pass through. It is made from 2 mm sheet metal to be able to bend it.

Part; Small T top

The Small T Top also has measurement from Geometry. It is made from 2 mm sheet metal to connect more securely to the Big T-Shape.

Part; Small T Bottom

This part corresponds to the Small T-Top, with the noticeable difference it has a bent part extending down to function as the pivot point. This was made in the front of the cart to keep the same shooting height for different angles.

Part; Small T Shaft

This shaft is there to secure a smooth rotating motion for the L-shapes; therefore, the outside diameter corresponds to a plate bearing.

Subassy; Visors

Part; Visor top

This part is there to simplify horizontal alignment. It looks the way it does to simplify the assembly.

Part; Visor Side

This part needs to connect to the Visor Top and to the Big T-Shape. Therefore, it has rivets to connect to the later and slots to connect to the first.

Subassy; Pulley

Part; Big Disc

On the university's prototyping website [2] the minimal size for pulleys for 4 mm rope is stated to be 38 mm in diameter. This Big Disc is 4 mm bigger to ensure the rope does not fall off. All pulley discs have an extra hole through them through which a bolt can be placed to secure and align these parts.

Part; Small disc

The small disc is the aforementioned 38 mm in diameter for the corresponding reason.

Part; Pulley visor

This visor is there to correspond to the visor on the front of the machine. The measurements are there to connect it to something.

Part; Pulley side

The side plates of the pulley subassembly are there to connect securely to the Big T-shape. Next to that holes are there to align the bottom of the pulley with the top of the cart, to ensure the rope will pull on the cart in the best direction possible. Next to that are threaded holes on both sides matched with the Big T-Shape to enable star knobs to be inserted and tightened for the angle changing mechanism.

Subassy; Release mechanism

Subassy; Casing

Part; Casing Bottom

The casing for the dart shooter was divided into two main parts - the side plates and the bottom part. This design decision was made due to the difficulty in bending the laser-cut parts as a single integrated casing. By separating the casing into sides and a bottom, the manufacturing process was simplified, and the parts could be more easily secured together using rivets.

The bottom part of the casing measures 46x80mm, ensuring it does not protrude width wise from the T-shape while lodging the entire release mechanism. It features two 23h6 holes on each longer side, where the sleeve bearings with inner diameter of 20mm will go. Additionally, it has 4.2mm connection holes on the flanges to secure this part to the Casing Sides using rivets. And at the bottom 5.5mm diameter holes for securing the spring plate to it.

Part; Casing Side

The sides of the casing have four 4.2mm diameter connection holes, as well as holes on the flanged edge facing upwards to connect the entire casing to the bottom of the T-shape.

Subassy; Pinplate

Part; Release pin

The primary function of the release pins, which protrude through the bottom of the Big T shape, is to interfere with the dart cart, holding it in place until the release mechanism is activated, at which point the release pin retracts, allowing the cart to slide forward. There are two release pins to ensure secure engagement with the dart cart, providing stability and redundancy in the mechanism.

Part; Spring plate

The spring plate at the bottom of the casing has dimensions of 36x78mm, designed to avoid interference with the casing's connection points and bends at the bottom. It features four 10mm diameter holes in the corners to hold 10 mm diameter, 30 mm long

springs. To prevent the spring plate from moving, it has four 5.5mm diameter tapped holes that align with the holes in the casing.

Part; Release plate top

The release plate, which is supported by springs, measures 36x80mm. This width accommodates the dimensions and rotation of the gears and bearing sleeves in the mechanism. The release plate is also constructed from 2x3mm thick laser-cut parts that are spot-welded. However, the top and bottom plates have distinct features.

The top release plate includes two 10mm diameter holes located in the middle of the 80mm length, spaced 20mm apart from each centre. These holes are intended for securing the pins.

Part; Release plate bottom

The bottom release plate features four 10mm diameter holes that align with those on the spring plate, securing the springs and further holding the release plate in place. Additionally, in the middle of the plate, countersunk and holes have been made for M4 screws, aligning with the centre holes on the top plate, to secure the release plate and the pins together.

Subassy; Acylindrical bars

Part; Lever

The lever in the release mechanism is designed to rotate the part big bar, thus actuating the release mechanism. It is made from a cold-formed square tube with dimensions of 25x25mm. A 20mm diameter hole is drilled at one end of the lever, allowing the Big bar to be inserted and welded, preventing any motion between them. The length of the lever was altered after initial assembly, to accommodate for a smaller than anticipated range of motion.

Part; Big bar

The Big bar has a diameter of 20mm and a length of 40mm, allowing it to sit within the casing. There are two such bars, each with 2xM4 tapped holes on each face of the ends. Both Big bars connect to the gear bars at the same end. At the other end, one Big bar connects to the simple bar, while the other connects to the lever bar. Using two bars to push the release plate ensures greater stability and even force distribution, preventing wobbling and damage.

All bars in the release mechanism have two identically aligned holes: one at the centre and another at a 5mm radius from the centre. The parts connect in such a way that the centre hole of the one aligns with the outer hole of the others, and vice versa. This acylindrical arrangement forms the basis of the release mechanism. The bars at the end function as the axles, staying concentric within the sleeve bearing, while the Big Bar parts, rotate downwards and push the pinplate down.

Part; Small bar Simple

The simple bar is also made from free machining steel with a diameter of 20mm and length 23mm, which is to make sure it is as small as possible while still protruding out of the sides of the casing.

Part; Small bar Gear

The gear bar has the same dimensions as the simple bar. However, 6mm from the top has been cut flat, aligning with a 45 degree angle relative to the holes, to easily connect the gears to the release mechanism.

Part; Small bar Lever

This bar has similar dimensions as the other bars in this subassembly, it is, however, longer than the other and has tapped, deep counterbored holes to connect to the big bar and avoid interference with the T-shape.

Part; Release Gear

The purpose of the gears is to ensure both big bars rotate down at the same moment with same rate. The gears are made from 2mm thick sheet metals. There are two of them and they feature a hole that fits the gear bar precisely.

Subassy spring modules

Part; trident lock springs

This part secures the springs in the longitudinal direction onto the springs grid locks. They are made from 3 mm sheet metal to be as strong as possible. If necessary, they can be welded onto the other parts to avoid the subassembly falling apart.

Part; pins connection

These pins stick through the grid lock part and have 2 holes, one for the trident to pass through with a friction fit and a hole for an M8 screw. These pins are designed to fit into the grid locks with a very tight fit to reduce the play between the parts.

Part; springs grid lock t frame

These parts and the next are very similar. They both have a 3x3 grid of rectangular slots through which the springs will be positioned. This part has two flanges, one on either side, both flanges have an aligned hole through them for an M12 bolt to connect this part to the T-frame.

Part; springs grid clear

This part has the same grid as the previous one. Next to that it has holes for the pins connection. This makes a connection to the L-shapes quick and easy.



Chapter Four - Implementation & Evaluation

After conceptualisation, detailing the design, making 3D models and technical drawings it was time to make the full-scale prototype.

This was done using the machines and materials available in the Hangar at the UT. For all machined parts work preparation sheets were made to simplify the machining process, these can be found in Appendix Six - Work Preparation Sheets. Next to the milling and turning the laser cutter was used for a lot of parts. Machined parts which needed to have a certain accuracy, in order for them to connect properly to other parts were measured and compared to their tolerances. The results of these measurements are shown in the measurement report in Appendix Seven - Measurement Report.

During the final days of the production process, a number of final improvements were made to the design, based on its practical performance. The improvements made have been ordered in this table, including the reason behind the corresponding improvement.

Improvement	Reason
Spring modules were replaced with 4 springs.	Spring modules were thought to be too strong; the risk of the prototype breaking was too big.
Small T Top was reinforced with extra plates.	Force with which the Cart slammed into this part was too big.
Cart Frame was reinforced with extra plates.	Force with which the Cart slammed into Small T Top was too big.
Cart Hook was redesigned to hold up to higher forces.	The part was not strong enough, it would break.
Lever of release mechanism was shortened.	Lever could not be pushed down far enough to release the cart.
Shape and height of both Release Pins were altered.	Points interfered with the releasing of the cart.
Small Lever Bar was lengthened.	Lever would collide with the T-shape.
Arm of winch was made longer.	Original winch lever had too little mechanical advantage.
Baseplate-Spacers were shortened.	Distance between insides of upright parts of Baseplate became smaller when bent than anticipated in model.
Square tube was added to strengthen Baseplate.	Baseplate would bend when cart was drawn back.
Joining method between the Big T and Small T-Top was changed from rivets to bolts.	Connection was not strong enough.
Instead of 2 knobs for angle adjustment, only one was used.	Second knob turned out to be redundant and inconsequential to functionality of device.

With the prototype fully assembled, it was necessary to verify if the shooter actually met all the requirements described in the list of requirements (Appendix One - List of Requirements).

The Material and Production, the Dimensional and the Time requirements were (almost) all met. The only exception was R03.03: 'The product shall 'shoot' the darts with enough speed and under such an angle that the darts stick on the board for at least 2 full minutes.', this is because the dart is not yet able to reach the distance consistently. The functional requirements are all estimated to be met, but this can only be verified properly after extensive testing. The first, second and fourth User/Safety requirement are also estimated to be met, the R05.03: 'The product shall be operable by one person, meaning there shall be two or less places where a manual force needs to be exerted at one time, whilst at the same time meaning that those places shall be reachable at the same time.' is not met due to the working of the lever of the release mechanism. This does not stay upright as anticipated, but needs to be held up manually. This causes the device to not be as safe as it should have been, even though it is still possible to operate the device individually.

Based on the evaluation, it can be concluded that there is still room for improvement. The most obvious ones are stated below.

- Requirement R03.03: ' The product shall 'shoot' the darts with enough speed and under such an angle that the darts stick on the board for at least 2 full minutes.' was not met. To improve this, stronger springs could be used to shoot the dart with a higher energy. Therefore, it would have a higher velocity upon impact.
- Requirement 05.03: ' The product shall be operable by one person, meaning there shall be two or less places where a manual force needs to be exerted at one time, whilst at the same time meaning that those places shall be reachable at the same time.' was not met as well. This is because the lever of the release mechanism keeps falling down. Something must be designed to keep the lever upright. A possible solution for this is altering the design to accommodate for another spring mechanism, which pulls the lever up.

Other improvements, independent to the LoR (Appendix One - List of Requirements), that could be implemented if this design were to be taken further include:

1. To improve the safety of the user while using the device, moving parts could be encased to protect the user from any potential harm caused by these moving parts.
2. The design could make use of handles on the baseplate for it to be easier to carry, on the same subject the materials could be altered to make the device have a smaller mass.
3. Improving ease of use could be done by considering ergonomics and human anatomy: changing the design of the levers to include a grip or handle or altering their dimensions for them to be more comfortable to use.



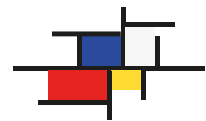
Conclusion

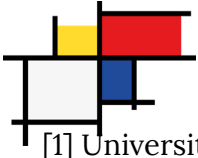
The design and development of the dart shooter have been marked by a rigorous process of conceptualization, detailed design, and iterative improvements, culminating in the production of a functional prototype. Throughout this project, various manufacturing techniques, including laser cutting, welding, milling, and turning, were utilized to produce the necessary components, ensuring both precision and reliability in the final assembly. The integration of these methods played a crucial role in achieving the required structural integrity and operational efficiency, which were key objectives of the design.

With limited time and resources, which restricted opportunities for extensive testing, some aspects of the prototype's performance were not fully evaluated. Notably, the spring system proved to be too powerful for the prototype, limiting its operational range and leading to structural concerns. Moreover, certain components required further reinforcement to enhance rigidity, while safety features were not fully implemented, given the prioritization of functional goals.

In light of these findings, several improvements are recommended to address the shortcomings observed during testing and evaluation. These include the adoption of springs with a more appropriate force rating to ensure the prototype can achieve the full operational range without compromising structural integrity. Additionally, incorporating safety features, such as shielding for moving parts and a more robust safety mechanism, will be crucial for ensuring user safety. Further refinement of the design is also recommended to improve usability and operational efficiency.

Overall, while the dart shooter prototype achieved its primary goal of functionality, there remains significant potential for optimization in future iterations. By addressing these identified improvements, the project can evolve into a more reliable and user-friendly product, meeting both performance and safety requirements.





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[2] Prototyping. (n.d.-b). Universiteit Twente. <https://www.utwente.nl/en/et/dpm/intranet/prototyping/>

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[5] Darts (2025). Wikipedia <https://en.wikipedia.org/wiki/Darts>



Appendices

Appendix One - List of Requirements

1	Material and Production Requirements	
R01.01	The product shall be made from materials available to us in the Hangar, and/or include up to 25 euros worth of additional materials including paints.	
R01.02	The product shall be made with production methods available to us in the Hangar.	
R01.03	The product shall contain at most two 3D printed parts. These parts will have to fit into a box with ribs of 0.6m each. These parts will have to be checked by our tutor before they are to be printed.	
R01.04	The product shall be produced and assembled in such a manner that it will stay securely together for the period of time up to and including the project exam.	
R01.05	The parts for the product shall be made within the timeframe of four workshop moments.	
2	Dimensional Requirements	
R02.01.01	Dart	The product shall be able to shoot darts with the below mentioned dimensions.
R02.01.02		The product shall have space for at least one dart with the below mentioned dimensions to be stored/loaded until release. Total length: 170mm Total width: 42mm Total height: 42mm Diameter grip: 20mm Back of fin till back of grip: 85mm Back of grip till back of suction cup: 75mm Depth suction cup: 10mm Frontal diameter suction cup: 30mm
R02.02.01	Table	The product shall be able to be secured onto a table with the below mentioned dimensions. Height of table: 0.9m Width of table: 0.5m Length of table: 1.0m
R02.03.01	Board	The product shall 'shoot' the darts at a target board with the below mentioned dimensions. Height of centre of target: 1.5m Radius of outer edge of target: 0.6m Distance from edge of table: 4.0 m up to and including 8.0 m.
R02.04.01	Machine	The product shall not be bigger than the top area of the table.
R02.04.02		The product shall shoot the darts from a height of 0.2 m above the table.
3	Time requirements	
R03.01	The product shall be able to be secured onto the table within the time constraints of 50 seconds.	
R03.02	The product shall be able to 'shoot' three darts at the dart board within the time constraint of 1 minute and 50 seconds in total.	

R03.03	The product shall 'shoot' the darts with enough speed and under such an angle that the darts stick on the board for at least 2 full minutes.
4	Functional requirements
R04.01	The angle under which the product shoots the darts shall be adjustable within the range of 7.5° till 15.2°.
R04.02	The product shall ensure that a velocity of 21.20 m/s can be achieved by the darts at the moment when they leave the product.
R04.03	The product shall 'shoot' the darts in such a manner that the angle under which the dart hits the board shall lie within a 15° range relative to the ground.
R04.04	The product shall 'shoot' the darts in such a manner that the velocity with which the dart hits the board shall be at least 5.15 m/s.
R04.05	The product shall 'shoot' the darts in such a manner that the darts hit the board within 0.3 m of the centre.
5	User/safety requirements
R05.01	The product shall have a mass smaller than 23 kg.
R05.02	The product shall take less than 115 N to operate.
R05.03	The product shall be operable by one person, meaning there shall be two or less places where a manual force needs to be exerted at one time, whilst at the same time meaning that those places shall be reachable at the same time.
R05.04	The product shall have no sharp edges on which a person can cut themselves while using the product.

Appendix Two - Morphological Chart

Clamping device	Glue clamp	Hook	Rope	Weight	Springs	Double sided tape	Suction cup	Glue	Velcro	Magnet
Human power input	One hand	Two hands	One foot	Two feet						
Energy input motion	Push	Pull	Squeeze	Rotating hands	Kicking forward	Kicking down	Rotating feet			
Charging system	Lever	Pulley	Threaded rod	Winch	Double lever	Crank	Ratchet	Cam mechanism		
Store/control energy	Compression spring	Tension spring	Torsion spring	Spiral torsion spring	Layered spring	Mass	Flywheel	Magnetic		
Loading dart	From top	From front	From side	From back	Carousel	Magazine	Slide holes			
Holding dart	Groove	Tube	Pin in back	Cart						
Height adjustment	Pin-holes	Rack	Lock	Ratchet	Support bar	Threaded rod	Pulley	Gear train	Scissors	
Aiming	Visor	Laser	Riflescope							
Release	Trigger	Ratchet	Pin-hole	Button	Lever	Clamp	Plank			
Shooting	Lever	Impulse	Wedges	Rack	Linear guide	Flywheel				

Appendix Three - Concept Choice

Three A - Selection Table

	Criteria	Priority	W/F	Concept 1 (Twang)	Concept 2 (Boing)	Concept 3 (Woosh)
	Material and Production					
C1.1	Uses Hangar materials and/or fits within the 25-euro budget	M	4	4	4	3
C1.2	Uses Hangar production methods	S	3	4	4	4
C1.3	Includes maximum of 2 3D printed parts	M	4	4	4	4
C1.4	Sturdy assembly lasting through the project exam	M	4	4	4	3
C1.5	Time management	S	4	3	4	4
	Dimension					
C2.1	Compatible with specified dart dimensions	M	4	4	4	4
C2.2	Includes loading space for at least one dart	M	4	4	4	4
C2.3	Securable on table	M	3	4	4	4
C2.4	Works with the target board dimensions	M	4	4	4	2
C2.5	Fits within the table surface area	C	1	3	4	4
C2.6	Maintains proper dart shooting height (0.2m)	C	2	4	2	4
	Time					
C3.1	Quick and easy to secure (50sec)	M	4	4	4	4
C3.2	Can shoot three darts within set amount of time (1min50sec)	M	4	4	4	4
C3.3	Darts stick to the board	S	3	3	3	3
	Functionality					
C4.1	Allows adjustment of shooting angle	S	3	3	2	1
C4.2	Hits the board with a minimum velocity of 5.15 m/s.	S	3	4	4	2
C4.3	Ensures darts impact within a 0.3 m radius from the centre of the board	C	2	2	2	1
C4.4	Shoots the darts to achieve specified velocities upon leaving the product.	W	1	4	4	2

C4.5	Launched darts have an impact angle on the board between 70° and 110° (0° = parallel to the board)	W	1	2	2	1
	User/Safety					
C5.1	Mass smaller than 23kg	C	3	3	3	3
C5.2	Takes maximum 115N to operate	C	3	4	3	3
C5.3	Ensure single-person operation of the product.	M	4	4	4	4
C5.4	Provide simultaneous reachability of all force points.	C	2	4	4	4
C5.5	Limit manual force exertion to two points or fewer simultaneously.	C	4	4	4	4
C5.6	No sharp edges	M	4	4	4	4

Three B - Grading Scales

Criteria	Performance scale	Description
C1.1	Availability materials	
		1 We need to exceed the budget by more than 15 euros to buy extra material.
		2 We need to exceed the budget by between 0 and 15 euros to buy extra material.
		3 We need to use the budget we have to buy extra material.
		4 All of the material we need is available in the Hangar.
C1.2	Production Method availability	
		1 Not 1 part of the product can be made with the production methods available.
		2 Some parts of the product can be made with the production methods available.
		3 All parts, except 1 or 2 parts cannot be made with the production methods available.
		4 The product can be completely made with the production methods available.
C1.3	Number of 3D Printed parts	
		1 The product cannot be made with less than 3 3D printed parts.
		2 The product needs 2 3D printed parts.
		3 The product needs 1 3D printed part
		4 The product does not need any 3D printed parts
C1.4	Product security	

		1 The product will fall apart when you use it.
		2 The product has a big chance that it falls apart when used.
		3 The product can fall apart, because not everything is completely fastened.
		4 The product is secure and there is no chance that it will fall apart
C1.5	Time management	
		1 It will most certainly take more time to make all the parts than we have available in the three workshop moments.
		2 It will likely take more time to make all the parts than we have available in the three workshop moments.
		3 All parts of the product can likely be made within the three workshop moments we have available
		4 All parts of the product can certainly be made within the three workshop moments we have available
C2.1	Shootability dart	
		1 The product is not able to shoot the dart
		2 The product is maybe able to shoot the dart.
		3 The product is probably able to shoot the dart.
		4 The product is 100% able to shoot the dart.
C2.2	Storability dart	
		1 The dart cannot be stored.
		2 The dart can maybe be stored.
		3 The chance is big that the dart can be stored.
		4 The dart can 100% be stored.
C2.3	Table dimensions	
		1 The product cannot be secured to the table, as it does not match the table's dimension
		2 The product can be secured to the table, but requires major adjustment
		3 The product can be secured to the table with minor adjustment.
		4 The product can be secured to the table perfectly, matching the table dimension
C2.4	Dart board dimension	
		1 The darts consistently miss the target board or fail to hit within the specified dimensions
		2 Darts occasionally miss or fail to align with the target board dimensions.
		3 Darts mostly hit the target board with minor deviations.

		Darts consistently hit the target board within target board dimensions
C2.5	Table surface area	
		1 The product exceeds the top surface area of the table
		2 The product barely fits within the top surface area of the table
		3 The product fits within the top surface area of the table with minimal overhang
		4 The product perfectly fits within the top surface area of the table
C2.6	0.2m above the table	
		1 The product cannot shoot from the height of 0.2m above the table
		2 The product can shoot from a height less than 0.2m above the table
		3 The product can shoot from a height close to 0.2m above the table
		4 The product can shoot from a height of 0.2m as specified
C3.1	Secured within 50 sec	
		1 The product takes more than 50 seconds to be secured to the table
		2 The product can be secured to the table within 50 seconds with major effort
		3 The product can be secured to the table within exactly 50 seconds
		4 The product can be secured to the table in less than 50 seconds
C3.2	Three darts in 1 min 50 sec total	
		1 The product takes more than 1min 50sec to shoot three darts
		2 The product can shoot three darts but exceeds the time limit slightly (e.g. by 10 sec)
		3 The product can shoot three darts within exactly 1min 50sec
		4 The product can shoot three darts in less than 1min 50sec
C3.3	Stick for at least 2min	
		1 The darts consistently fail to stick to the board for 2 full minutes
		2 The darts occasionally stick to the board for slightly less than 2 minutes
		3 The darts consistently stick to the board for at least 2 full minutes under normal conditions
		4 The darts consistently stick to the board for significantly longer than 2 minutes under all conditions

C4.1	Shooting angle of 0°-45°	
		1 The product does not allow any adjustment to the shooting angle
		2 The product allows partial adjustment of the shooting angle but does not cover the entire range of 0° to 45°
		3 The product allows adjustment of the shooting angle within the range of 0° to 45°, with minor limitations
		4 The product allows precise and smooth adjustment of the shooting angle across the full range of 0° to 45°
C4.3	Impact angle of 70° and 110° (0°= parallel to the board)	
		1 The darts fail to hit the board at an angle within the range of 70° to 110°
		2 The darts hit the board at an angle less than 70° or greater than 110°, but some shots are within the range
		3 The darts mostly hit the board within the range of 70° to 110°, with occasional minor deviations (<10° outside the range)
		4 The darts always hit the board at an angle exactly within the range of 70° and 110°
C4.4	hit velocity>5.15m/s	
		1 The darts frequently or consistently hit the board at a velocity below 5.15 m/s
		2 The darts hit the board at a velocity close to 5.15 m/s but less
		3 The darts hit the board at a velocity of at least 5.15 m/s
		4 The darts hit the board at a velocity more than 5.15 m/s
C4.5	hit the board within 0.3m from its centre	
		1 The darts frequently or consistently hit the board outside the 0.3 m radius
		2 The darts hit the board within the 0.3 m radius, but some shots land outside
		3 The darts hit the board just around the 0.3 m radius
		4 The darts consistently and accurately hit the board well within the 0.3 m radius of the centre
C5.1	Mass smaller than 23 kg.	
		1 The product has a mass bigger than 23 kg.
		2 The product has a mass between 18 kg and 23 kg.
		3 The product has a mass between 18 kg and 13 kg.
		4 The product has a mass smaller than 13 kg.

C5.2	Takes maximum 115N to operate	
		1 It certainly takes more than 115N to operate the product.
		2 It likely takes more than 115N to operate the product.
		3 It likely takes less than 115N to operate the product.
		4 It certainly takes less than 115N to operate the product.
C5.3	Ensure single-person operation of the product.	
		1 The product does not ensure a single person operation for the product.
		2 The product will be operable with a little help (i.e. holding a lever) from a second person.
		3 The product will take some effort (i.e. walking around the device) to be able to be manually operable by one person.
		4 The product can easily be used by a single person.
C5.4	Provide simultaneous reachability of all force points.	
		1 The product has more than four points where a force needs to be exerted, and they are not easily reachable
		2 The product has three or four points where force needs to be exerted, and they cannot be reached from the same point
		3 The product has two or three points where force needs to be exerted, and they cannot be reached from the same point.
		4 The product has one or two points where force needs to be exerted, and all of them are easily accessible.
C5.5	Operability product	
		1 The product cannot be operated by only 1 person
		2 The product can maybe be operated by one person
		3 The product can be difficult to operate by 1 person.
		4 The product can easily be operated by 1 person.
C5.6	Sharp edges	
		1 The product has only sharp edges
		2 The product has some sharp edges
		3 The product has 1 sharp edge
		4 The product has no sharp edges

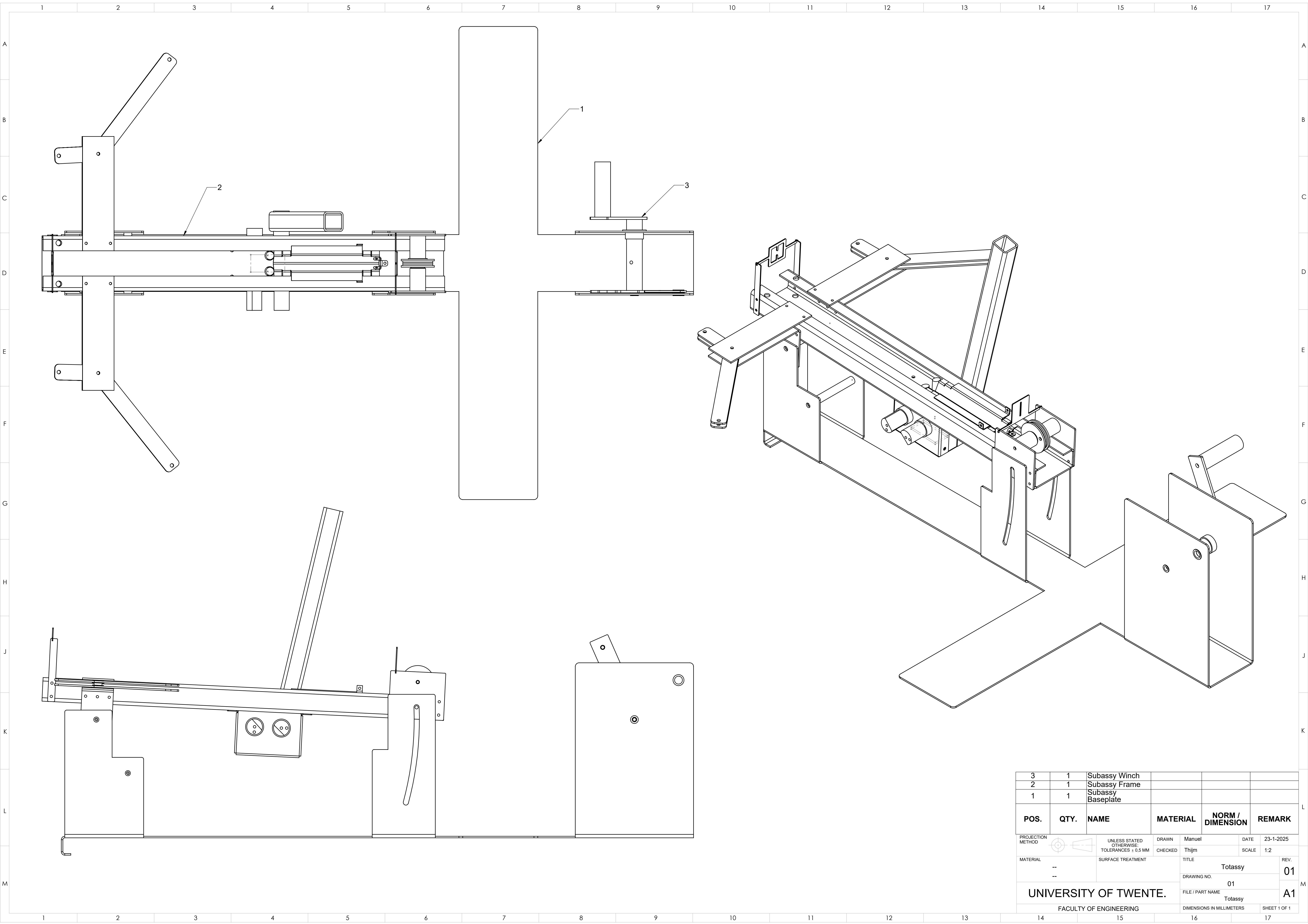
Appendix Four – Tree of Parts

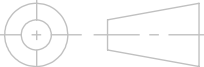
Totassy

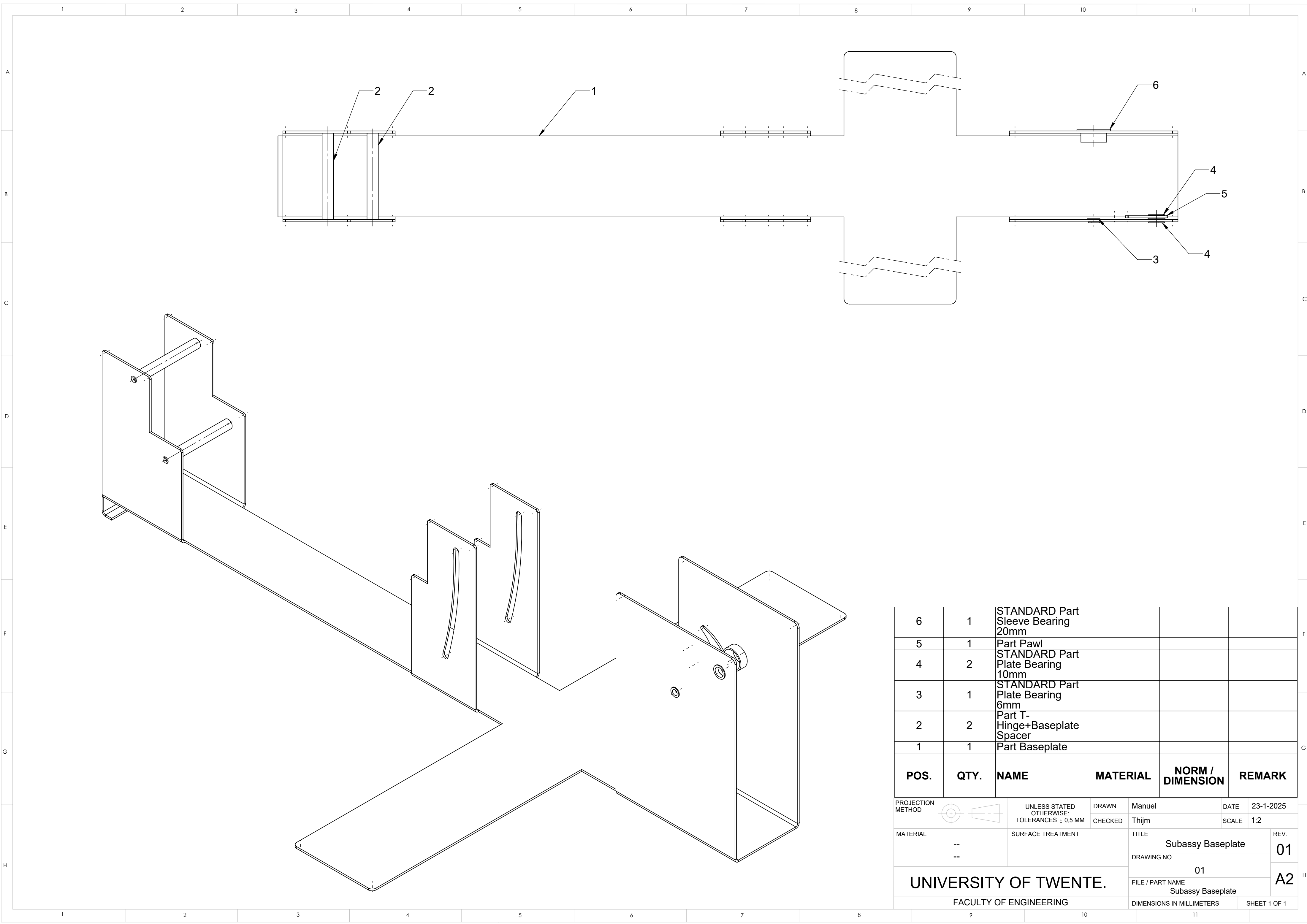
- Subassy; Baseplate
 - Part; Baseplate
 - Part; Pawl
 - Part; T-Hinge/Baseplate Spacer
- Subassy; Winch
 - Subassy; Crank
 - Part; Winch Lever
 - Part; Winch Handle
 - Part; Ratchet wheel
 - Part; Winch Axle
 - Part; cart hook
- Subassy; Frame
 - Subassy; Cart
 - Part; Cart Frame
 - Part; Cart Backplate
 - Subassy; L-Shape
 - Part; L-Shape
 - Subassy; Spring Modules bottom
 - Part; Connection Plate Spring Modules L-Shapes “connection pin”
 - Part; Spring Retainment Trident
 - Part; Connection Spring Plate T-Frame
- Part; Spring Spacing Plate L-Shape
 - Subassy; T-Shape
 - Part; Big T-Shape
 - Part; Small T Top
 - Part; Small T Bottom
 - Part; Small T Shaft
 - Subassy; Visors
 - Part; Visor Top
 - Part; Visor Side
 - Subassy; Pulley
 - Part; Big Disc
 - Part; Small Disc
 - Part; Pulley Visor
 - Part; Pulley Side
 - Part; Pulley Axle
 - Part; Pulley Axle Spacer
 - Subassy; Release Mechanism
 - Subassy; Casing
 - Part; Release mechanism Casing Side
 - Part; Release mechanism Casing Bottom
 - Subassy; Pinplate
 - Part; Release Pin
 - Part; Release Plate Bottom
 - Part; Release Pate Top
 - Part; Spring Plate

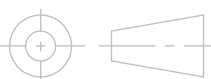
- Subassy; Acylindrical Bars
 - Part; Lever release mechanism
 - Part; Big Bar "Big asymmetric part"
 - Part; Small Bar Gear "Small asymmetric bar 2"
 - Part; Small Bar Simple "Small asymmetric bar 1"
 - Part; Small Bar Lever "Small asymmetric bar 3"
 - Part; Gear Release mechanism

Appendix Five - Technical Drawings



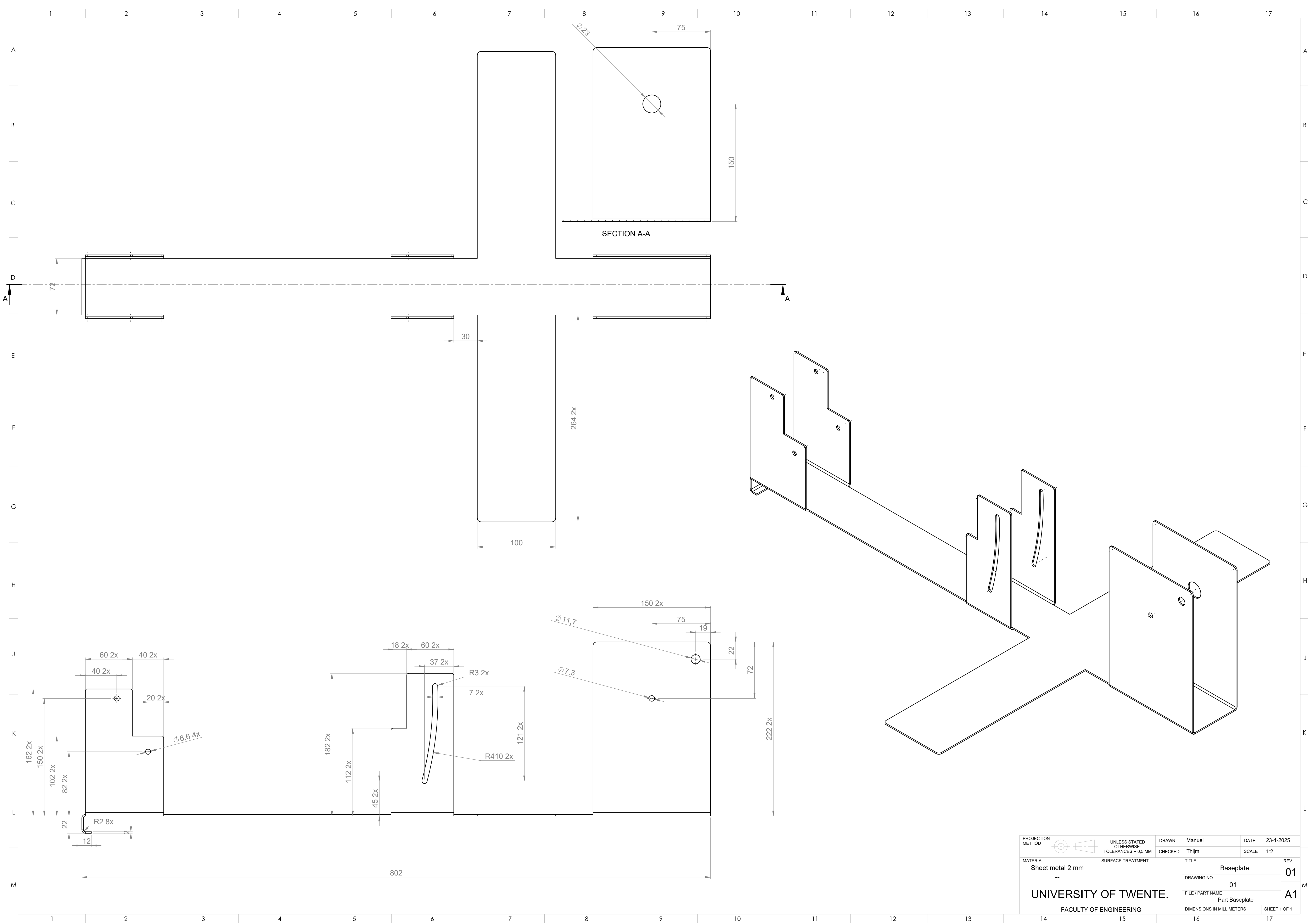
3	1	Subassy Winch						
2	1	Subassy Frame						
1	1	Subassy Baseplate						
POS.	QTY.	NAME	MATERIAL	NORM / DIMENSION	REMARK			
PROJECTION METHOD			UNLESS STATED OTHERWISE: TOLERANCES : 0,5 MM		DRAWN	Manuel	DATE	23-1-2025
			CHECKED	Thijn	SCALE	1:2		
MATERIAL		SURFACE TREATMENT	TITLE				REV.	
--			Totassy				01	
--			DRAWING NO.				A1	
			FILE / PART NAME					
UNIVERSITY OF TWENTE.			Totassy					
FACULTY OF ENGINEERING			DIMENSIONS IN MILLIMETERS		SHEET 1 OF 1			

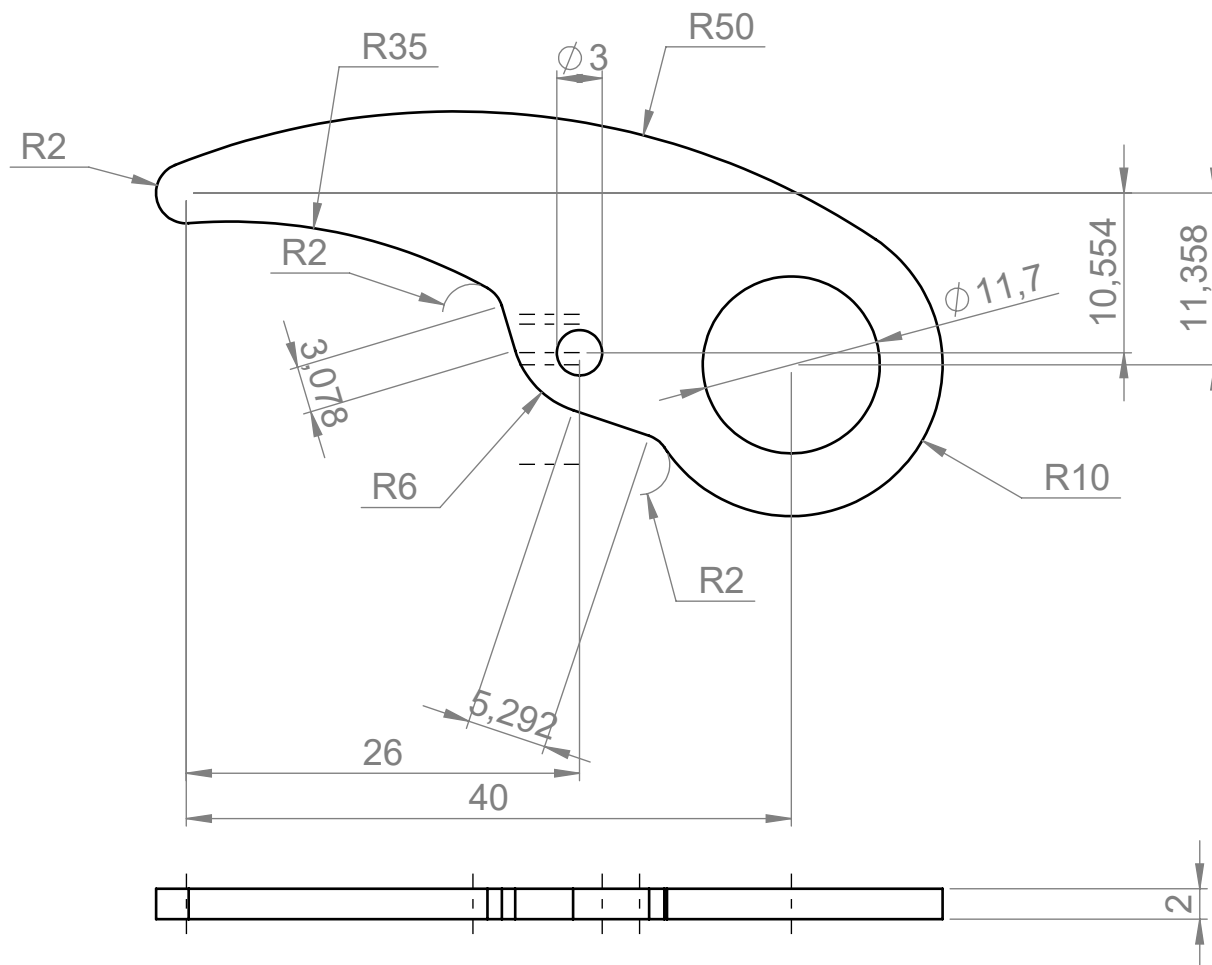


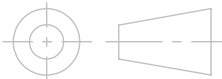
6	1	STANDARD Part Sleeve Bearing 20mm						
5	1	Part Pawl						
4	2	STANDARD Part Plate Bearing 10mm						
3	1	STANDARD Part Plate Bearing 6mm						
2	2	Part T-Hinge+Baseplate Spacer						
1	1	Part Baseplate						
POS.	QTY.	NAME	MATERIAL	NORM / DIMENSION	REMARK			
PROJECTION METHOD			UNLESS STATED OTHERWISE: TOLERANCES ± 0,5 MM		DRAWN	Manuel	DATE	23-1-2025
			CHECKED	Thijm	SCALE	1:2		
MATERIAL		SURFACE TREATMENT	TITLE				REV.	
--			Subassy Baseplate				01	
--			DRAWING NO.				A2	
			01					
UNIVERSITY OF TWENTE.			FILE / PART NAME					
			Subassy Baseplate					
FACULTY OF ENGINEERING			DIMENSIONS IN MILLIMETERS			SHEET 1 OF 1		

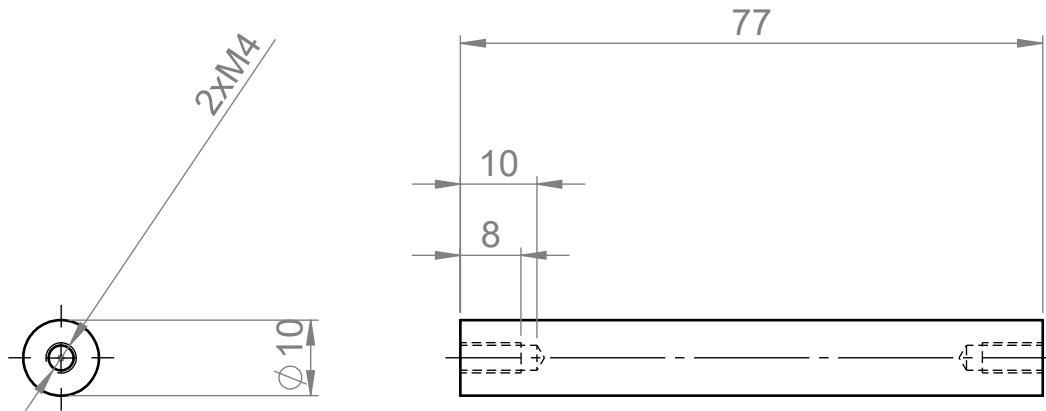
UNIVERSITY OF TWENTE.

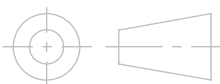
FACULTY OF ENGINEERING

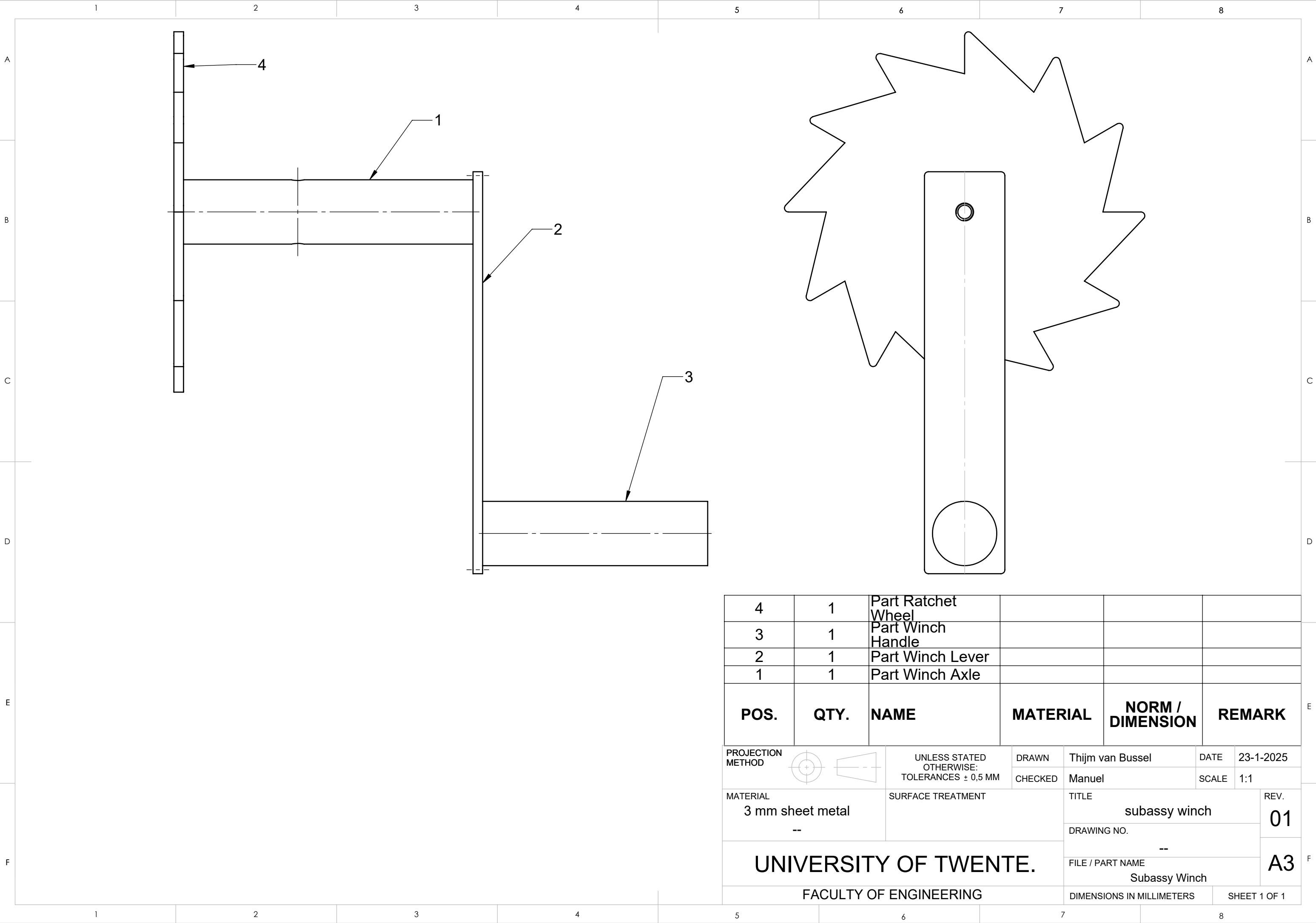


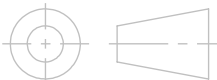


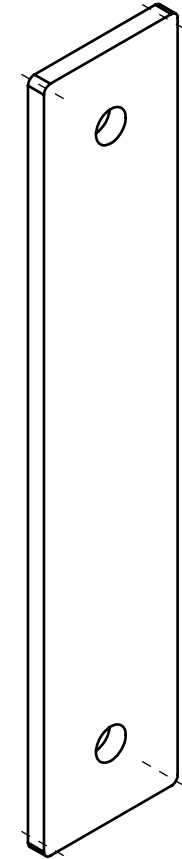
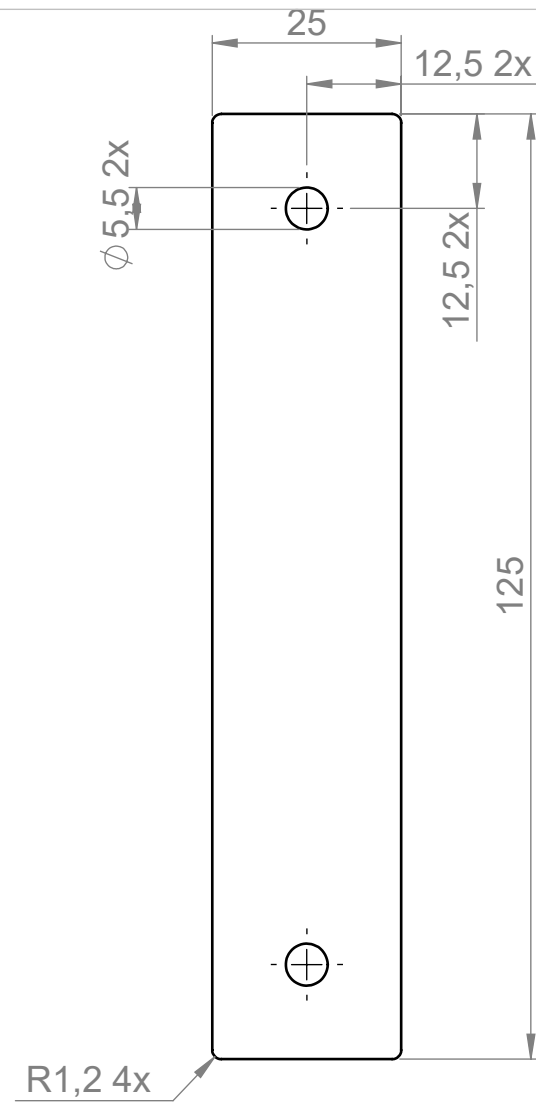
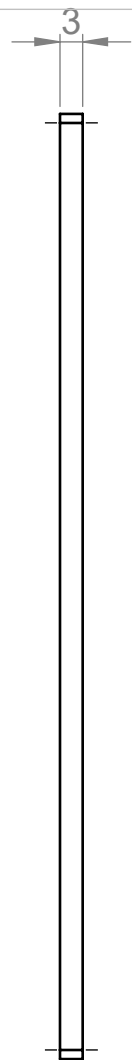
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			CHECKED	Thijm van Bussel	SCALE	2:1	
MATERIAL 2mm sheet metal --		SURFACE TREATMENT		TITLE Part pawl			REV. 01
				DRAWING NO. --			
UNIVERSITY OF TWENTE.				FILE / PART NAME Part Pawl			A4
				FACULTY OF ENGINEERING			
DIMENSIONS IN MILLIMETERS				SHEET 1 OF 1			

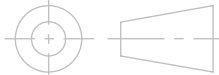


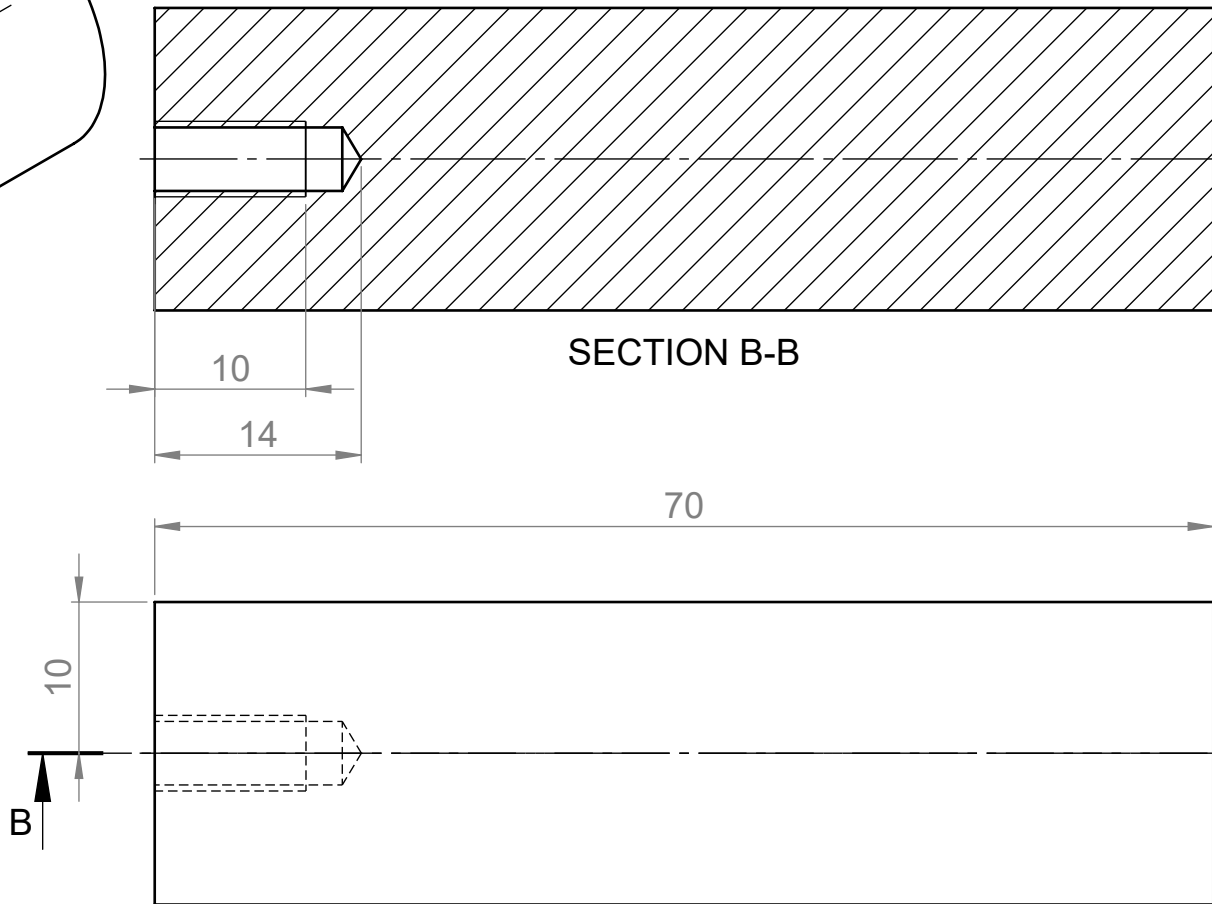
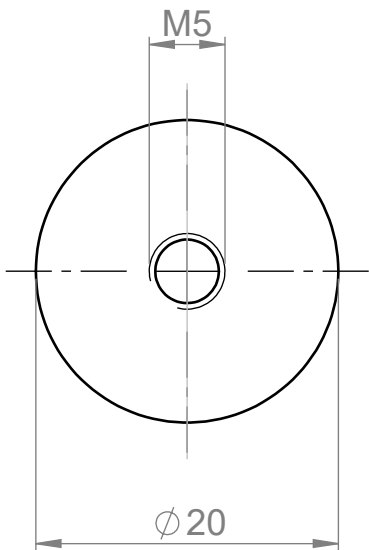
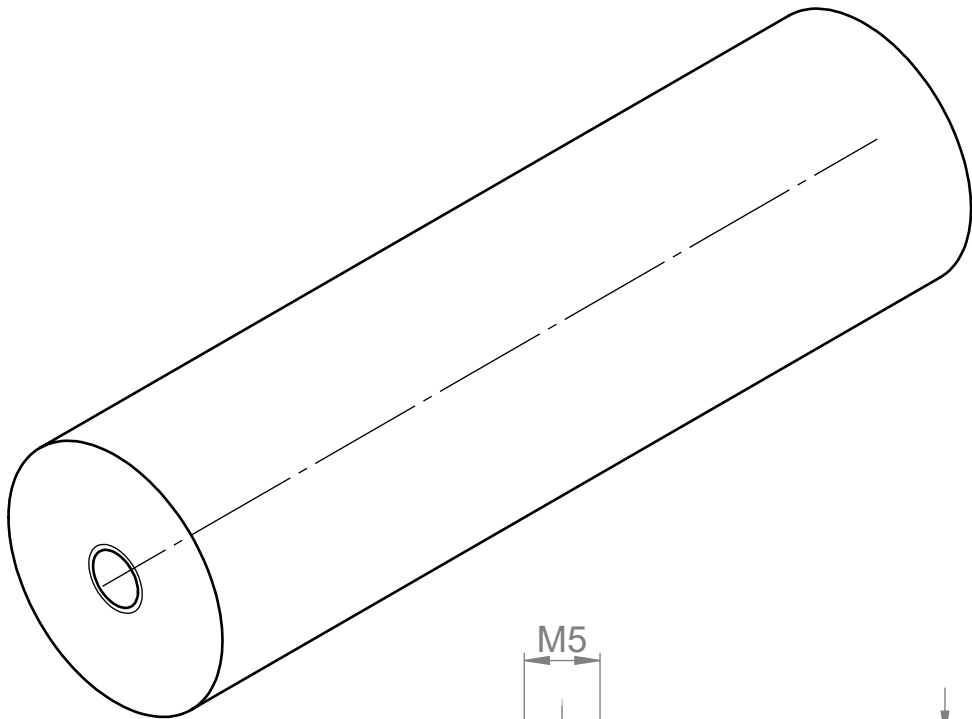
PROJECTION METHOD 	UNLESS STATED OTHERWISE: TOLERANCES ± 0,5 MM	DRAWN	Simon Hengeveld	DATE	12/01/2025
		CHECKED	--	SCALE	1:1
MATERIAL Bright Drawn Steel --	SURFACE TREATMENT	TITLE Part T-Hinge+Baseplate Spacer			REV. 01
		DRAWING NO. 04			A4
UNIVERSITY OF TWENTE.		FILE / PART NAME Technical Drawing - Part T-Hinge+Baseplate Spacer			
		FACULTY OF ENGINEERING		DIMENSIONS IN MILLIMETERS	SHEET 1 OF 1

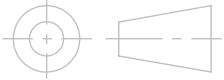


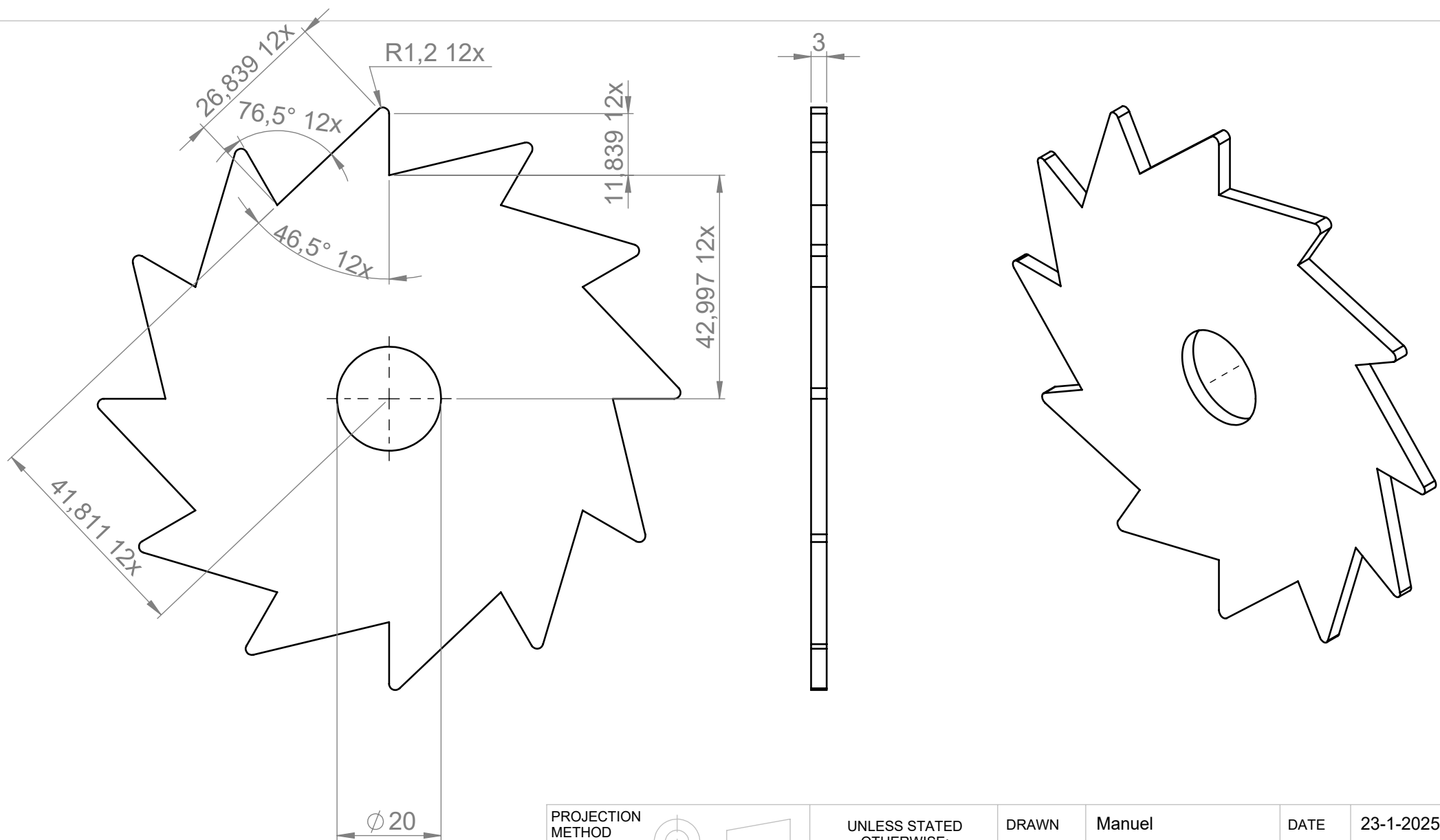
4	1	Part Ratchet Wheel						
3	1	Part Winch Handle						
2	1	Part Winch Lever						
1	1	Part Winch Axle						
POS.	QTY.	NAME	MATERIAL	NORM / DIMENSION	REMARK			
PROJECTION METHOD		UNLESS STATED OTHERWISE: TOLERANCES ± 0,5 MM		DRAWN	Thijm van Bussel	DATE	23-1-2025	
				CHECKED	Manuel	SCALE	1:1	
		MATERIAL		SURFACE TREATMENT		TITLE		REV.
		3 mm sheet metal				subassy winch		01
		--				DRAWING NO.		A3
UNIVERSITY OF TWENTE.				--				
				FILE / PART NAME				
FACULTY OF ENGINEERING				Subassy Winch				
				DIMENSIONS IN MILLIMETERS		SHEET 1 OF 1		

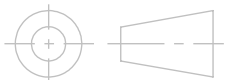


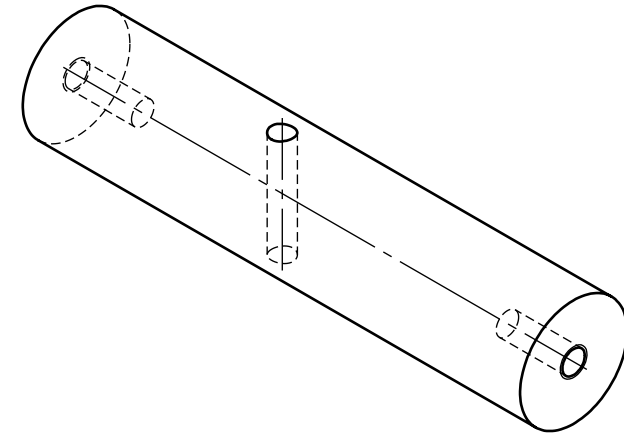
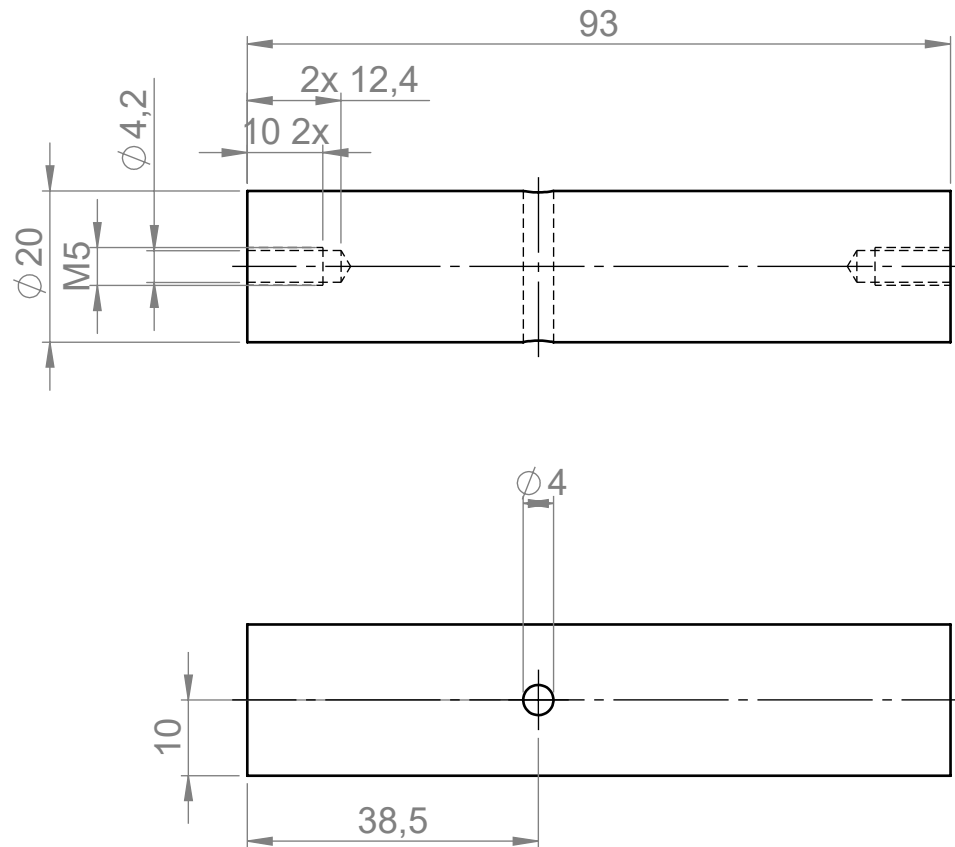
PROJECTION METHOD 	UNLESS STATED OTHERWISE: TOLERANCES ± 0,5 MM	DRAWN	Manuel	DATE	23-1-2025
		CHECKED	Thijm van Bussel	SCALE	1:1
MATERIAL Sheet metal 3 mm --	SURFACE TREATMENT	TITLE Part Winch Lever			REV. 01
		DRAWING NO. 01			A4
UNIVERSITY OF TWENTE.					
FACULTY OF ENGINEERING					
		DIMENSIONS IN MILLIMETERS		SHEET 1 OF 1	

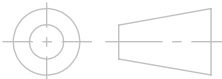


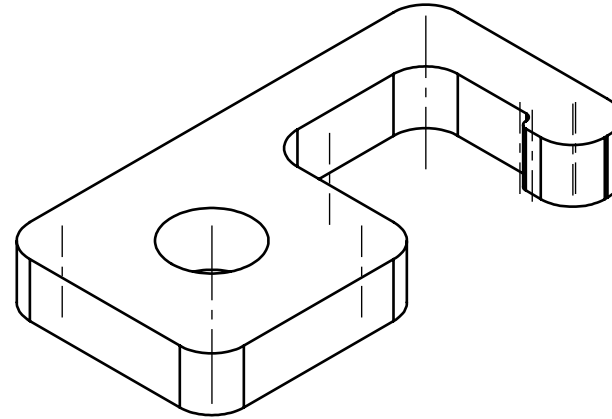
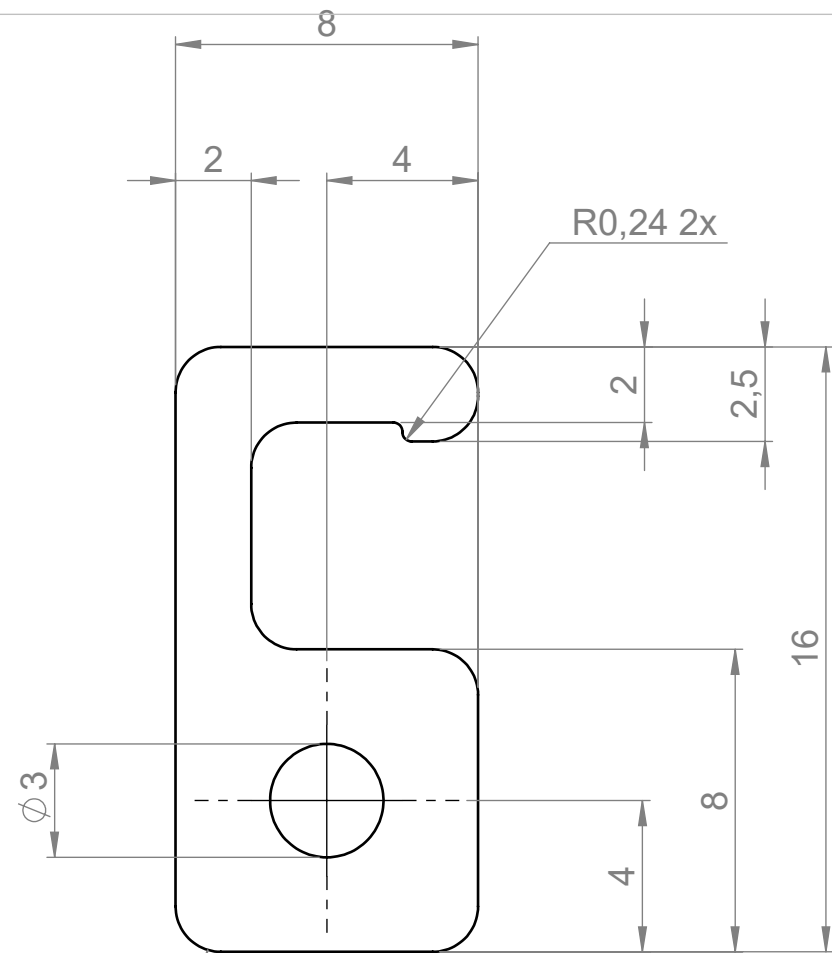
PROJECTION METHOD		UNLESS STATED OTHERWISE: TOLERANCES ± 0,5 MM	DRAWN	Manuel	DATE	23-1-2025
			CHECKED	Thijm van Bussel	SCALE	2:1
MATERIAL		SURFACE TREATMENT	TITLE		REV.	
solid iron bar diameter of 20 mm			Part winch handle		01	
--		UNIVERSITY OF TWENTE.	DRAWING NO.		A4	
			--			
		FACULTY OF ENGINEERING	FILE / PART NAME			
			Part Winch Handle			
			DIMENSIONS IN MILLIMETERS		SHEET 1 OF 1	



PROJECTION METHOD 	UNLESS STATED OTHERWISE: TOLERANCES ± 0,5 MM	DRAWN	Manuel	DATE	23-1-2025
		CHECKED	Thijm van Bussel	SCALE	1:1
MATERIAL Sheet metal 3mm --	SURFACE TREATMENT	TITLE Part Ratchet wheel			REV. 01
		DRAWING NO. 01			A4
FILE / PART NAME Part Ratchet Wheel					
FACULTY OF ENGINEERING		DIMENSIONS IN MILLIMETERS		SHEET 1 OF 1	

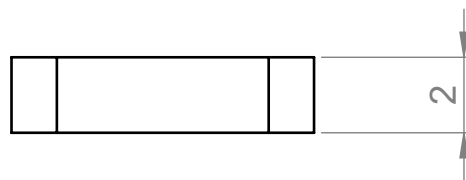


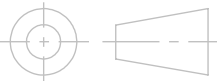
PROJECTION METHOD		UNLESS STATED OTHERWISE: TOLERANCES ± 0,5 MM	DRAWN	Manuel	DATE	23-1-2025	
			CHECKED	Thijm van Bussel	SCALE	1:1	
MATERIAL Bright drawn steel --		SURFACE TREATMENT	TITLE Part Winch Axle				REV. 01
			DRAWING NO. 01				A4
UNIVERSITY OF TWENTE.			FILE / PART NAME Part Winch Axle				
FACULTY OF ENGINEERING			DIMENSIONS IN MILLIMETERS			SHEET 1 OF 1	

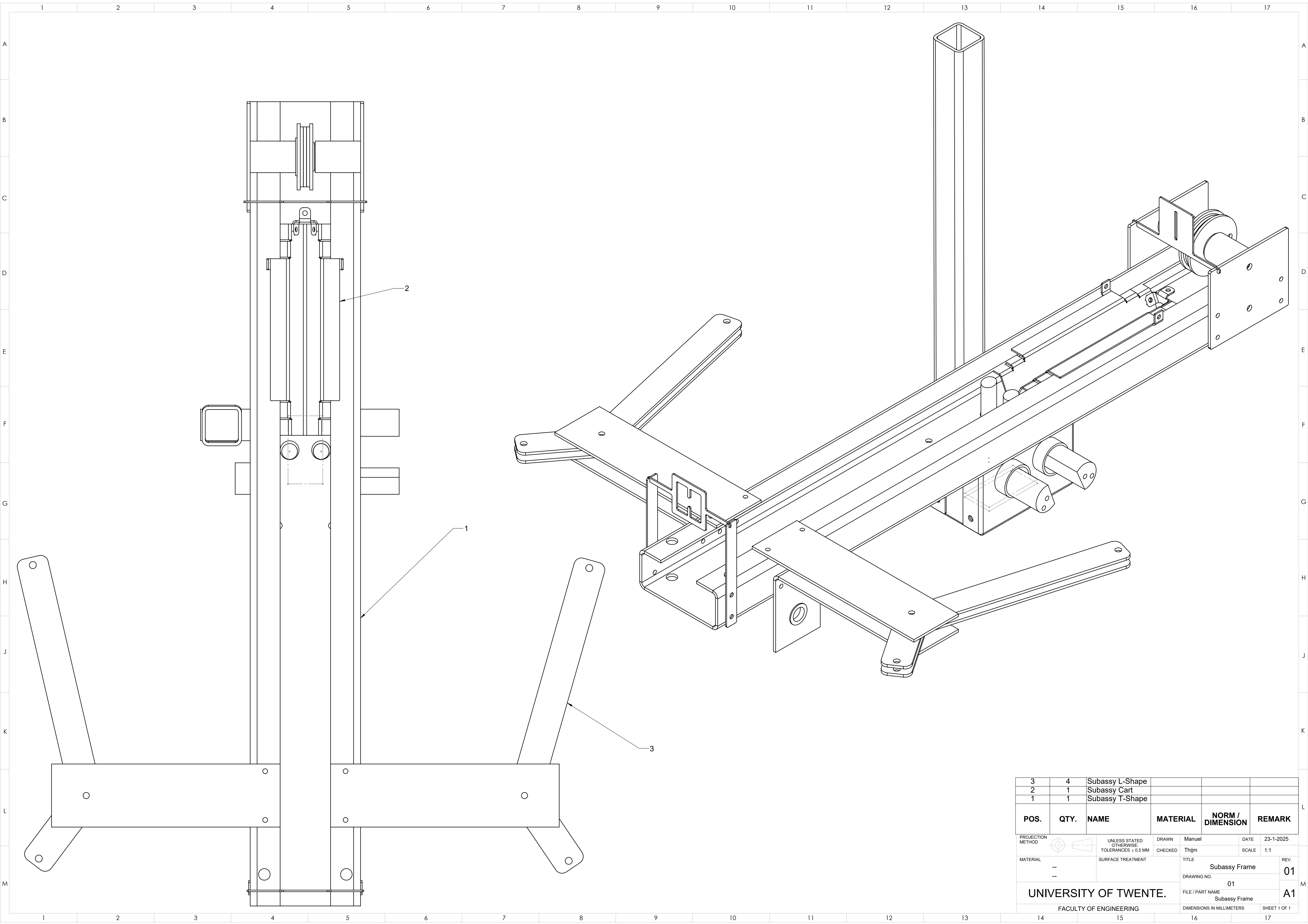


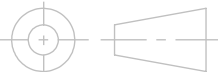
R1,2 7x

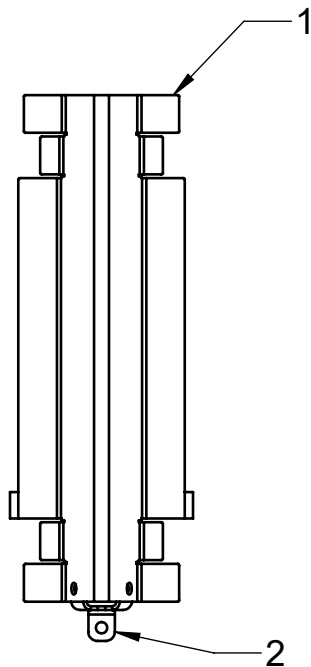
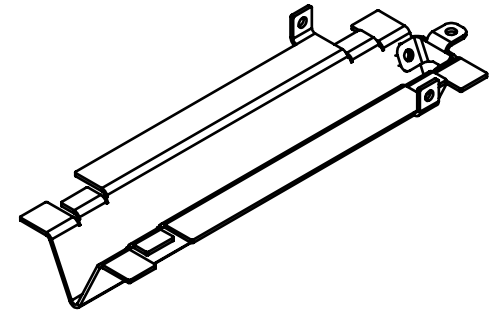
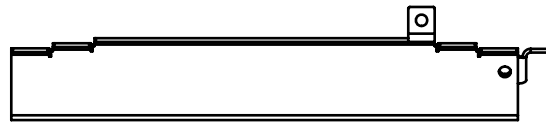
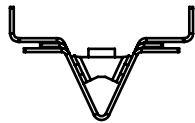
R0,24 2x

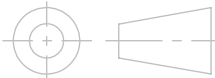


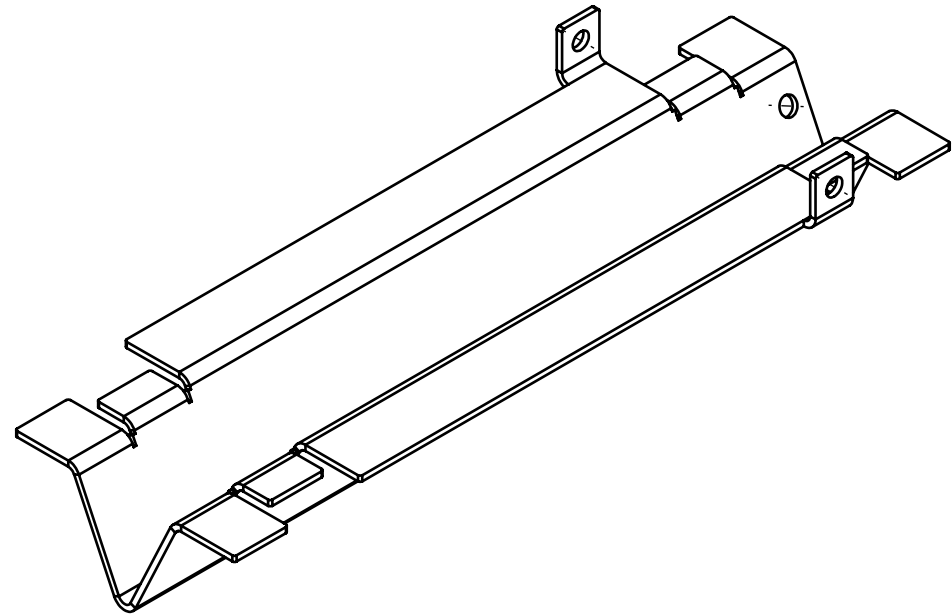
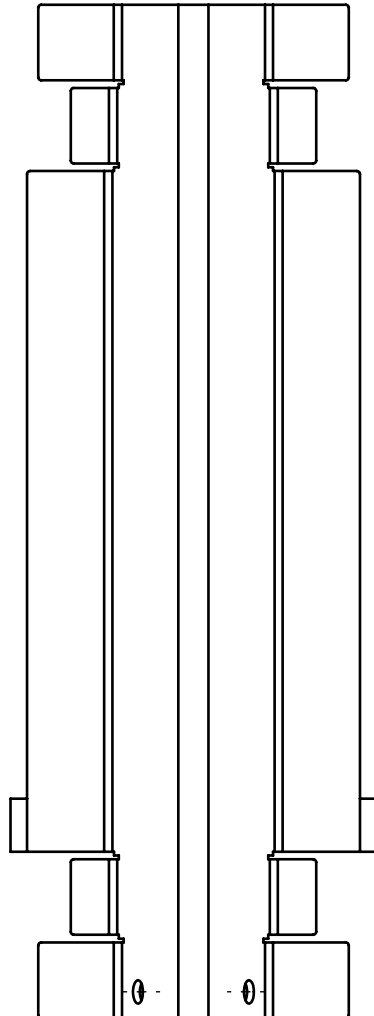
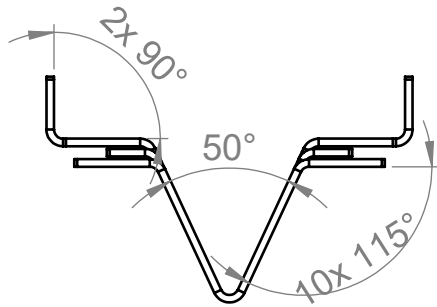
PROJECTION METHOD		UNLESS STATED OTHERWISE: TOLERANCES ± 0,5 MM	DRAWN	Manuel	DATE	23-1-2025	
			CHECKED	Thijm van Bussel	SCALE	5:1	
MATERIAL Sheet metal 2mm --		SURFACE TREATMENT	TITLE Part Cart Hook				REV. 01
			DRAWING NO. 01				A4
UNIVERSITY OF TWENTE.		FILE / PART NAME Part Cart Hook					
FACULTY OF ENGINEERING			DIMENSIONS IN MILLIMETERS			SHEET 1 OF 1	

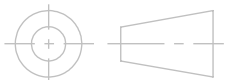


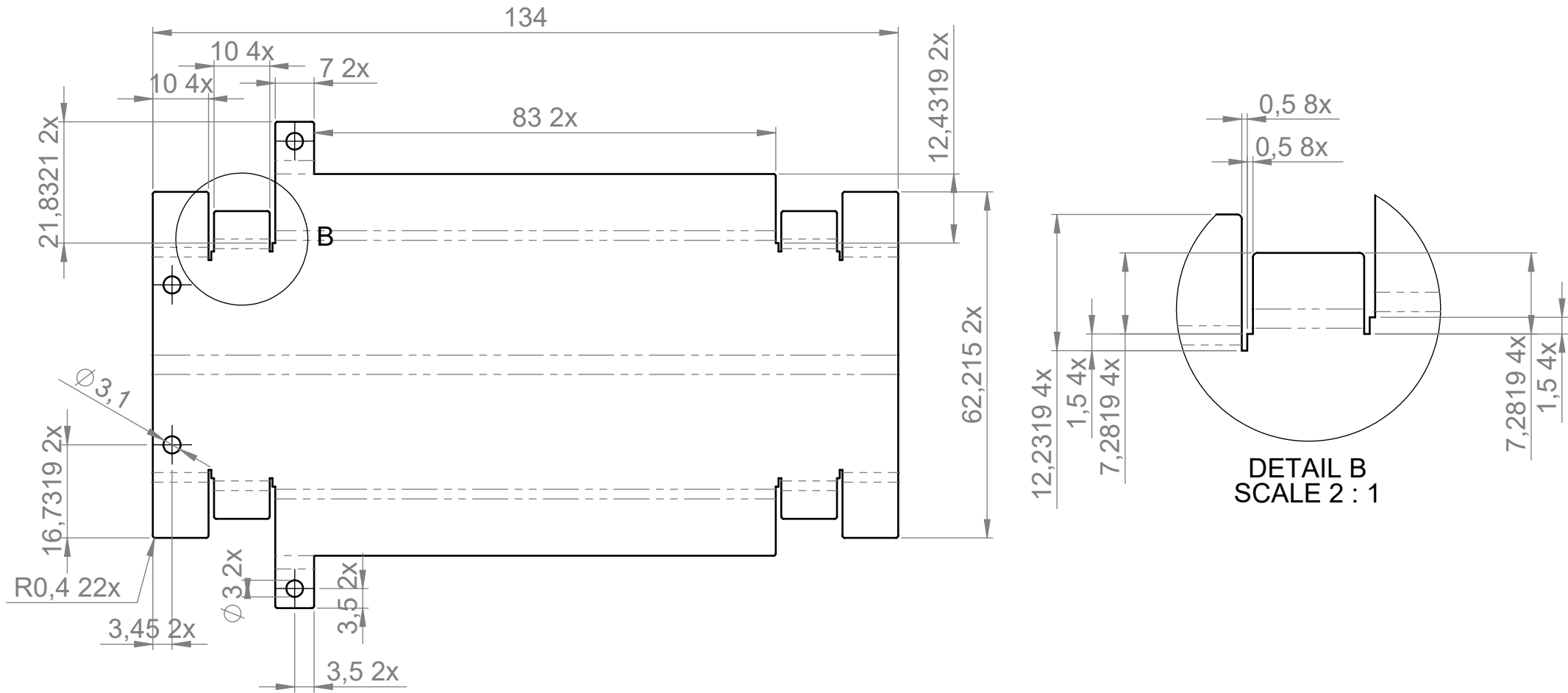
3	4	Subassy L-Shape			
2	1	Subassy Cart			
1	1	Subassy T-Shape			
POS.	QTY.	NAME	MATERIAL	NORM / DIMENSION	REMARK
PROJECTION METHOD			UNLESS STATED OTHERWISE TOLERANCES ± 0.5 MM	DRAWN Manuel	DATE 23-1-2025
			CHECKED Thijm	SCALE 1:1	
MATERIAL		SURFACE TREATMENT	TITLE Subassy Frame		REV.
--			DRAWING NO. 01		01
--			FILE / PART NAME Subassy Frame		A1
UNIVERSITY OF TWENTE.			DIMENSIONS IN MILLIMETERS		SHEET 1 OF 1
FACULTY OF ENGINEERING					



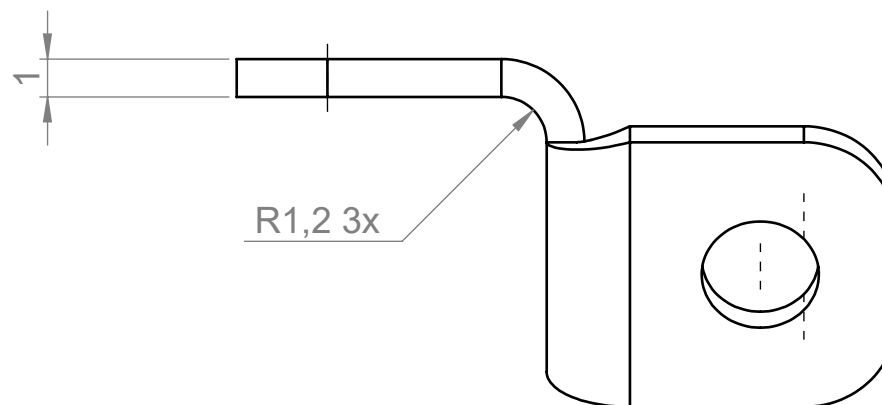
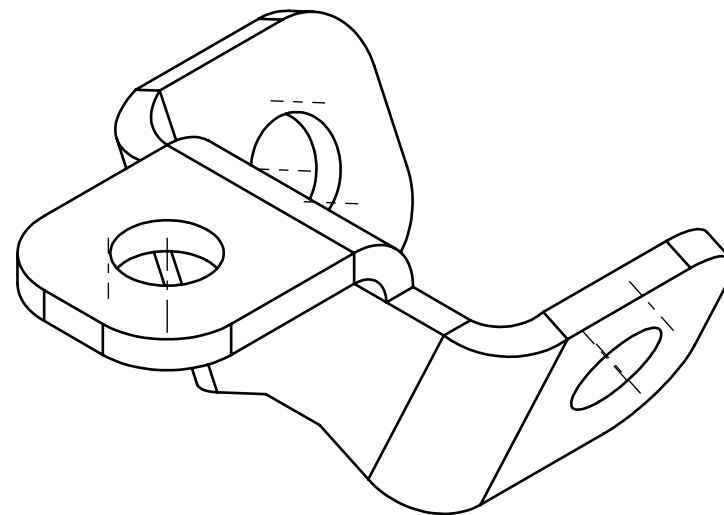
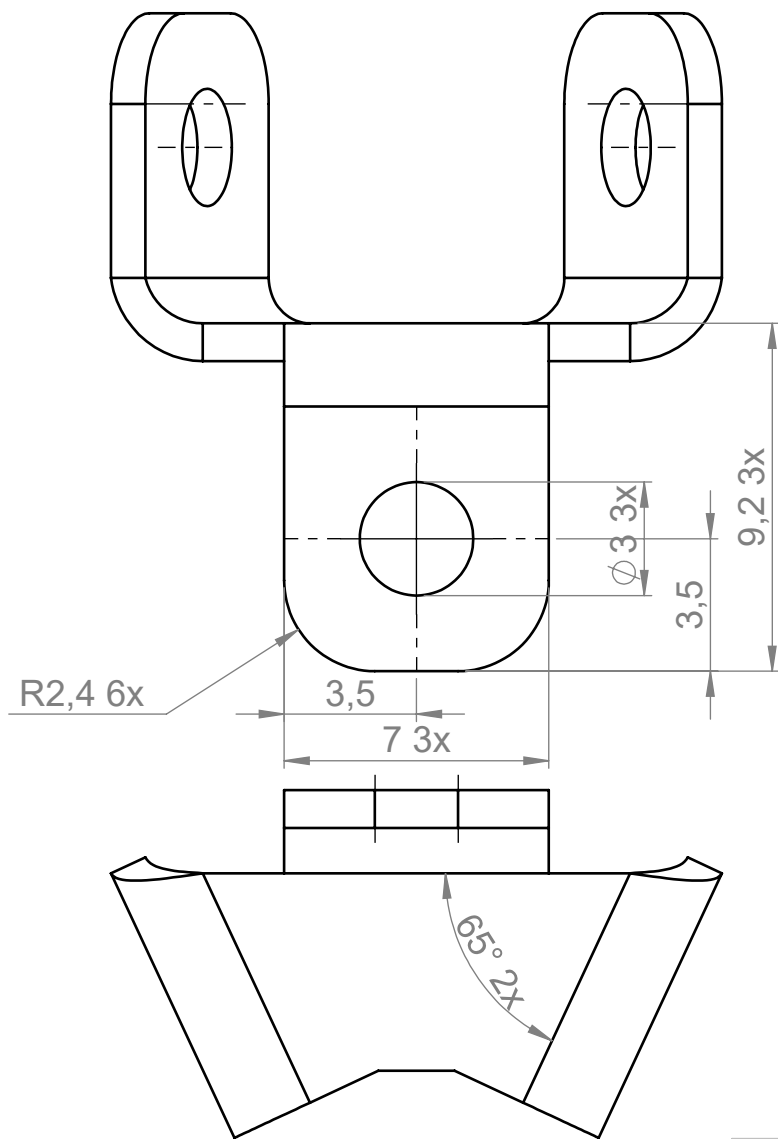
2	1	Part Cart Backplate					
1	1	Part Cart Frame					
POS.	QTY.	NAME	MATERIAL	NORM / DIMENSION	REMARK		
PROJECTION METHOD			UNLESS STATED OTHERWISE: TOLERANCES ± 0,5 MM	DRAWN	Manuel	DATE	23-1-2025
			CHECKED	Thijm	SCALE	1:2	
MATERIAL		SURFACE TREATMENT		TITLE		REV.	
--				Subassy Cart		01	
--				DRAWING NO.			
UNIVERSITY OF TWENTE.				01			
				FILE / PART NAME			
				Subassy Cart		A4	
FACULTY OF ENGINEERING				DIMENSIONS IN MILLIMETERS		SHEET 1 OF 1	

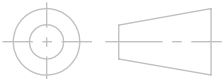


PROJECTION METHOD		UNLESS STATED OTHERWISE: TOLERANCES ± 0,5 MM	DRAWN	Manuel	DATE	24-1-2025	
			CHECKED	Thijm	SCALE	1:1	
MATERIAL	-- --	SURFACE TREATMENT	TITLE			REV.	
UNIVERSITY OF TWENTE. FACULTY OF ENGINEERING			Part Cart Folded			01	
			DRAWING NO.			A4	
			01				
			FILE / PART NAME				
			Part Cart Frame				
			DIMENSIONS IN MILLIMETERS			SHEET 1 OF 1	



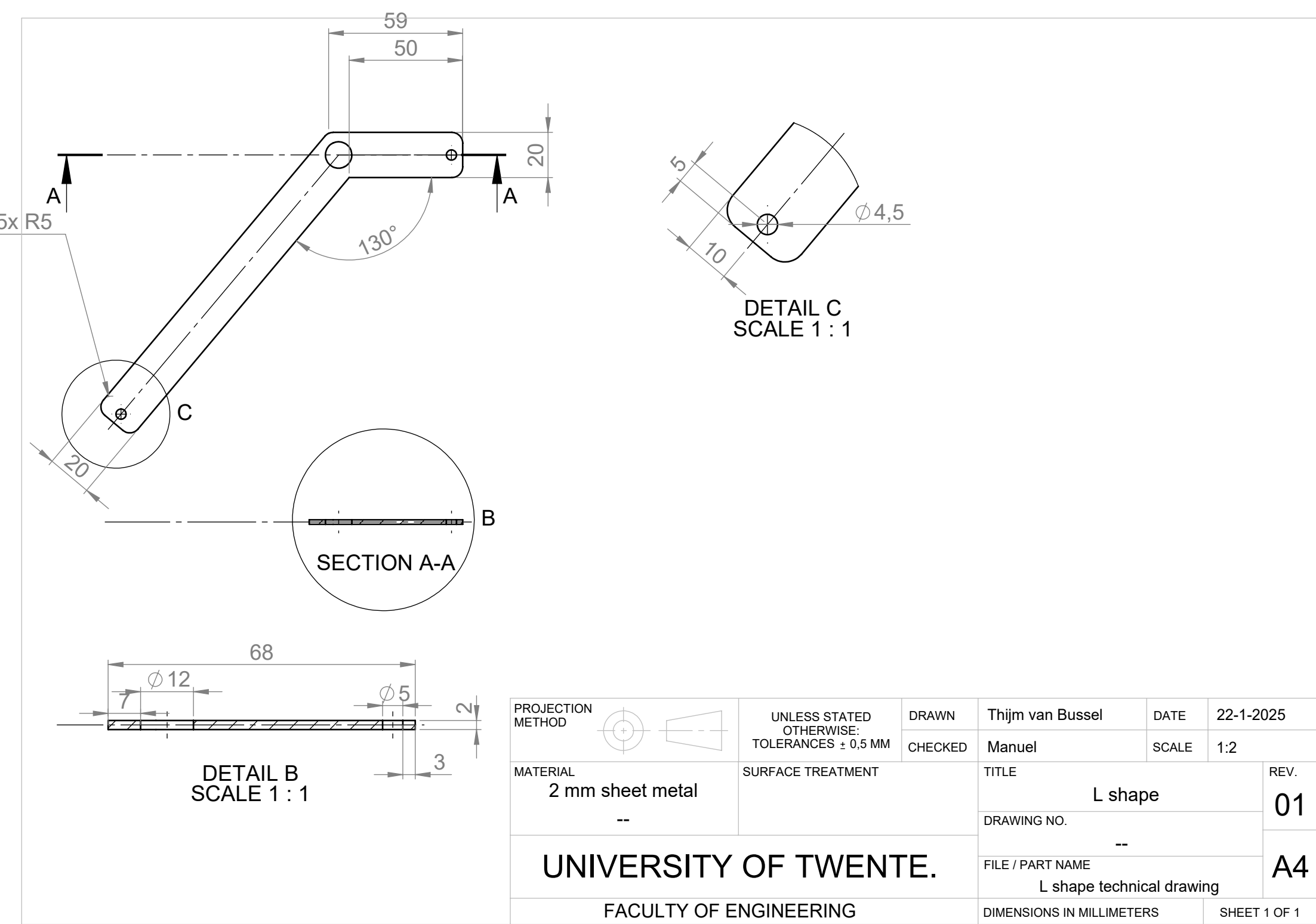
PROJECTION METHOD		UNLESS STATED OTHERWISE: TOLERANCES ± 0,5 MM	DRAWN	Manuel	DATE	24-1-2025	
			CHECKED	Thijm	SCALE	1:1	
MATERIAL Sheet metal 1mm --		SURFACE TREATMENT	TITLE Part Cart Frame			REV. 01	
			DRAWING NO. 01			A4	
FILE / PART NAME Part Cart Frame Flat							
UNIVERSITY OF TWENTE.			DIMENSIONS IN MILLIMETERS			SHEET 1 OF 1	
FACULTY OF ENGINEERING							

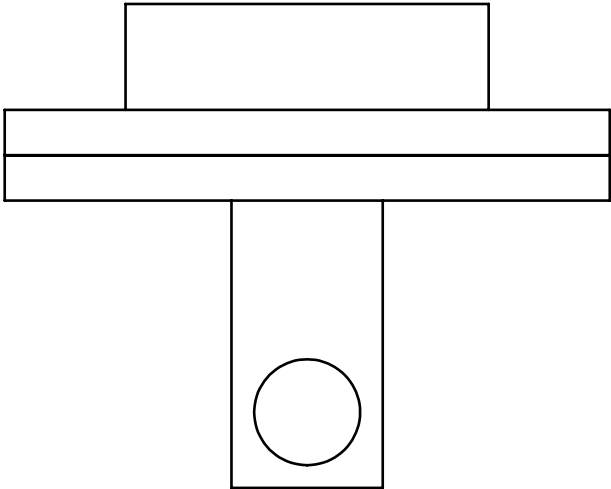
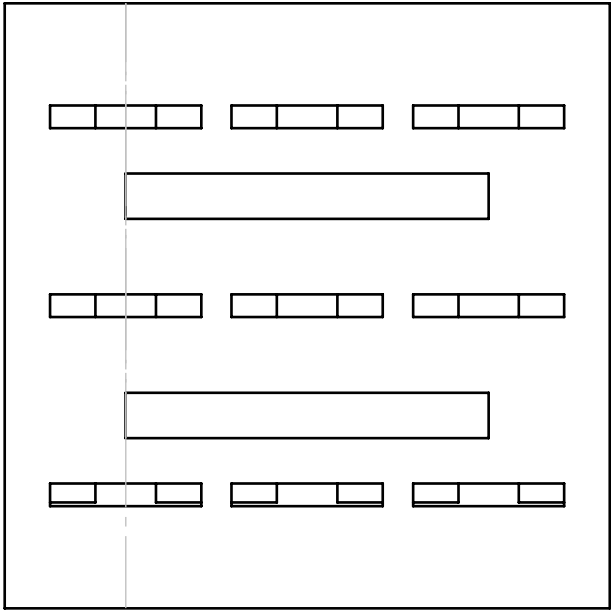


PROJECTION METHOD		UNLESS STATED OTHERWISE: TOLERANCES ± 0,5 MM	DRAWN	Manuel	DATE	23-1-2025
			CHECKED	Thijm van Bussel	SCALE	5:1
MATERIAL		SURFACE TREATMENT		TITLE		REV.
Sheet metal 1 mm				Part Cart Backplate		01
--		UNIVERSITY OF TWENTE.		DRAWING NO.		A4
				01		
				FILE / PART NAME		
				Part Cart Backplate		
				DIMENSIONS IN MILLIMETERS		SHEET 1 OF 1

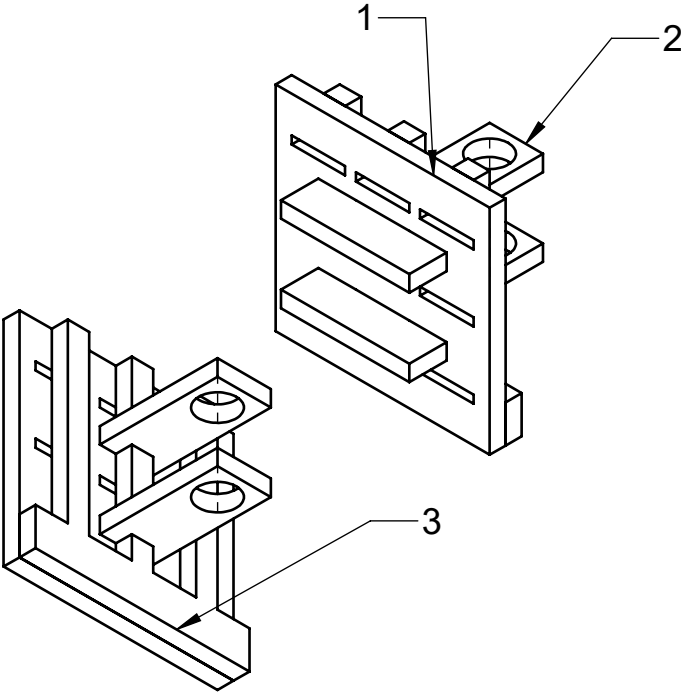
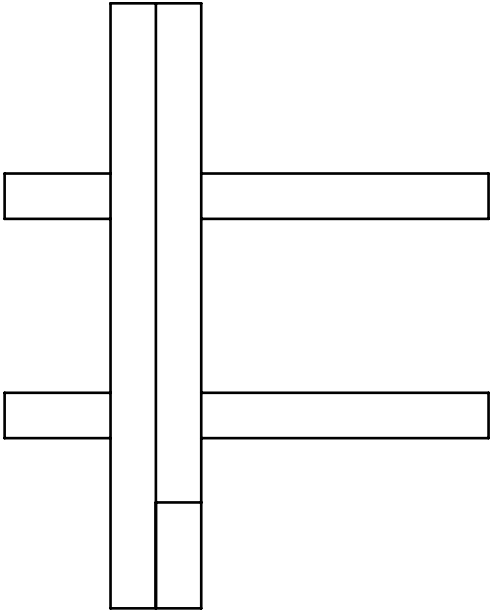
UNIVERSITY OF TWENTE.

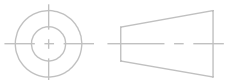
FACULTY OF ENGINEERING

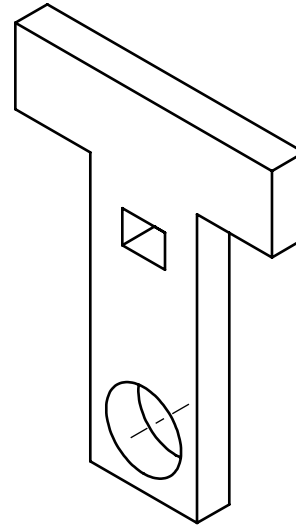
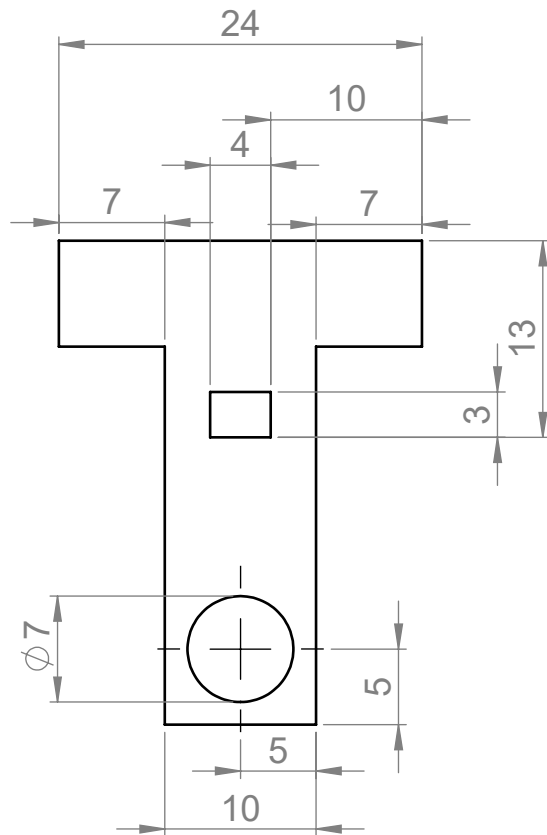




ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	Part; Spring Spacing Plate L-Shape		1
2	connection pin		2
3	trident		1



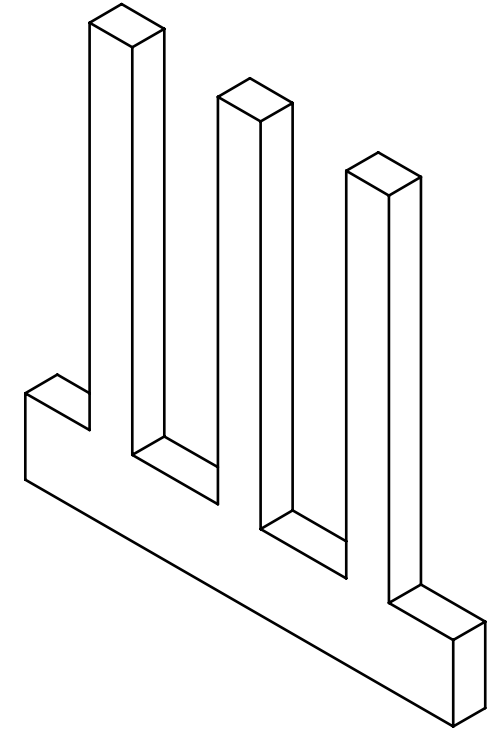
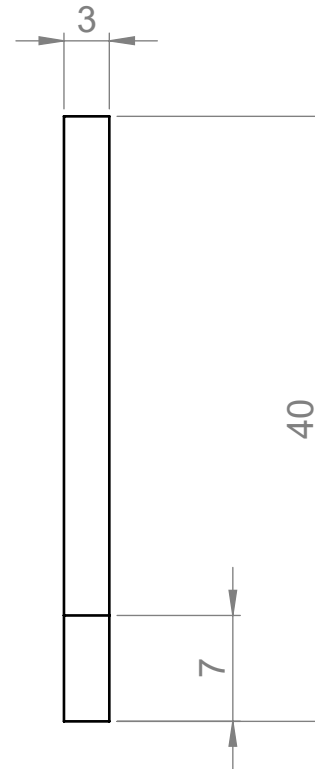
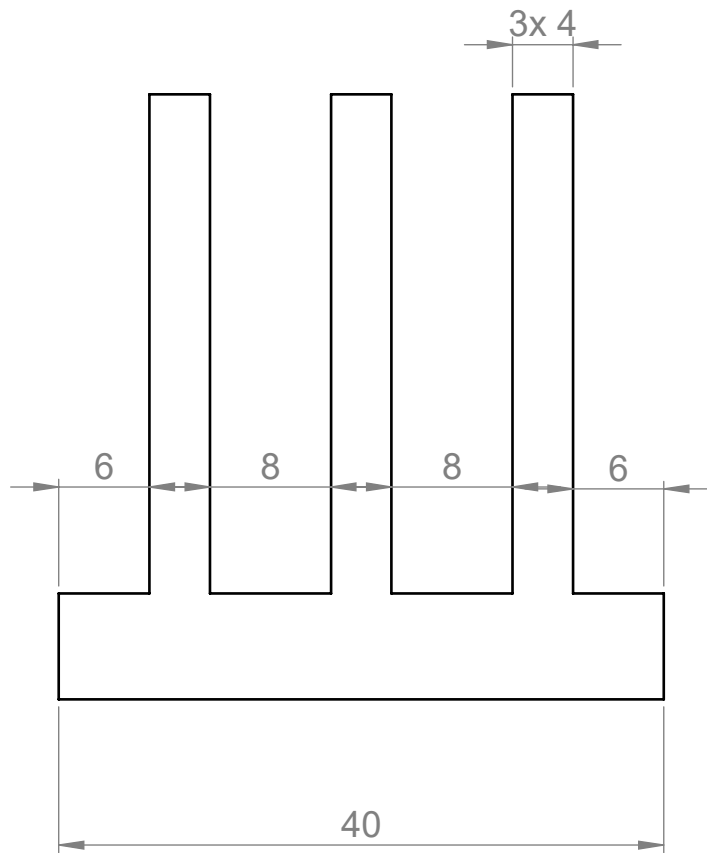
PROJECTION METHOD 	UNLESS STATED OTHERWISE: TOLERANCES ± 0,5 MM	DRAWN	Thijm van Bussel	DATE	23-1-2025
		CHECKED	Manuel	SCALE	1:1
MATERIAL 3 mm sheet metal --	SURFACE TREATMENT	TITLE spring modulus assy bottem			REV. 01
		DRAWING NO. --			A4
UNIVERSITY OF TWENTE.		FILE / PART NAME spring modulus assy bottem			
FACULTY OF ENGINEERING		DIMENSIONS IN MILLIMETERS		SHEET 1 OF 1	



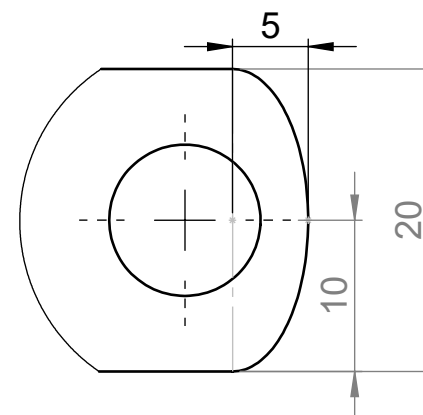
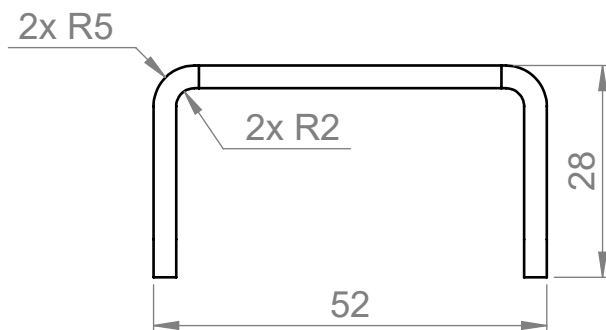
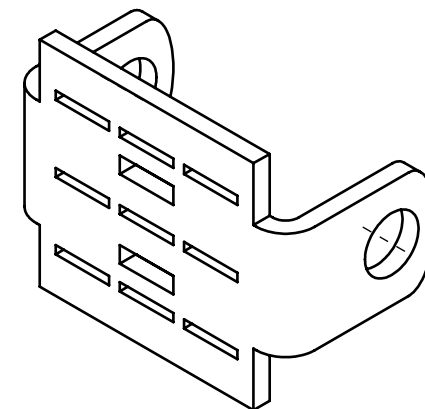
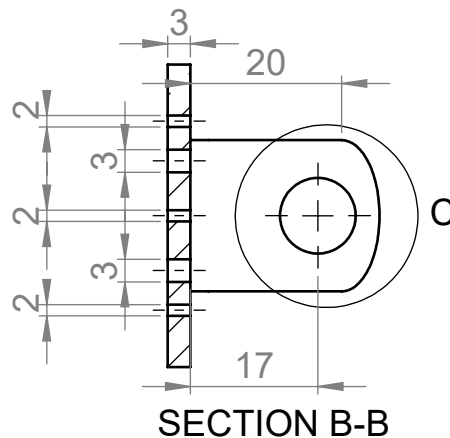
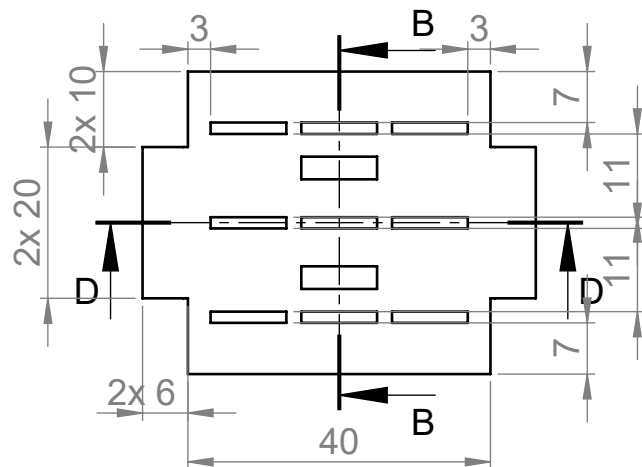
PROJECTION METHOD		UNLESS STATED OTHERWISE: TOLERANCES ± 0,5 MM	DRAWN	Thijm van Bussel	DATE	22-1-2025	
			CHECKED	Manuel	SCALE	2:1	
MATERIAL		SURFACE TREATMENT		TITLE		REV.	
3 mm sheet metal				connection pin		01	
				DRAWING NO.			
				FILE / PART NAME		A4	
				Part; Connection Plate Spring Modules L-Shapes			
				DIMENSIONS IN MILLIMETERS		SHEET 1 OF 1	

UNIVERSITY OF TWENTE.

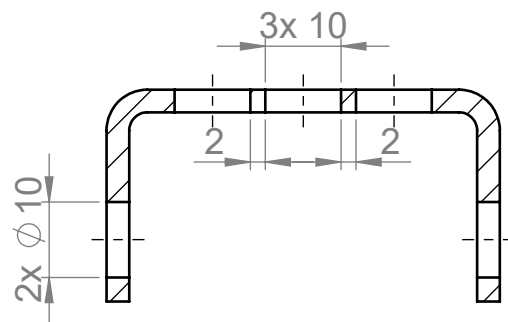
FACULTY OF ENGINEERING



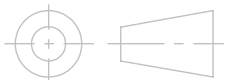
PROJECTION METHOD		UNLESS STATED OTHERWISE: TOLERANCES ± 0,5 MM	DRAWN	Thijm van Bussel	DATE	22-1-2025	
			CHECKED	Manuel	SCALE	2:1	
MATERIAL 3 mm sheet metal --		SURFACE TREATMENT	TITLE trident spring modulus			REV.	01
			DRAWING NO. --			A4	
UNIVERSITY OF TWENTE.		FILE / PART NAME Part; Spring Retainment Trident					
FACULTY OF ENGINEERING		DIMENSIONS IN MILLIMETERS			SHEET 1 OF 1		

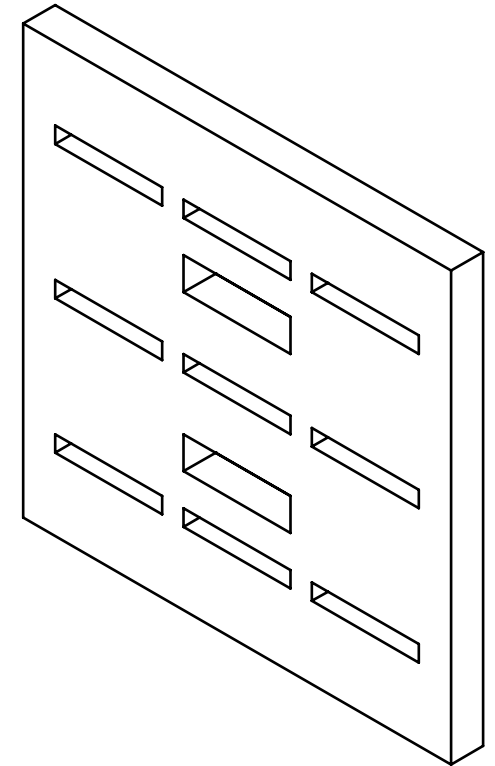
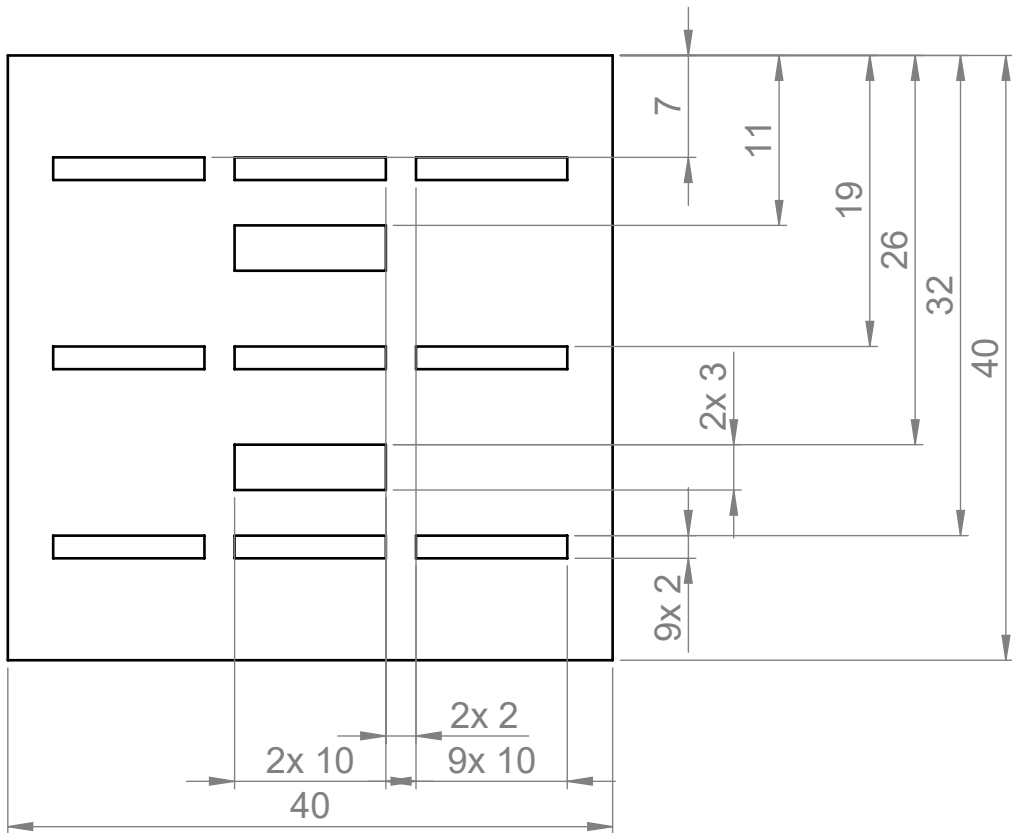


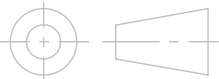
DETAIL C
SCALE 2 : 1

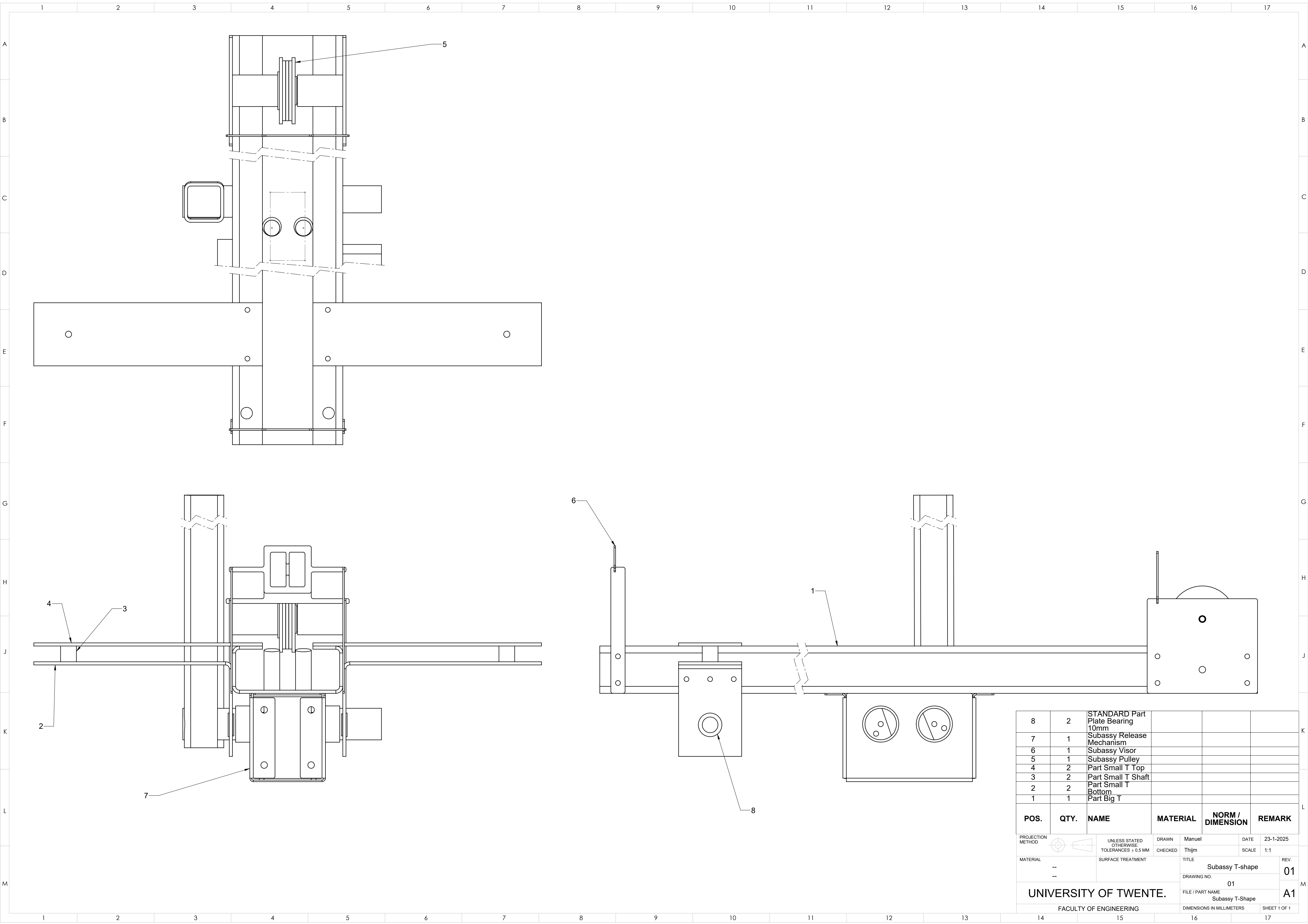


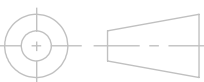
SECTION D-D

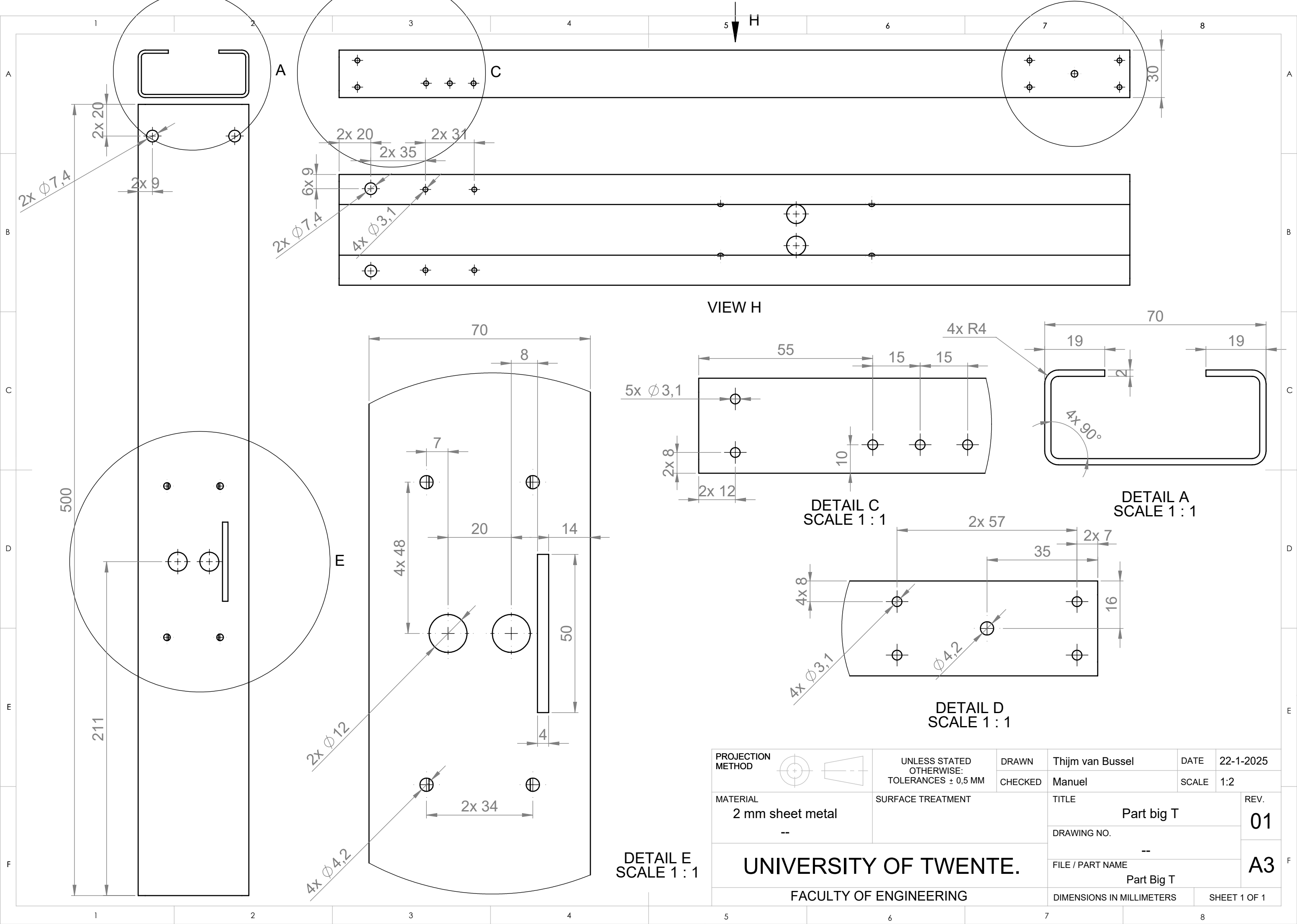
PROJECTION METHOD 	UNLESS STATED OTHERWISE: TOLERANCES ± 0,5 MM	DRAWN	Thijm van bussel	DATE	23-1-2025
		CHECKED	manuel	SCALE	1:1
MATERIAL 3 mm sheet metal --	SURFACE TREATMENT	TITLE connection plate t frame			REV. 01
		DRAWING NO. --			A4
FILE / PART NAME connection plate t frame					
DIMENSIONS IN MILLIMETERS			SHEET 1 OF 1		
FACULTY OF ENGINEERING					

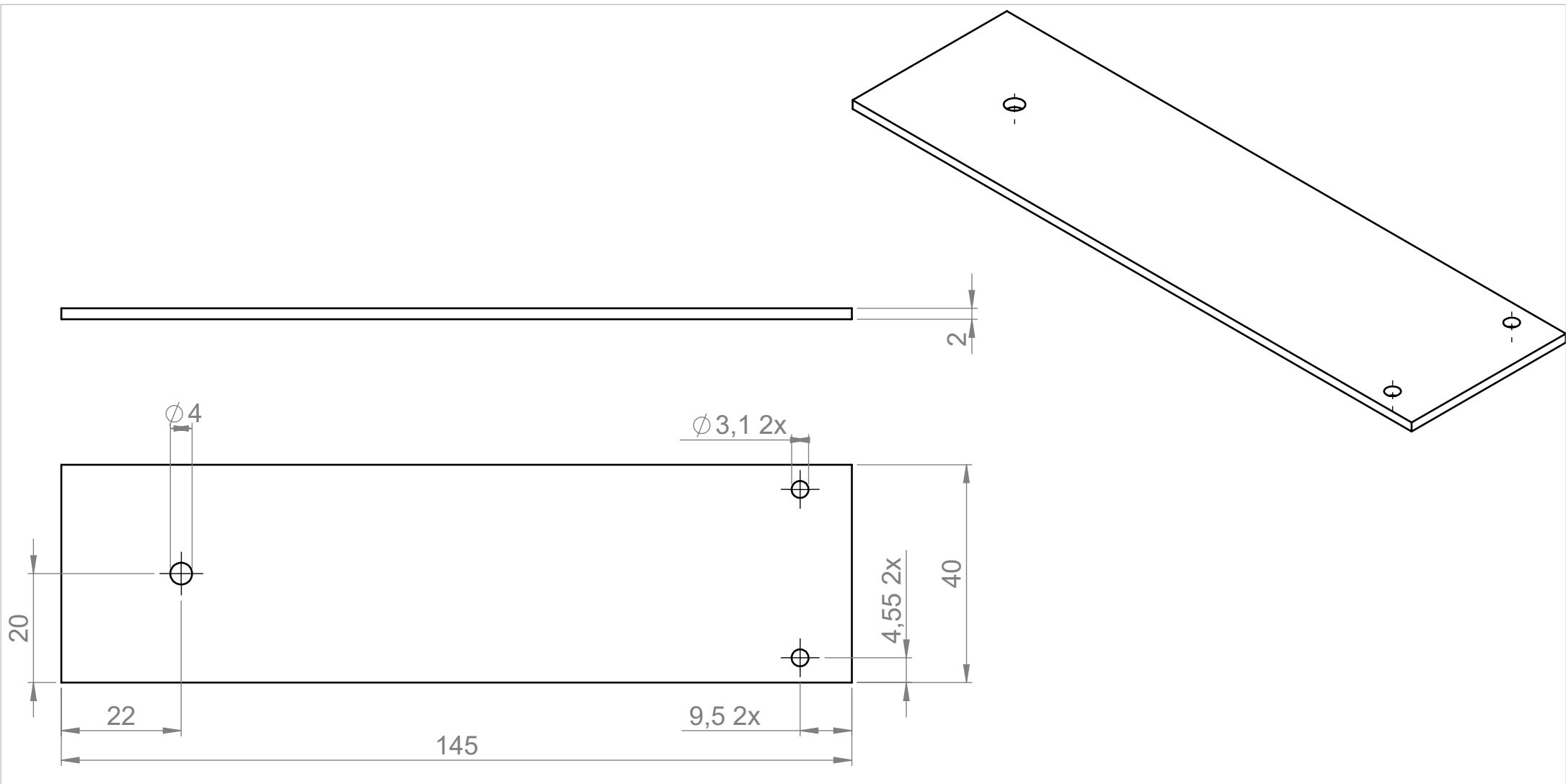


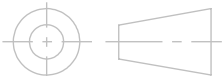
PROJECTION METHOD		UNLESS STATED OTHERWISE: TOLERANCES ± 0,5 MM	DRAWN	Thijm van Bussel	DATE	22-1-2025
			CHECKED	Manuel	SCALE	2:1
MATERIAL 3 mm sheet metal --		SURFACE TREATMENT	TITLE connection plate spring modulus			REV. 01
			DRAWING NO. --			A4
UNIVERSITY OF TWENTE.			FILE / PART NAME Part; Spring Spacing Plate L-Shape			
FACULTY OF ENGINEERING			DIMENSIONS IN MILLIMETERS	SHEET 1 OF 1		

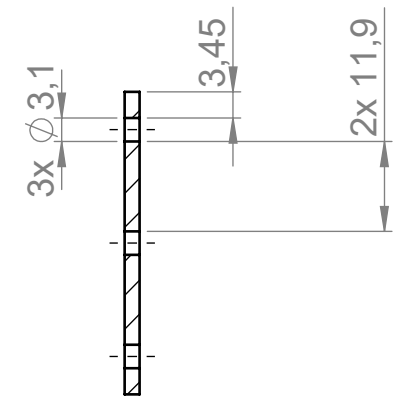
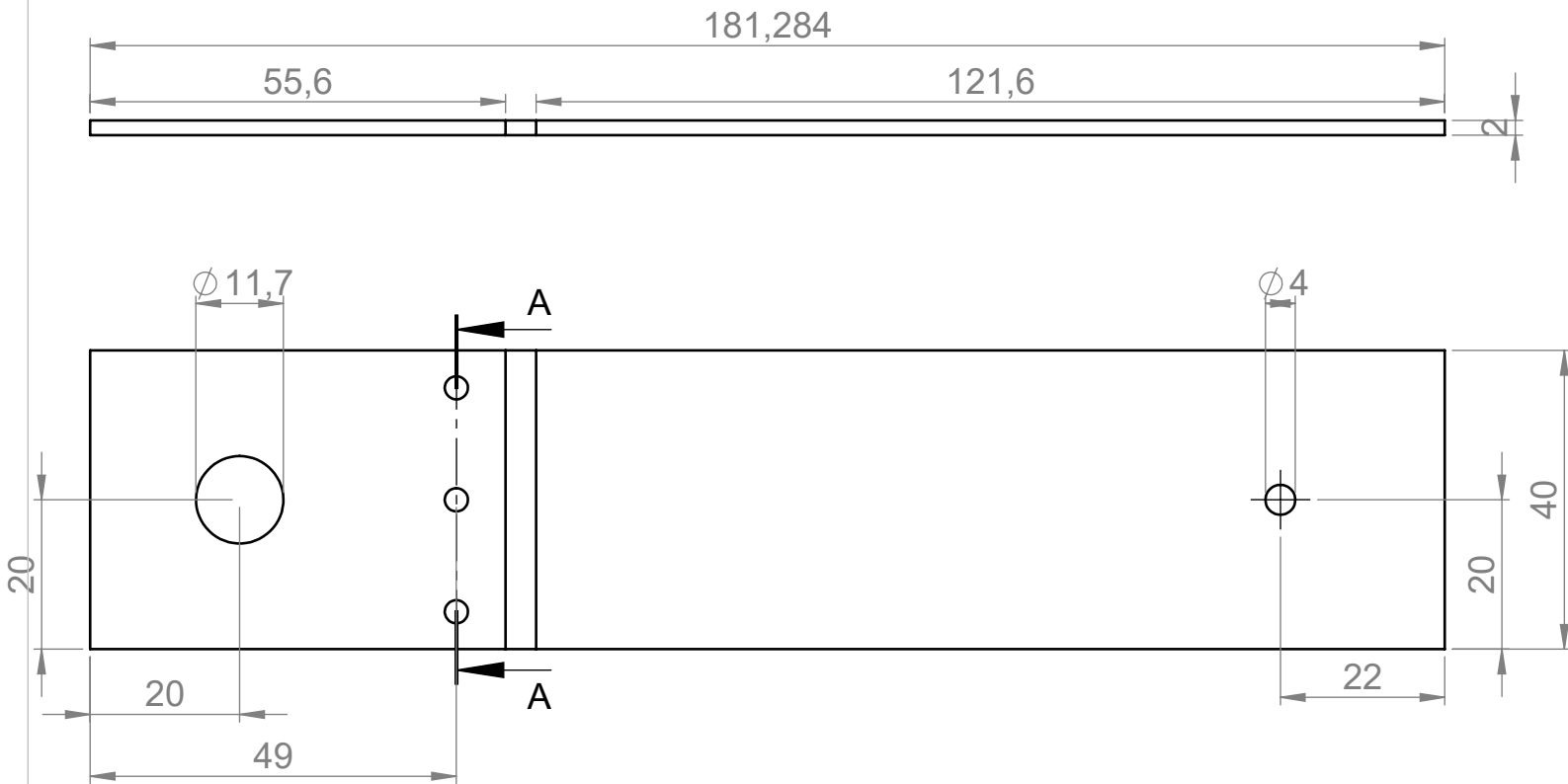


8	2	STANDARD Part Plate Bearing 10mm					
7	1	Subassy Release Mechanism					
6	1	Subassy Visor					
5	1	Subassy Pulley					
4	2	Part Small T Top					
3	2	Part Small T Shaft					
2	2	Part Small T Bottom					
1	1	Part Big T					
POS.	QTY.	NAME	MATERIAL	NORM / DIMENSION	REMARK		
PROJECTION METHOD			UNLESS STATED OTHERWISE: TOLERANCES : 0,5 MM	DRAWN	Manuel	DATE	23-1-2025
			CHECKED	Thijn	SCALE	1:1	
MATERIAL			SURFACE TREATMENT		TITLE		
--				Subassy T-shape			REV. 01
--				DRAWING NO.			
UNIVERSITY OF TWENTE.				01			A1
				FILE / PART NAME			
				Subassy T-Shape			
FACULTY OF ENGINEERING				DIMENSIONS IN MILLIMETERS			SHEET 1 OF 1

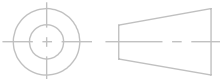


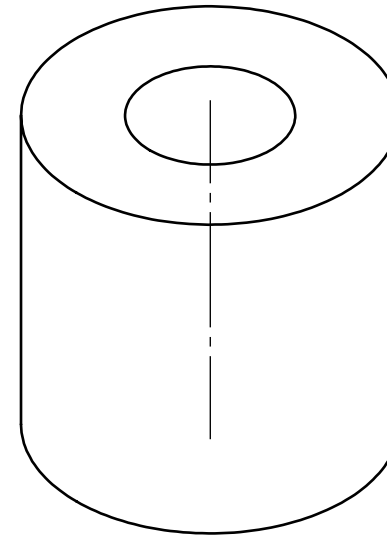
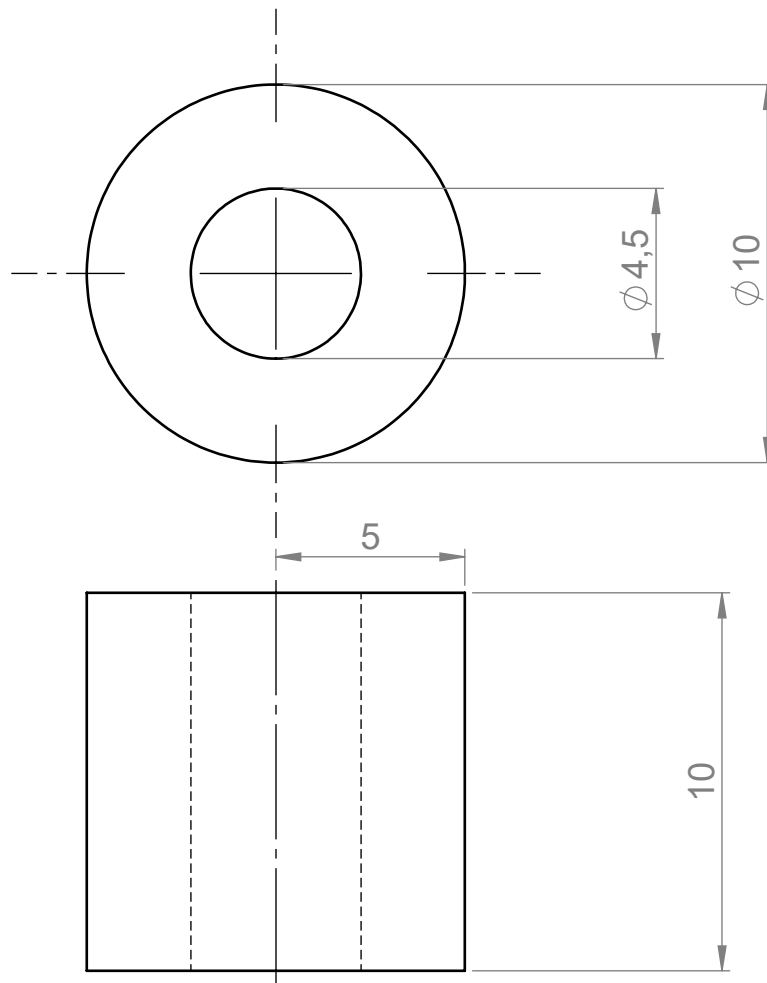


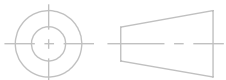
PROJECTION METHOD		UNLESS STATED OTHERWISE: TOLERANCES ± 0,5 MM	DRAWN	Manuel	DATE	23-1-2025
			CHECKED	Thijm van Bussel	SCALE	1:1
MATERIAL Sheet metal 2mm --	SURFACE TREATMENT	TITLE Part Smaal T top	DRAWING NO. 01			REV. 01
UNIVERSITY OF TWENTE.		DIMENSIONS IN MILLIMETERS			SHEET 1 OF 1	
FACULTY OF ENGINEERING						

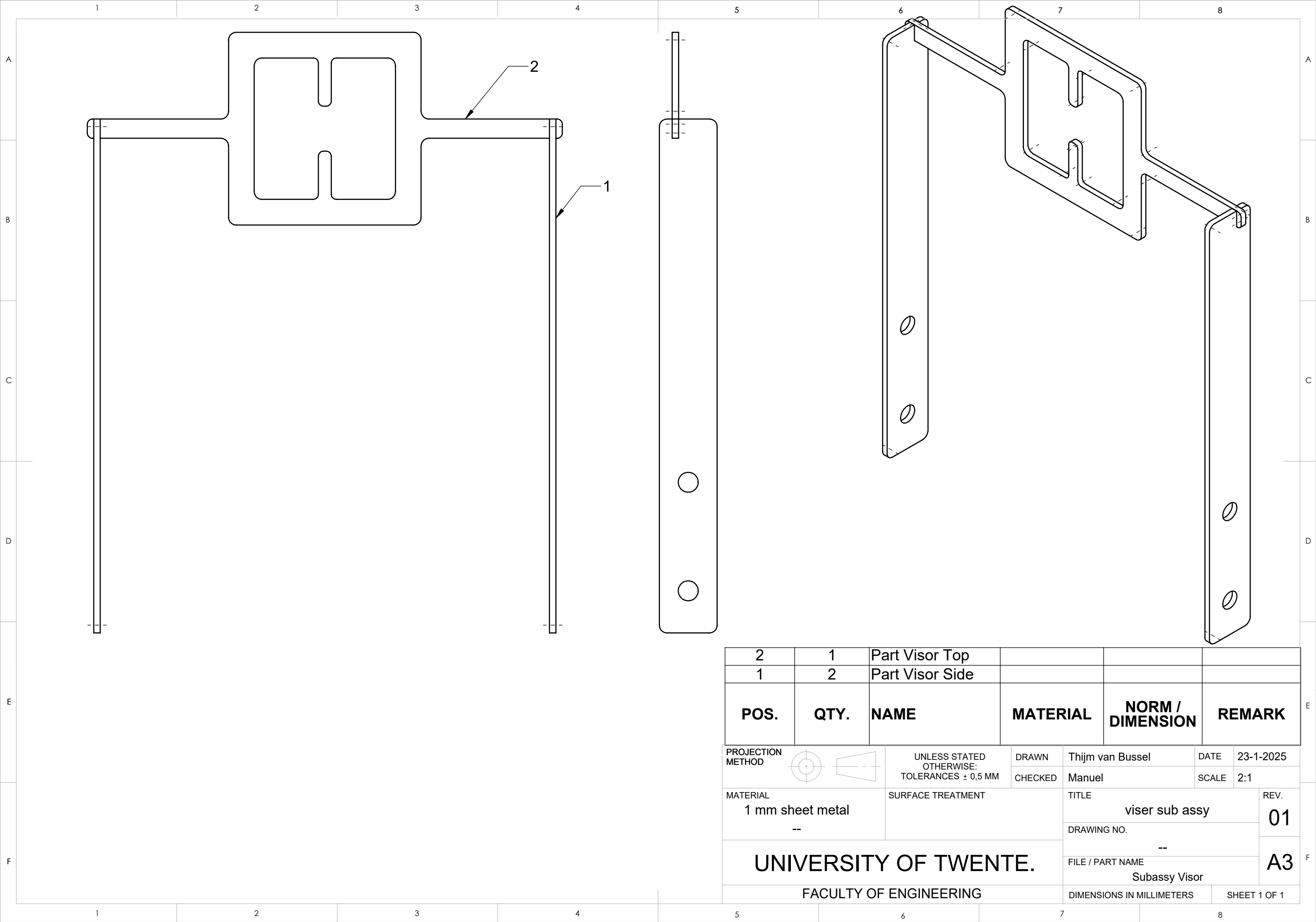


SECTION A-A

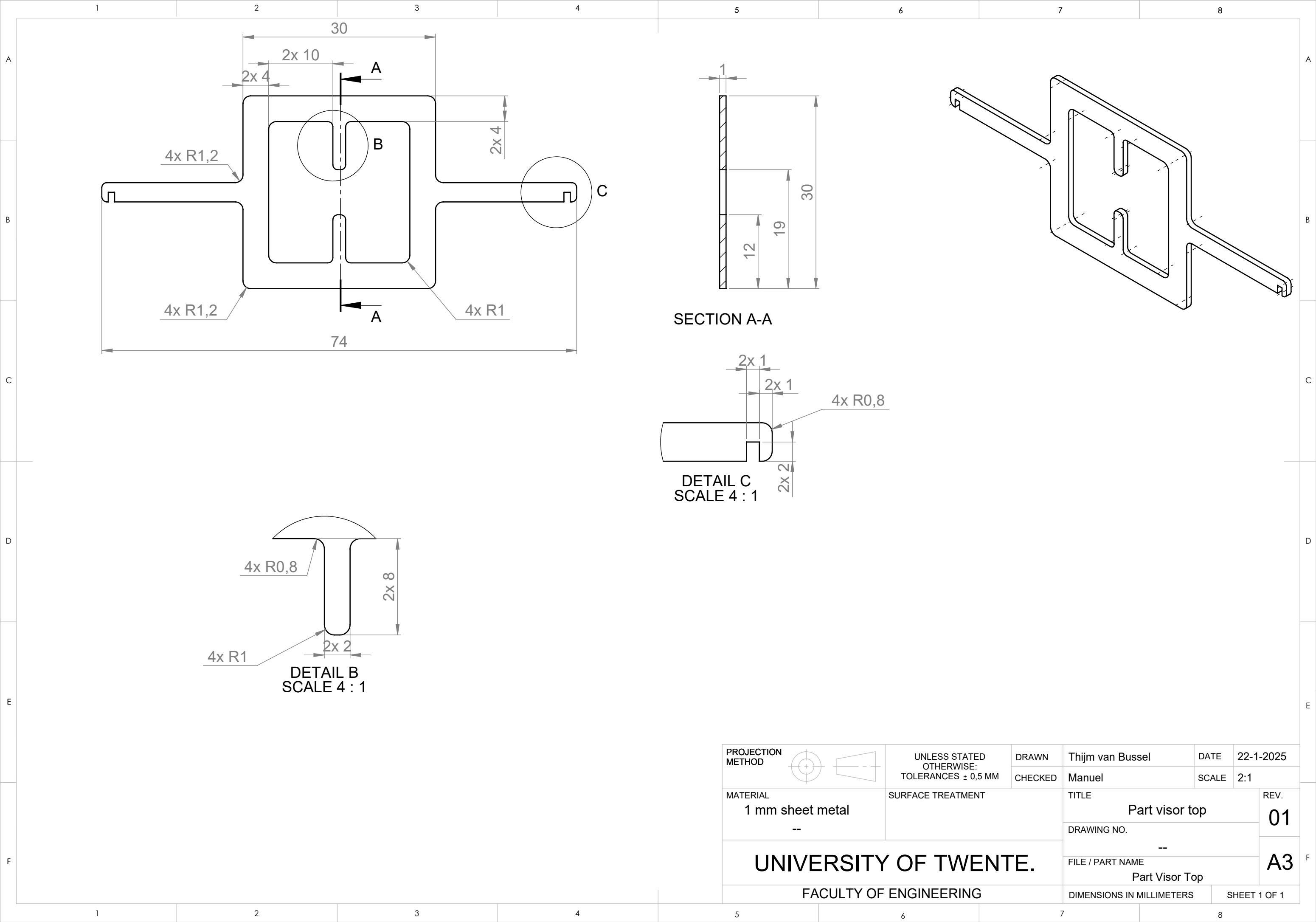
PROJECTION METHOD		UNLESS STATED OTHERWISE: TOLERANCES ± 0,5 MM	DRAWN	Manuel	DATE	23-1-2025	
			CHECKED	Thijm van Bussel	SCALE	1:1	
MATERIAL Sheet metal 2mm --		SURFACE TREATMENT	TITLE Part Small T bottom				REV. 01
			DRAWING NO. 01				A4
UNIVERSITY OF TWENTE.		FILE / PART NAME Part Small T Bottom					
		FACULTY OF ENGINEERING		DIMENSIONS IN MILLIMETERS		SHEET 1 OF 1	

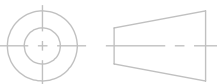


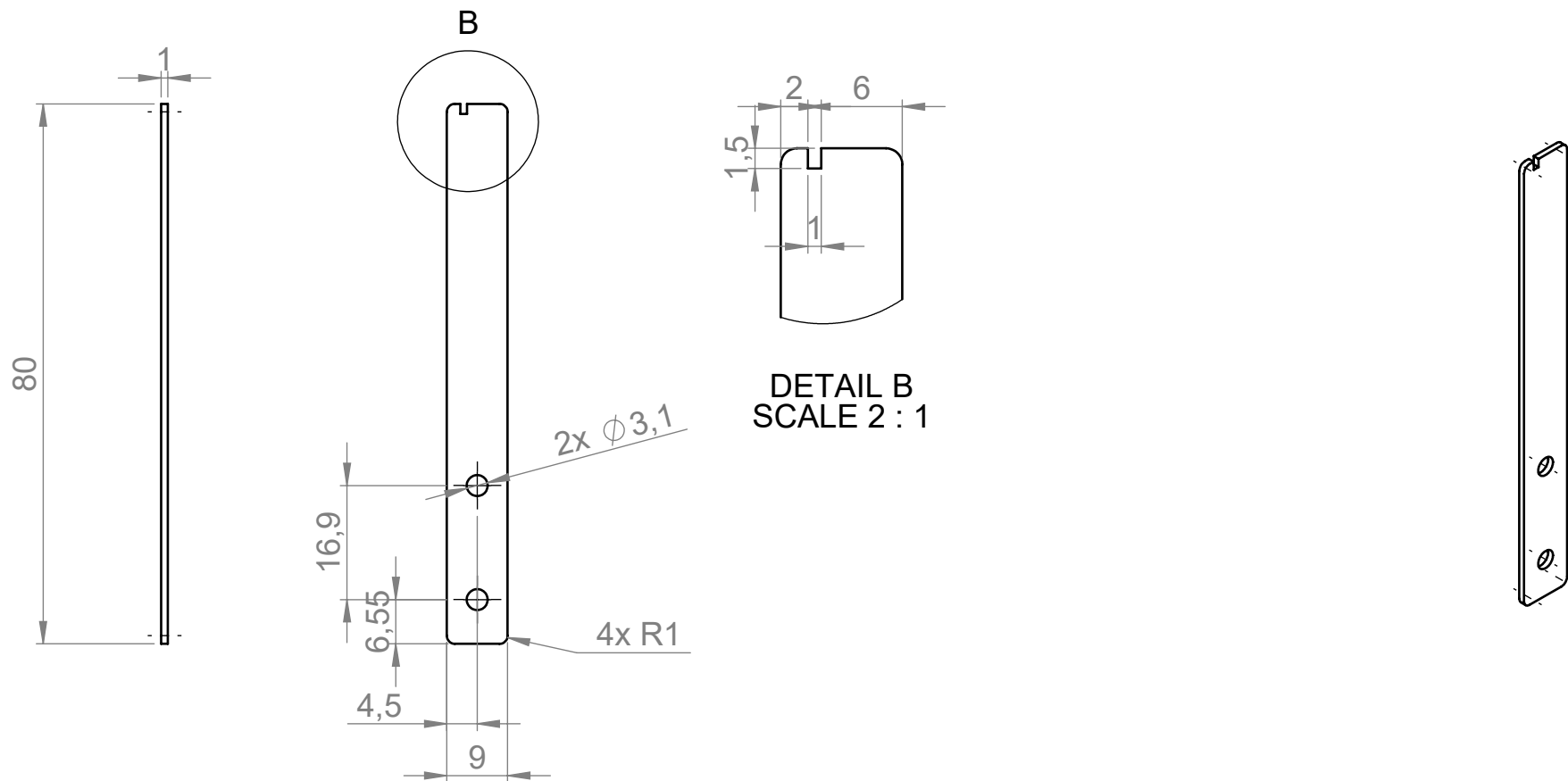
PROJECTION METHOD 	UNLESS STATED OTHERWISE: TOLERANCES ± 0,5 MM	DRAWN	Manuel	DATE	23-1-2025
		CHECKED	Thijm van Bussel	SCALE	5:1
MATERIAL Bright drawn steel (10 mm) --	SURFACE TREATMENT	TITLE Small T shaft			REV. 01
		DRAWING NO. 01			A4
UNIVERSITY OF TWENTE.					
FACULTY OF ENGINEERING		DIMENSIONS IN MILLIMETERS		SHEET 1 OF 1	

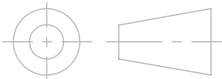


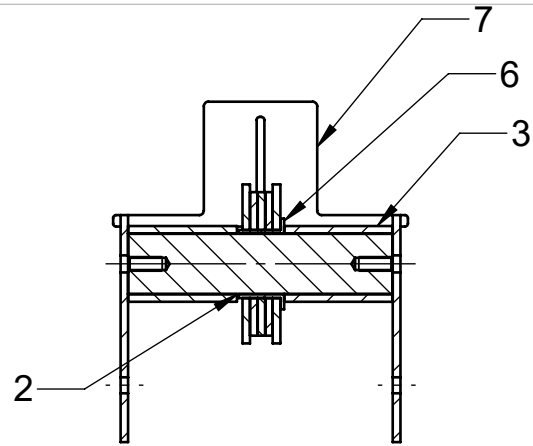
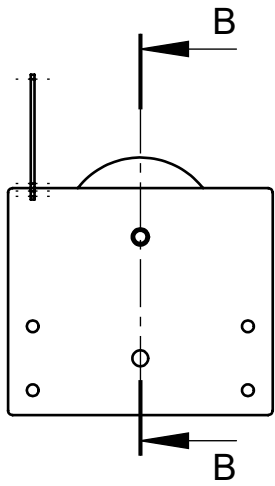
2		1	Part Visor Top			
1		2	Part Visor Side			
POS.	QTY.	NAME	MATERIAL	NORM / DIMENSION	REMARK	
PROJECTION METHOD			UNLESS STATED OTHERWISE: TOLERANCES ± 0,5 MM	DRAWN	Thijm van Bussel	DATE 23-1-2025
				CHECKED	Manuel	SCALE 2:1
MATERIAL			SURFACE TREATMENT		TITLE	REV.
1 mm sheet metal					viser sub assy	01
					DRAWING NO.	
					FILE / PART NAME	A3
					Subassy Visor	
					DIMENSIONS IN MILLIMETERS	SHEET 1 OF 1



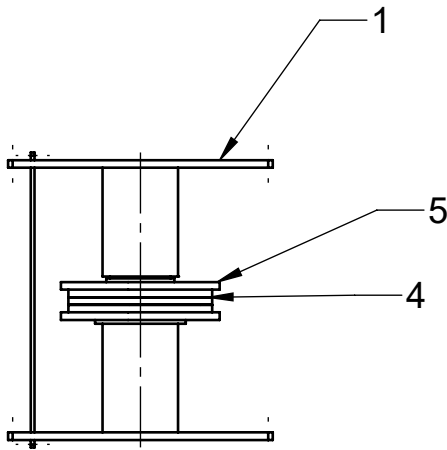
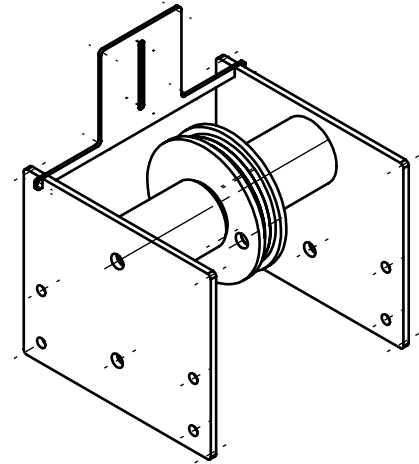
PROJECTION METHOD		UNLESS STATED OTHERWISE: TOLERANCES ± 0,5 MM	DRAWN	Thijm van Bussel	DATE	22-1-2025
			CHECKED	Manuel	SCALE	2:1
MATERIAL 1 mm sheet metal --	SURFACE TREATMENT		TITLE Part visor top			REV. 01
			DRAWING NO. --			A3
UNIVERSITY OF TWENTE.			FILE / PART NAME Part Visor Top			
FACULTY OF ENGINEERING			DIMENSIONS IN MILLIMETERS		SHEET 1 OF 1	



PROJECTION METHOD		UNLESS STATED OTHERWISE: TOLERANCES ± 0,5 MM	DRAWN	Thijm van Bussel	DATE	22-1-2025
			CHECKED	Manuel	SCALE	1:1
MATERIAL 1 mm sheet metal --		SURFACE TREATMENT	TITLE Part Visor Side			REV. 01
			DRAWING NO. --			A4
UNIVERSITY OF TWENTE.						
FACULTY OF ENGINEERING						
			DIMENSIONS IN MILLIMETERS		SHEET 1 OF 1	

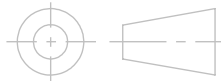


SECTION B-B



7	1	Part Pulley Visor			
6	1	STANDARD Part Sleeve Bearing 16mm			
5	2	Part Pulley Big Disc			
4	3	Part Pulley Small Disc			
3	2	Part Pulley Axle Spacer			
2	1	Part Pulley Axle			
1	2	Part Pulley Side			
POS.	QTY.	NAME	MATERIAL	NORM / DIMENSION	REMARK

PROJECTION METHOD



UNLESS STATED OTHERWISE:
TOLERANCES $\pm 0,5$ MM

DRAWN

Manuel

DATE

23-1-2025

CHECKED

Thijm

SCALE

1:2

MATERIAL

--

--

SURFACE TREATMENT

TITLE

Subassy Pulley

REV.

01

DRAWING NO.

--

FILE / PART NAME

Subassy Pulley

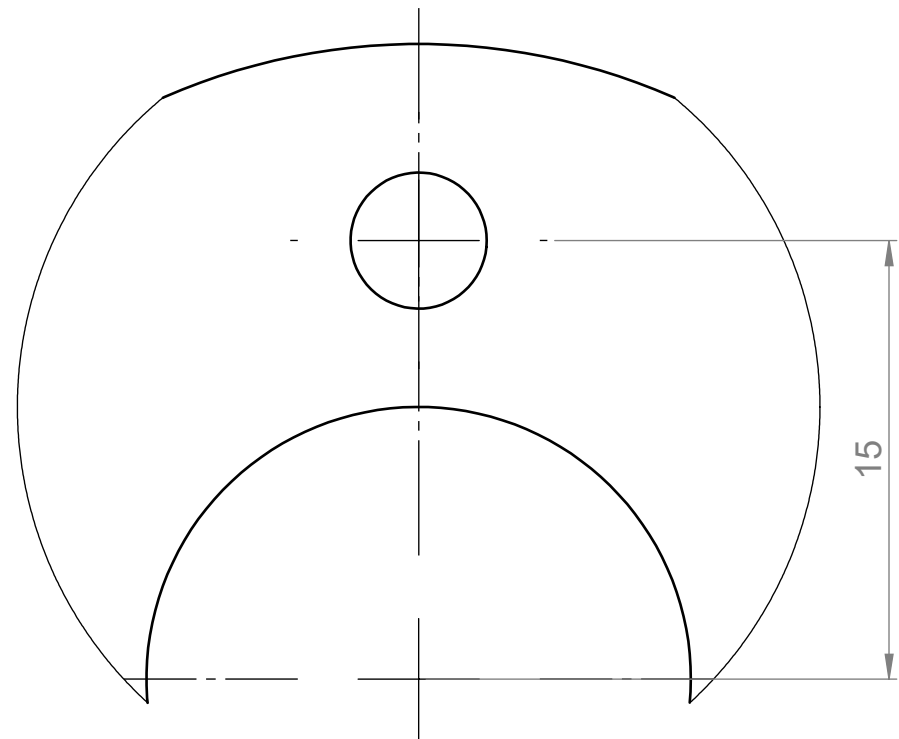
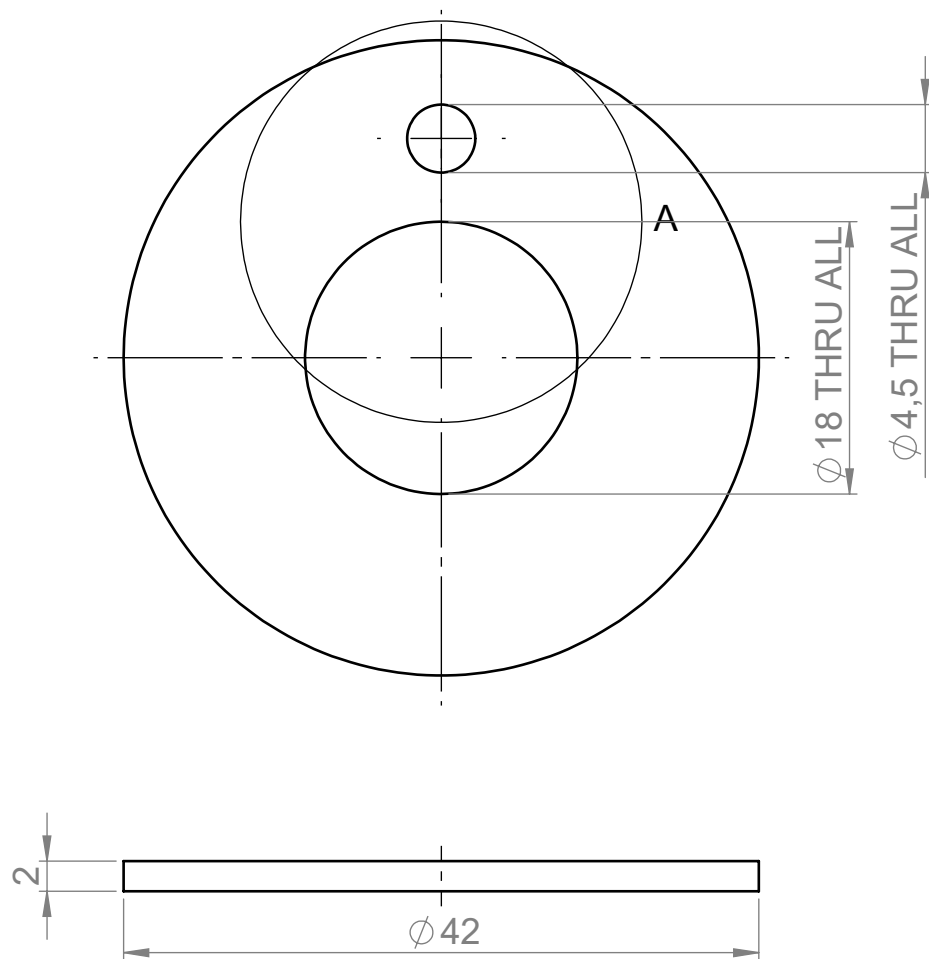
A4

UNIVERSITY OF TWENTE.

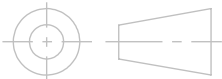
FACULTY OF ENGINEERING

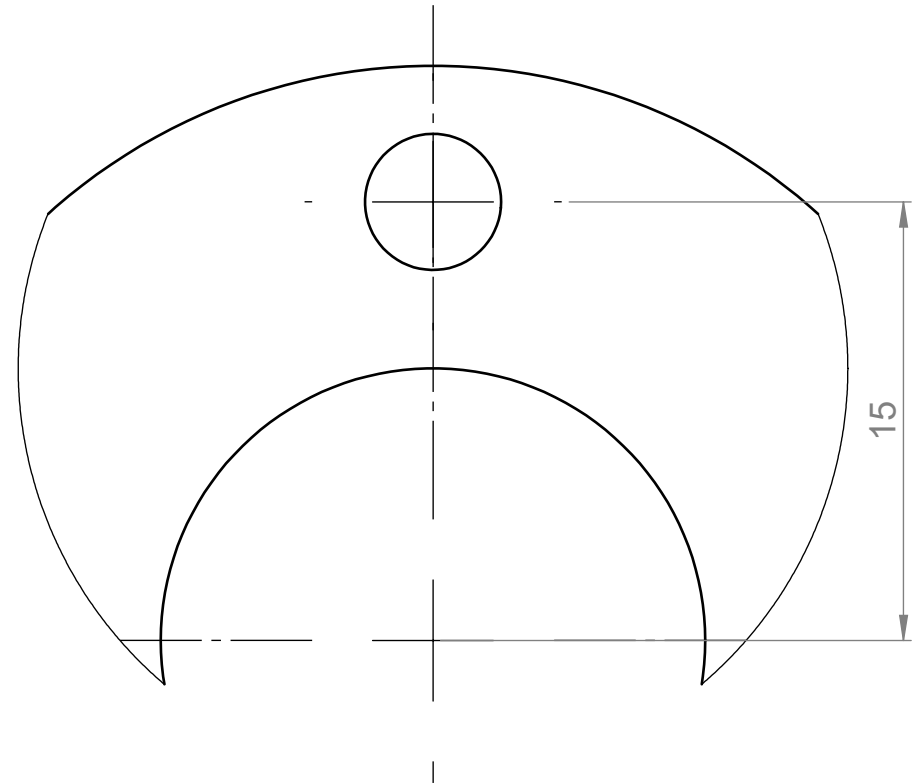
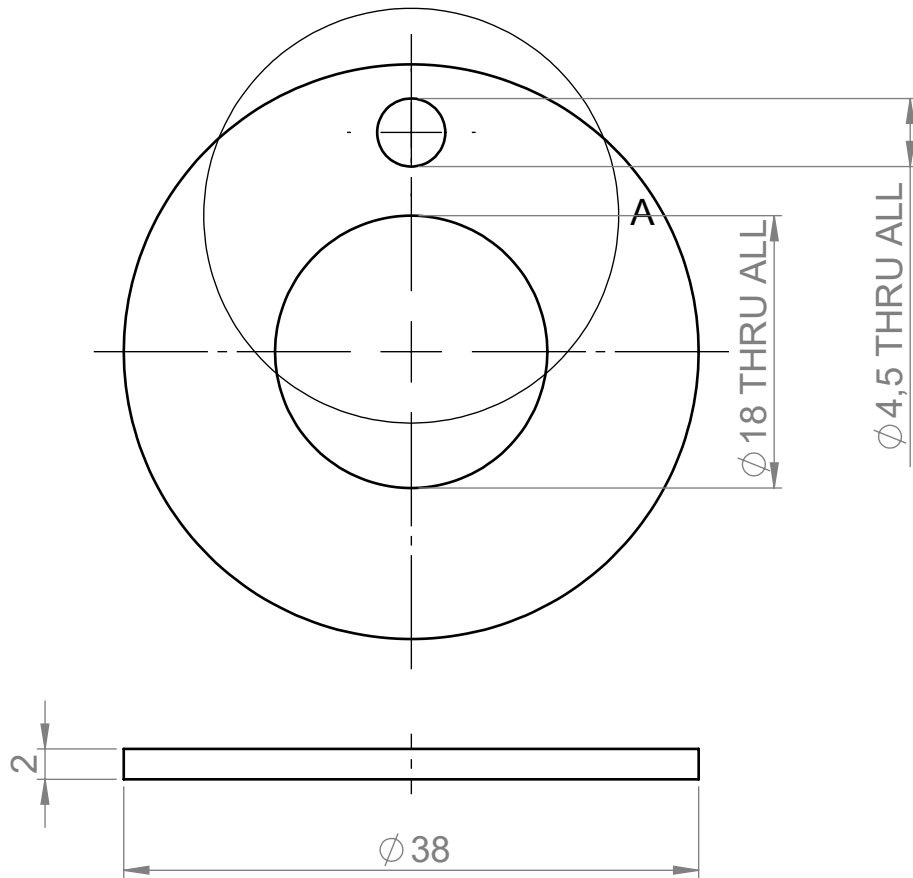
DIMENSIONS IN MILLIMETERS

SHEET 1 OF 1



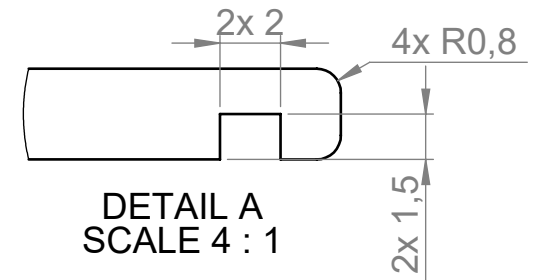
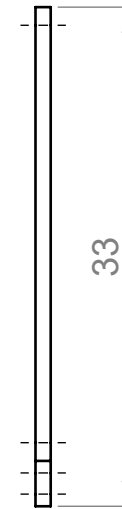
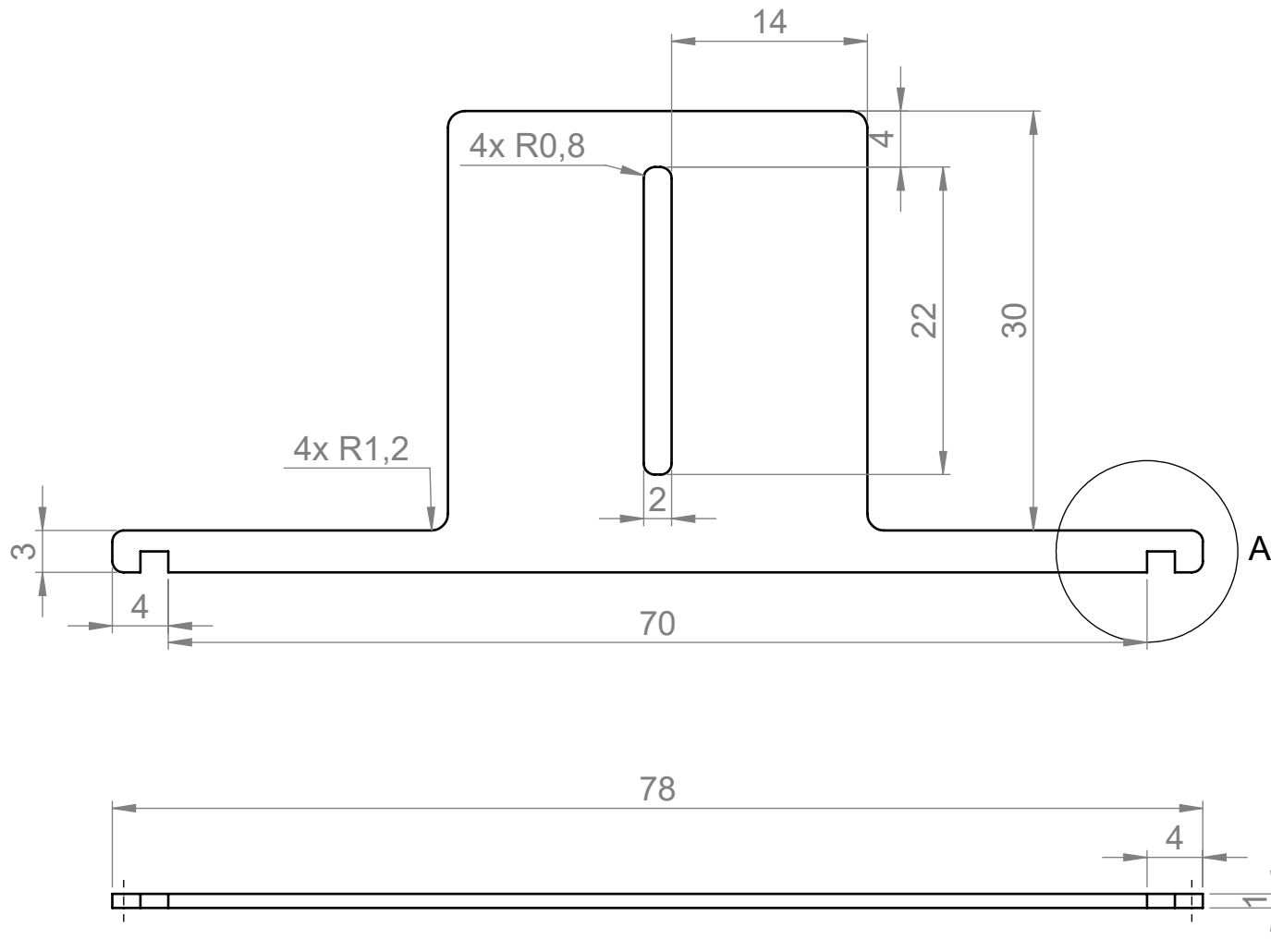
DETAIL A
SCALE 4 : 1

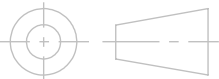
PROJECTION METHOD		UNLESS STATED OTHERWISE: TOLERANCES ± 0,5 MM	DRAWN	Thijm van Bussel	DATE	23-1-2025	
			CHECKED	Manuel	SCALE	2:1	
MATERIAL		SURFACE TREATMENT		TITLE			REV.
2 mm sheet metal		--		part pulley big disc			01
				DRAWING NO.			
				--			A4
				FILE / PART NAME			
				Part Pulley Big Disc			
FACULTY OF ENGINEERING				DIMENSIONS IN MILLIMETERS		SHEET 1 OF 1	

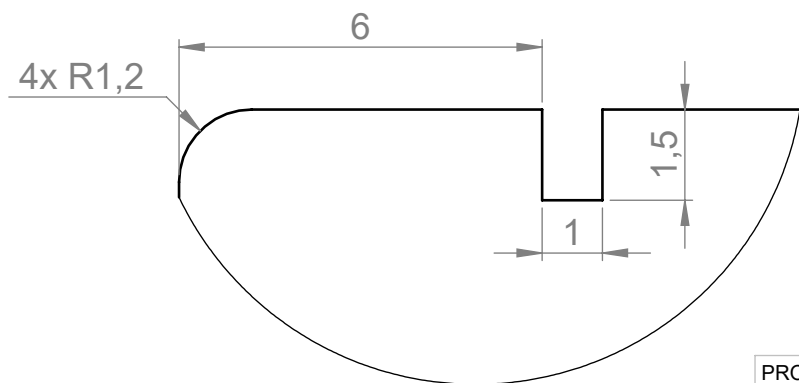
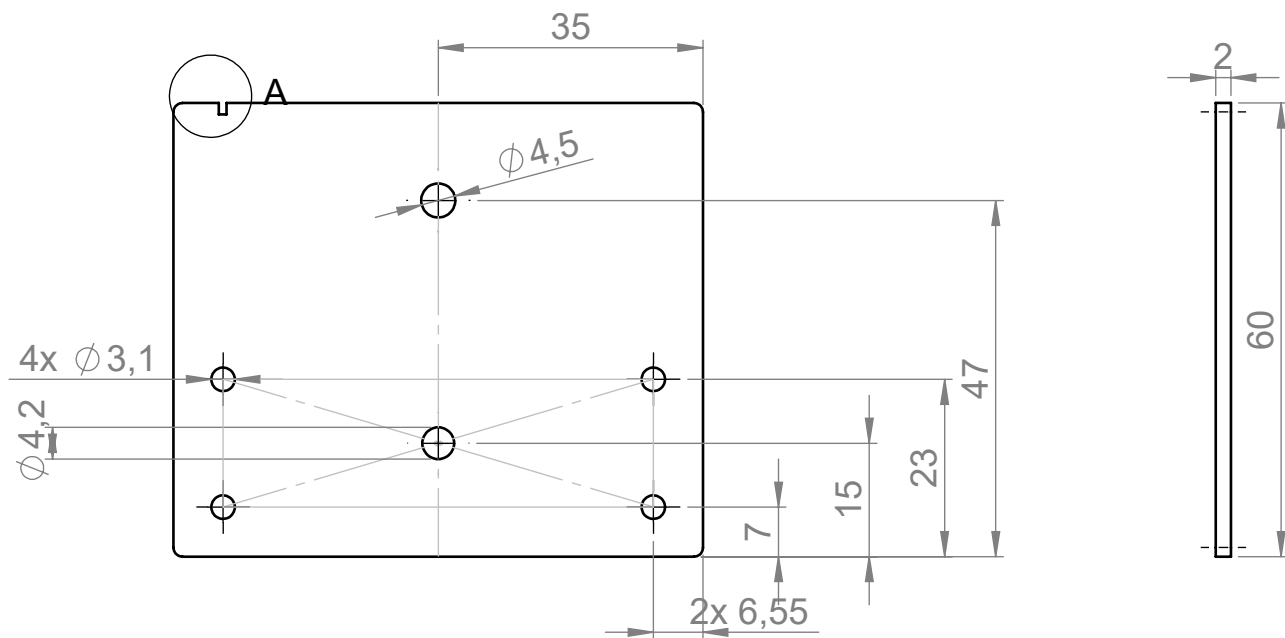


DETAIL A
SCALE 4 : 1

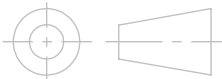
PROJECTION METHOD		UNLESS STATED OTHERWISE: TOLERANCES ± 0,5 MM	DRAWN	Thijm van Bussel	DATE	23-1-2025
			CHECKED	Manuel	SCALE	2:1
MATERIAL		SURFACE TREATMENT		TITLE		REV.
sheet metal 2mm				Part pulley small disc		01
--				DRAWING NO.		
UNIVERSITY OF TWENTE.				--		A4
				FILE / PART NAME		
				Part Pulley Small Disc		
FACULTY OF ENGINEERING			DIMENSIONS IN MILLIMETERS		SHEET 1 OF 1	

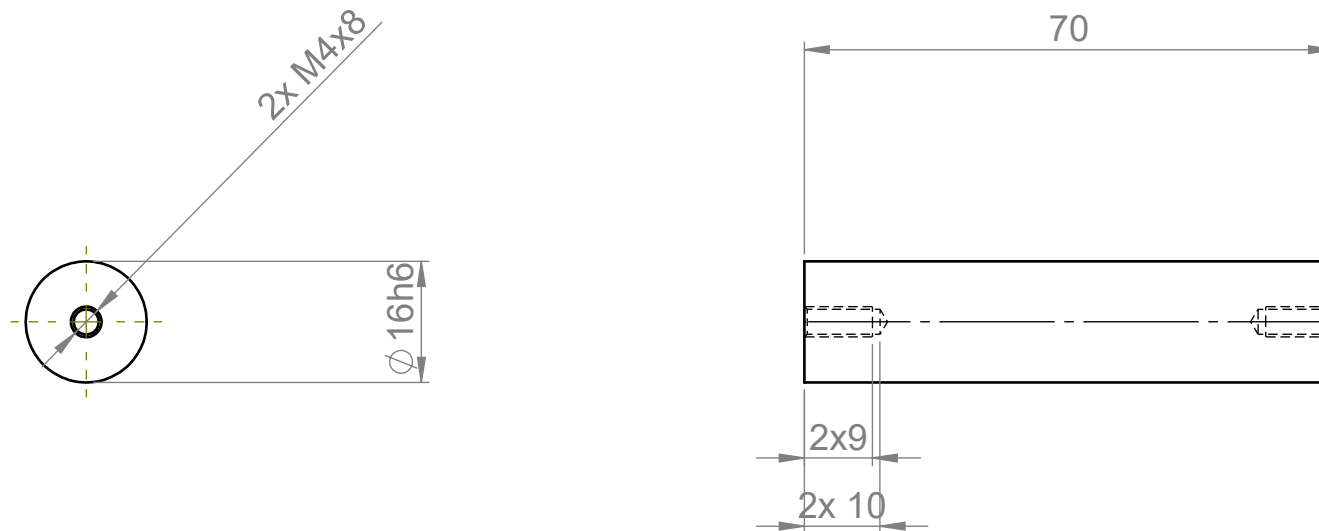


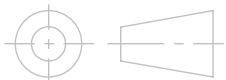
PROJECTION METHOD		UNLESS STATED OTHERWISE: TOLERANCES ± 0,5 MM	DRAWN	Thijm van Bussel	DATE	23-1-2025
			CHECKED	Manuel	SCALE	2:1
MATERIAL 1 mm sheet metal --		SURFACE TREATMENT	TITLE part pulley visor			REV. 01
			DRAWING NO. --			A4
UNIVERSITY OF TWENTE.		FILE / PART NAME Part Pulley Visor				
		FACULTY OF ENGINEERING		DIMENSIONS IN MILLIMETERS		SHEET 1 OF 1

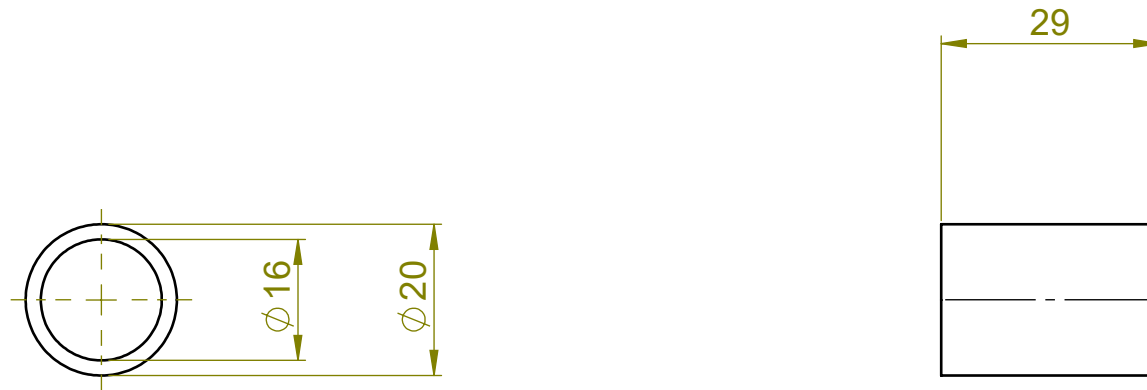


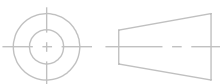
DETAIL A
SCALE 8 : 1

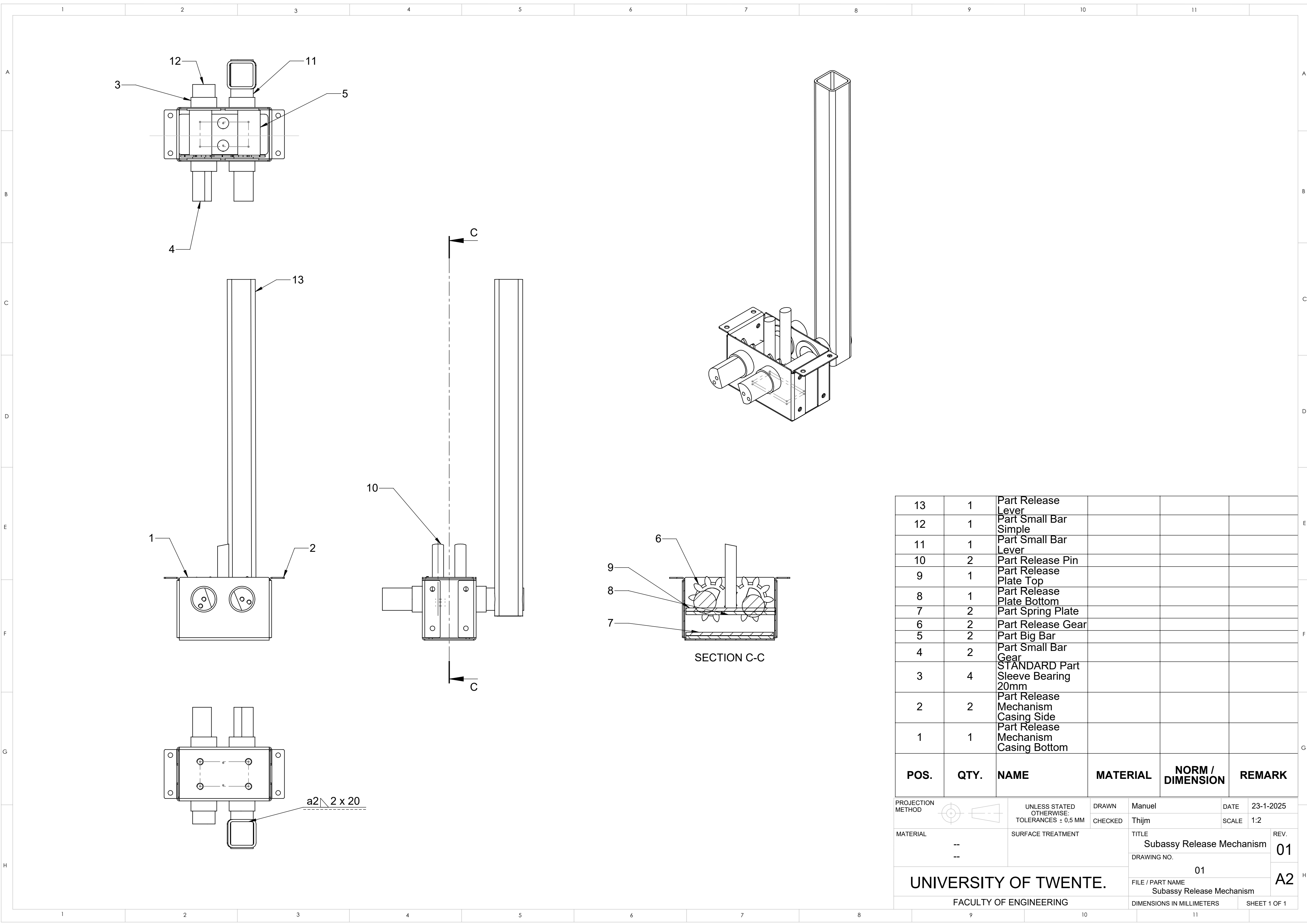
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			CHECKED	--	SCALE	1:1		
MATERIAL 2 mm sheet metal --		SURFACE TREATMENT	TITLE part pulley side				REV. 01	
			DRAWING NO. --					
UNIVERSITY OF TWENTE.			FILE / PART NAME Part Pulley Side				A4	
			DIMENSIONS IN MILLIMETERS					
FACULTY OF ENGINEERING			SHEET 1 OF 1					



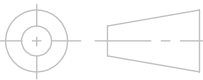
PROJECTION METHOD		UNLESS STATED OTHERWISE: TOLERANCES ± 0,5 MM	DRAWN	Simon Hengeveld	DATE	22-1-2025
			CHECKED	Manuel	SCALE	1:1
MATERIAL Bright Drawn Steel Roundbar Ø20	SURFACE TREATMENT		TITLE Part Pulley Axle			REV. 01
			DRAWING NO. 02			A4
FILE / PART NAME Technical Drawing - Part Pulley Axle						
FACULTY OF ENGINEERING			DIMENSIONS IN MILLIMETERS		SHEET 1 OF 1	

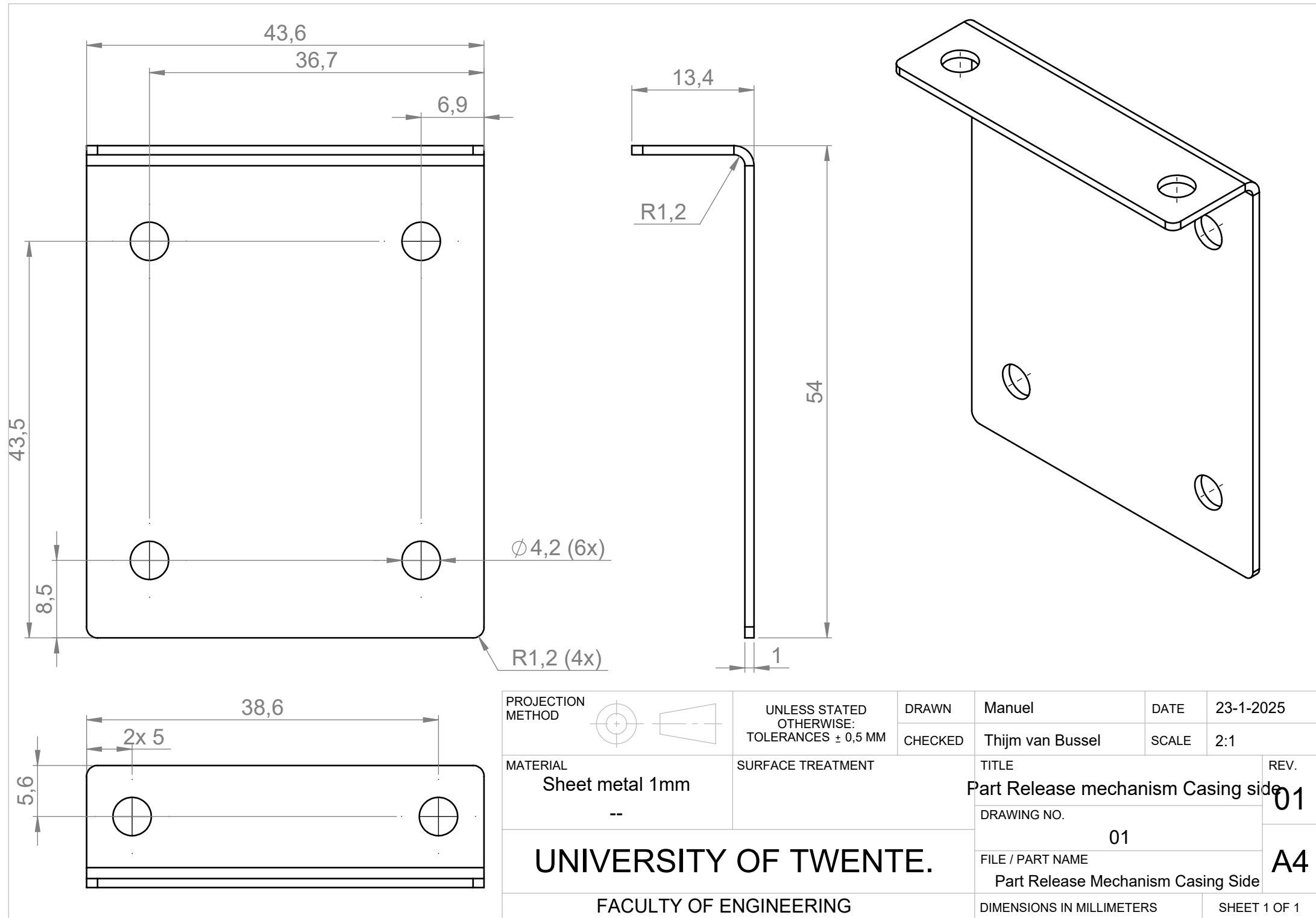


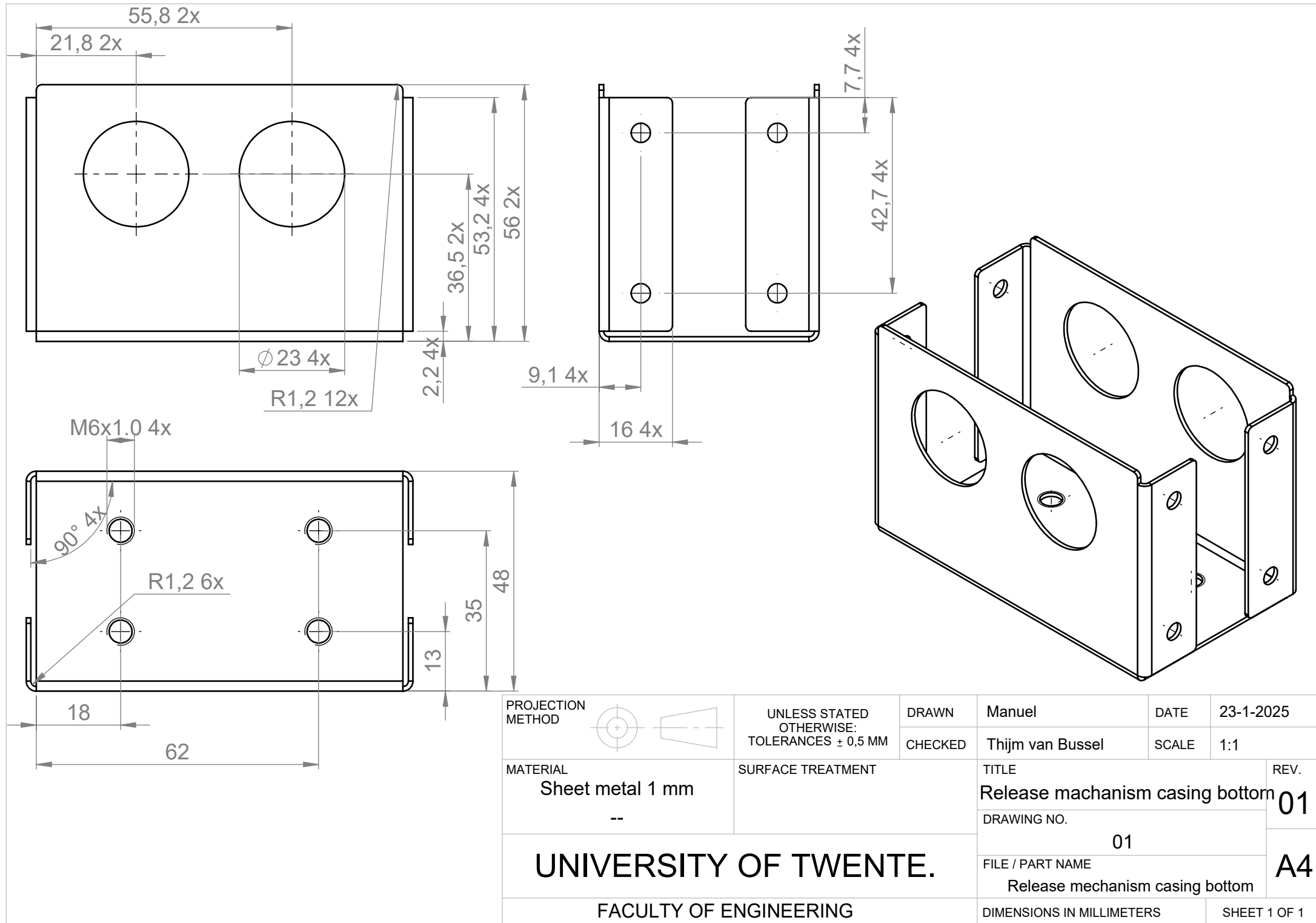
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			CHECKED	Thijm van Bussel	SCALE	1:1
MATERIAL Bright Welded Round E235 CR2; Ø20 x 2.0 mm	SURFACE TREATMENT	TITLE Part Pulley Axle Spacer				REV. 01
		DRAWING NO. 01				A4
FILE / PART NAME Technical Drawing - Part Pulley Axle Spacer						
UNIVERSITY OF TWENTE.						
FACULTY OF ENGINEERING						
DIMENSIONS IN MILLIMETERS				SHEET 1 OF 1		

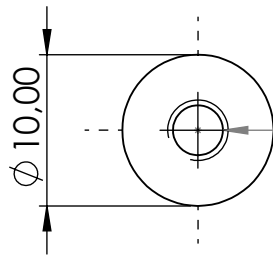


13	1	Part Release Lever			
12	1	Part Small Bar Simple			
11	1	Part Small Bar Lever			
10	2	Part Release Pin			
9	1	Part Release Plate Top			
8	1	Part Release Plate Bottom			
7	2	Part Spring Plate			
6	2	Part Release Gear			
5	2	Part Big Bar			
4	2	Part Small Bar Gear			
3	4	STANDARD Part Sleeve Bearing 20mm			
2	2	Part Release Mechanism Casing Side			
1	1	Part Release Mechanism Casing Bottom			
POS.	QTY.	NAME	MATERIAL	NORM / DIMENSION	REMARK

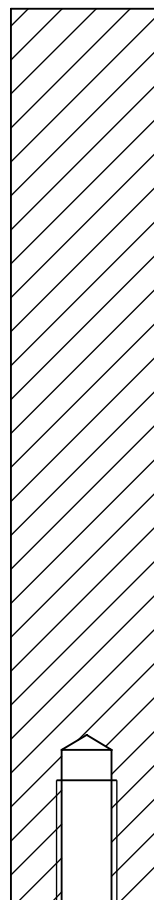
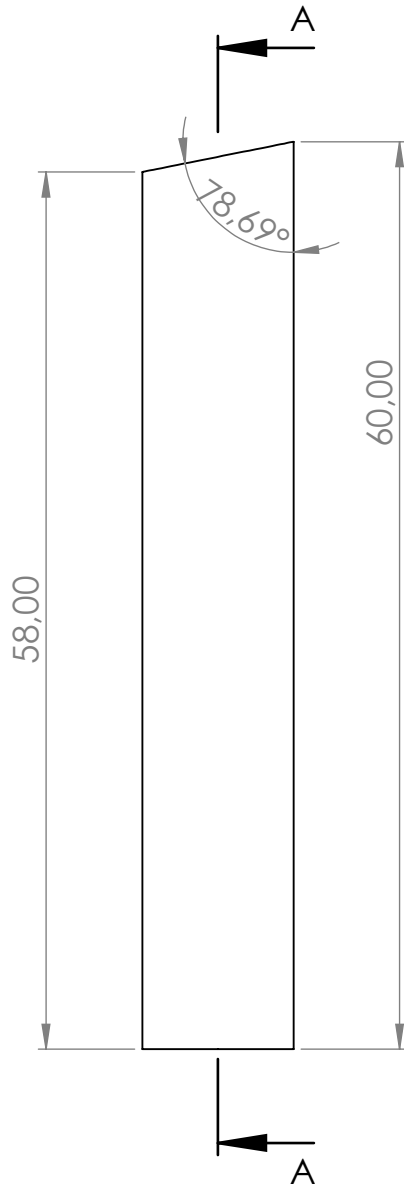
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MATERIAL	--	SURFACE TREATMENT	CHECKED	Thijm	SCALE	1:2
UNIVERSITY OF TWENTE.			TITLE			REV.
FACULTY OF ENGINEERING			Subassy Release Mechanism			01
			DRAWING NO.			A2
			FILE / PART NAME			
			Subassy Release Mechanism			
			DIMENSIONS IN MILLIMETERS			
			SHEET 1 OF 1			





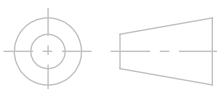


Ø 3,30 ± 10,00
M4x0.7 - 6H ± 8,00



SECTION A-A

PROJECTION
METHOD



UNLESS STATED
OTHERWISE:
TOLERANCES ± 0,5 MM

DRAWN

Naagie

DATE

22-1-2025

CHECKED

Manuel

SCALE

2:1

MATERIAL

Bright drawn steel Round fit h9

SURFACE TREATMENT

--

TITLE

RELEASE PIN

REV.

01

DRAWING NO.

1

FILE / PART NAME

Technical Drawing- Part Release Pin

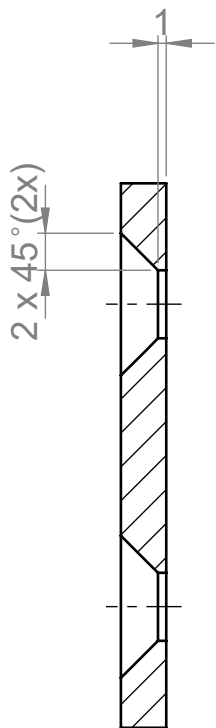
A4

UNIVERSITY OF TWENTE.

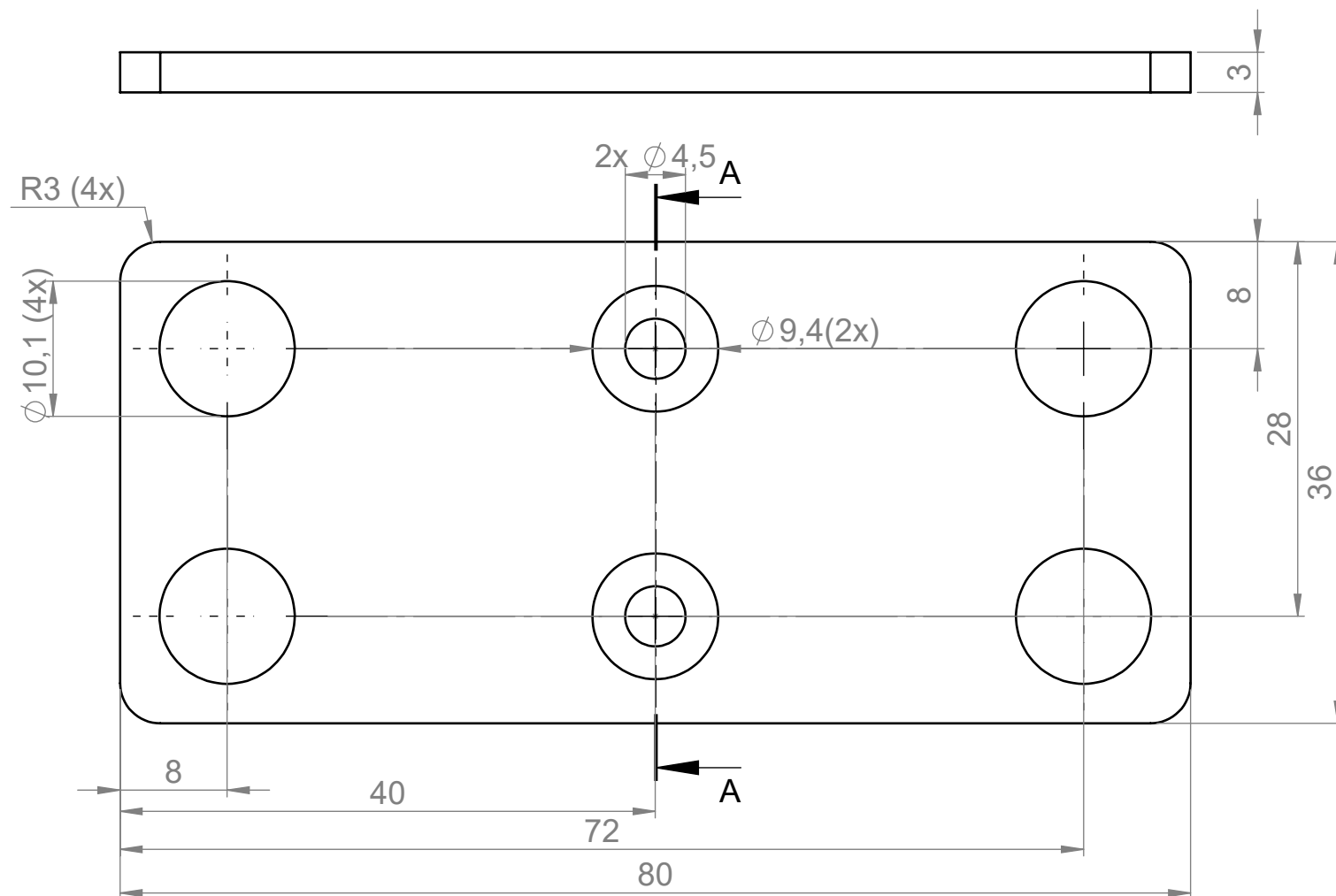
FACULTY OF ENGINEERING

DIMENSIONS IN MILLIMETERS

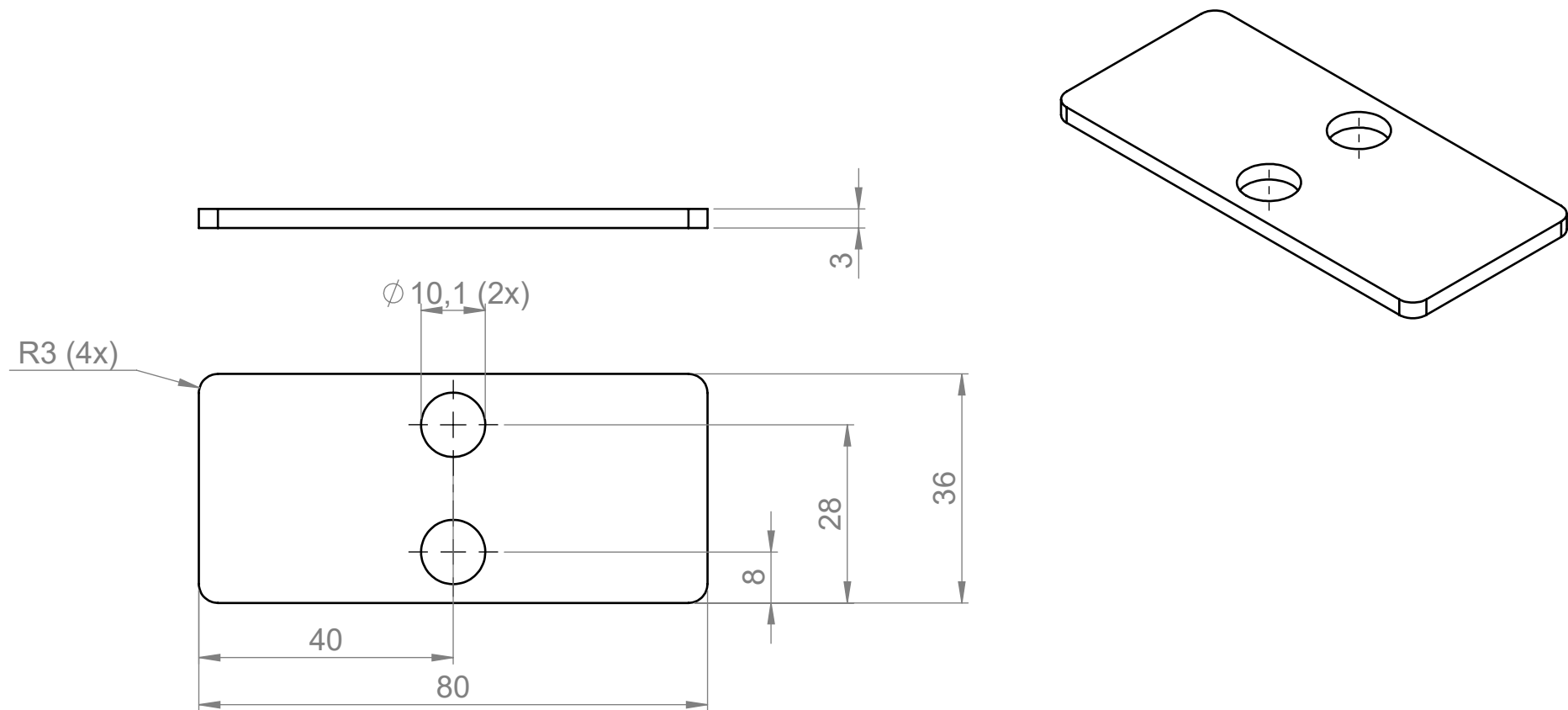
SHEET 1 OF 1

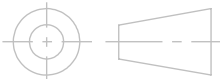


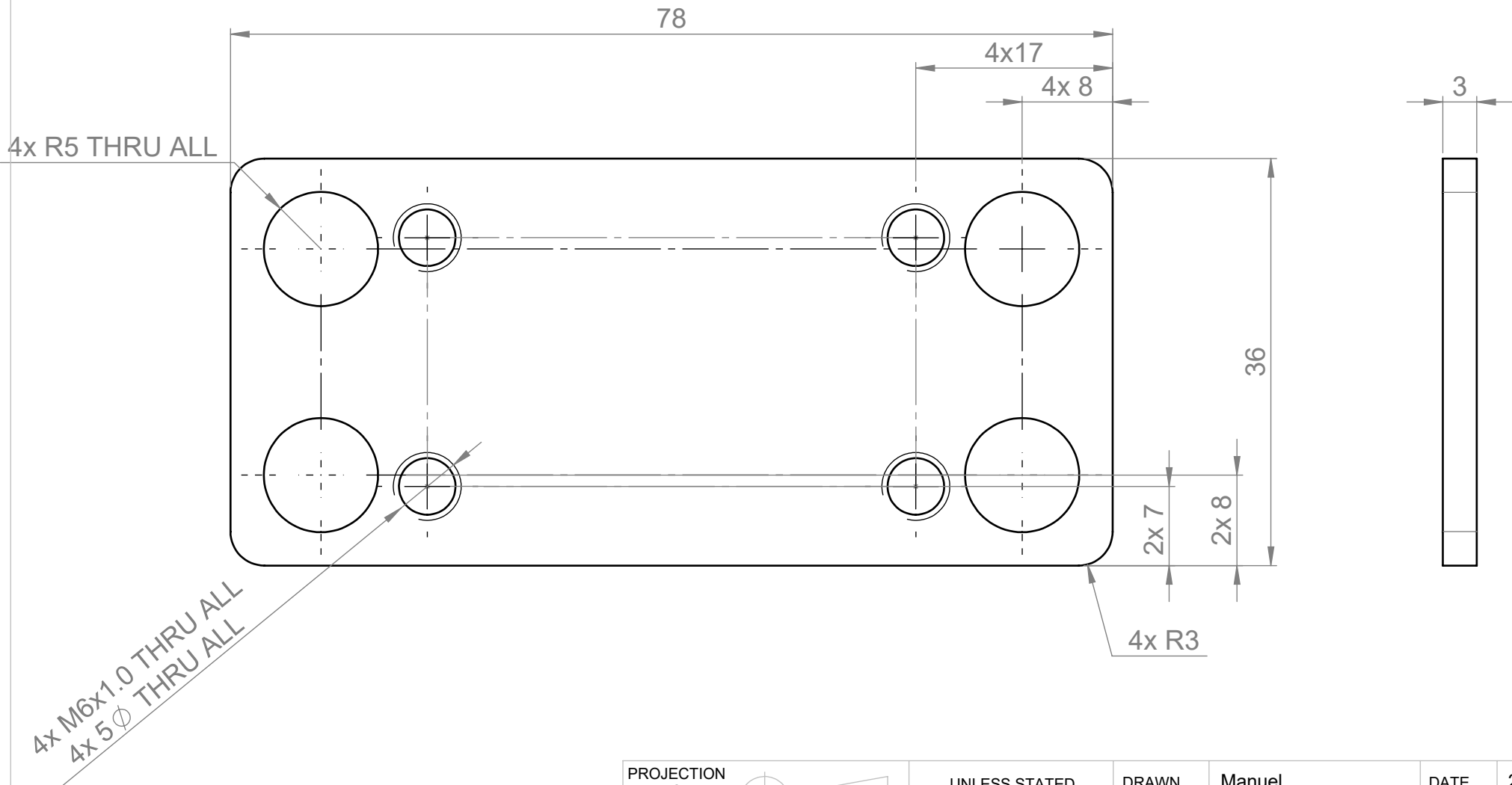
SECTION A-A



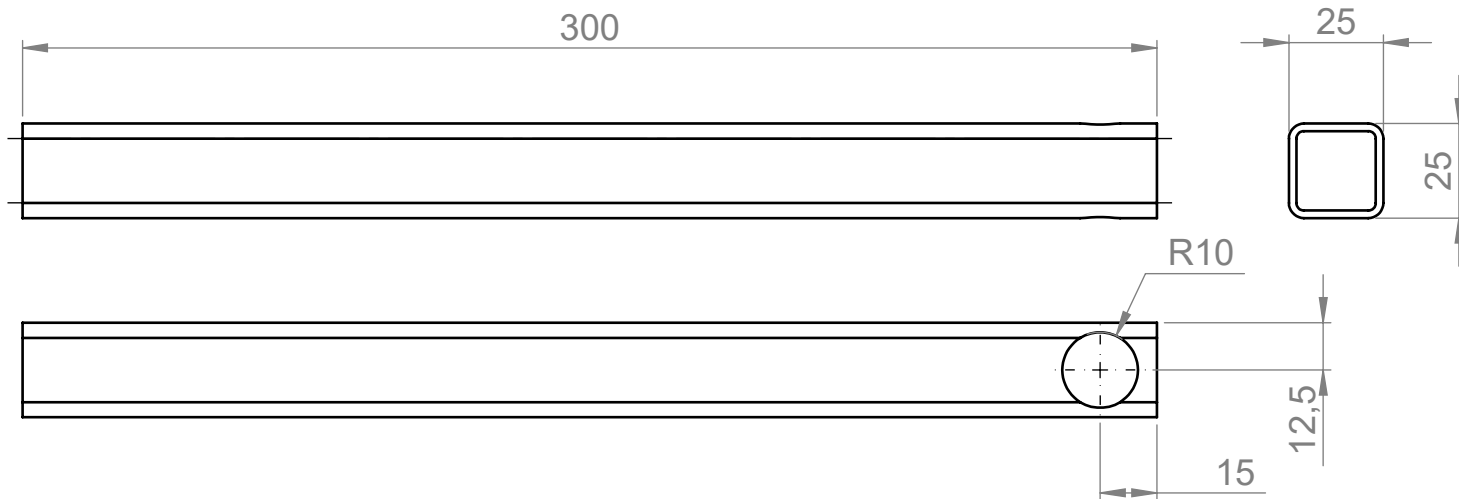
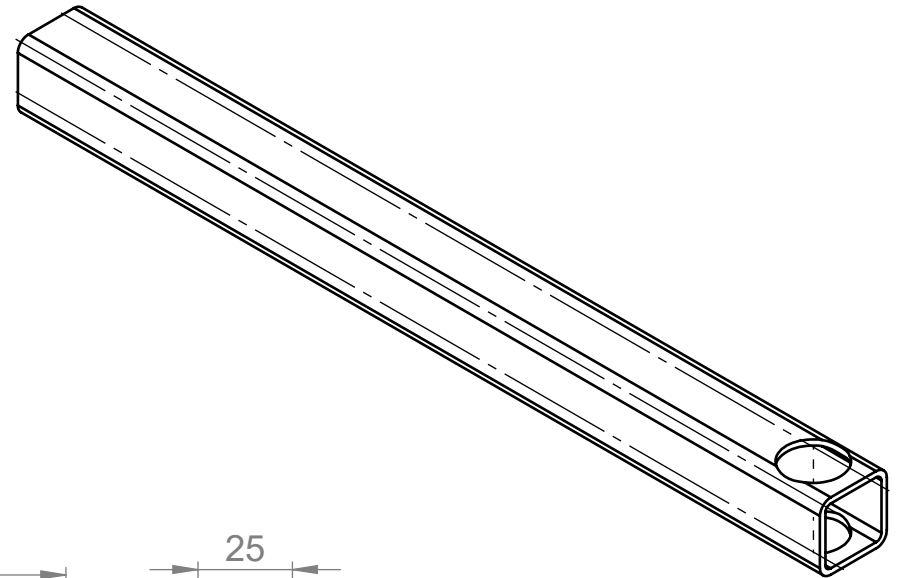
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			CHECKED	Thijm van Bussel	SCALE	2:1	
MATERIAL Sheet metal 3 mm --		SURFACE TREATMENT	TITLE Part Release Plate bottom				REV. 01
			DRAWING NO. 01				A4
FILE / PART NAME Part Release Plate Bottom							
FACULTY OF ENGINEERING			DIMENSIONS IN MILLIMETERS		SHEET 1 OF 1		

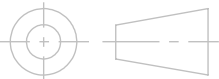


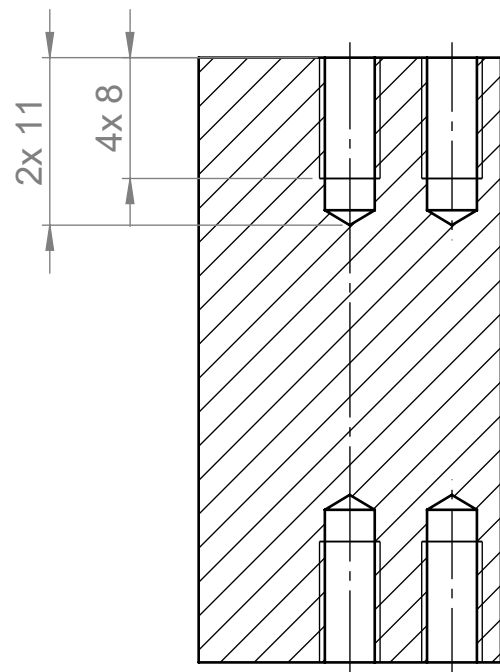
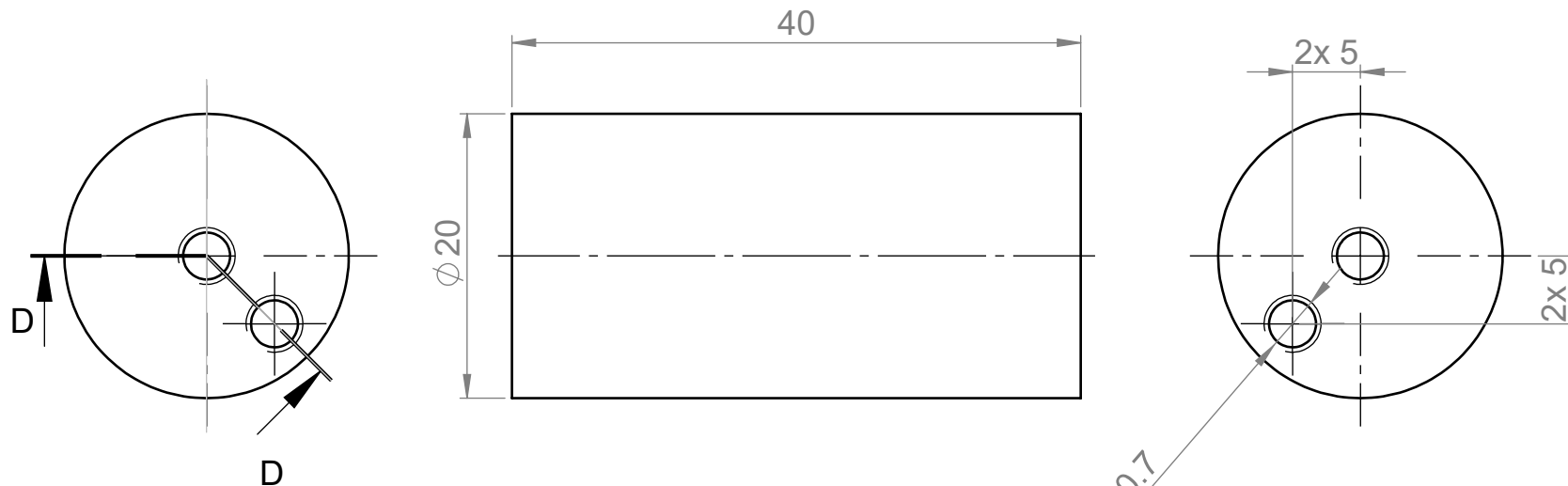
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			CHECKED	Thijm van Bussel	SCALE	1:1
MATERIAL Sheet metal 3mm --	SURFACE TREATMENT		TITLE Part Release plate top			REV. 01
			DRAWING NO. 01			A4
UNIVERSITY OF TWENTE.			FILE / PART NAME Part Release Plate Top			
			DIMENSIONS IN MILLIMETERS			SHEET 1 OF 1
FACULTY OF ENGINEERING						



PROJECTION METHOD		UNLESS STATED OTHERWISE: TOLERANCES ± 0,5 MM	DRAWN	Manuel	DATE	23-1-2025		
			CHECKED	Thijm van Bussel	SCALE	2:1		
MATERIAL		SURFACE TREATMENT		TITLE		REV.		
Sheet metal 3mm		--		Part Spring Plate		01		
UNIVERSITY OF TWENTE.		DRAWING NO.		01			A4	
		FILE / PART NAME		Part Spring Plate				
FACULTY OF ENGINEERING			DIMENSIONS IN MILLIMETERS		SHEET 1 OF 1			

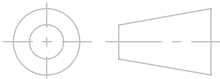


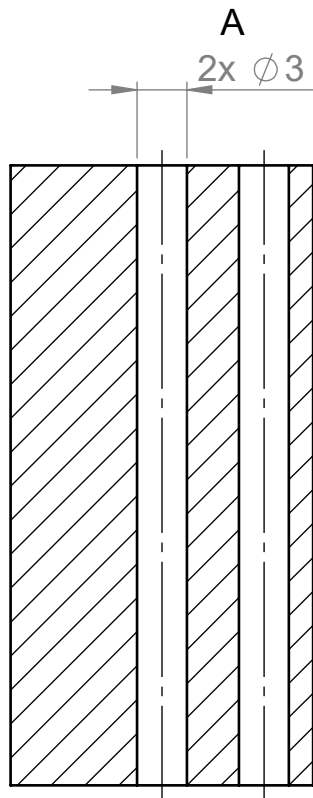
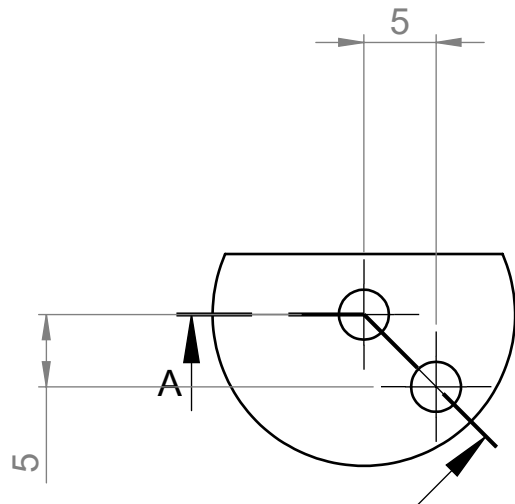
PROJECTION METHOD 	UNLESS STATED OTHERWISE: TOLERANCES ± 0,5 MM	DRAWN	Manuel	DATE	23-1-2025
		CHECKED	Thijm	SCALE	1:2
MATERIAL Cold-formed square tube --	SURFACE TREATMENT	TITLE Lever release mechanism			REV. 01
		DRAWING NO. 01			A4
UNIVERSITY OF TWENTE.		FILE / PART NAME Project Lever release mechanism			
FACULTY OF ENGINEERING		DIMENSIONS IN MILLIMETERS		SHEET 1 OF 1	



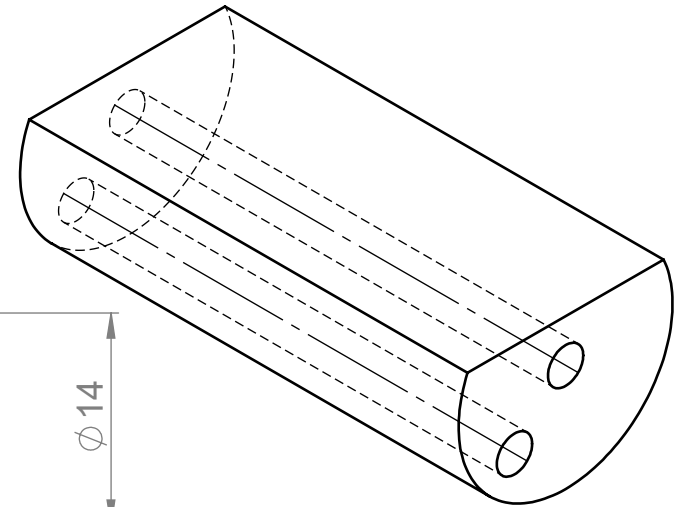
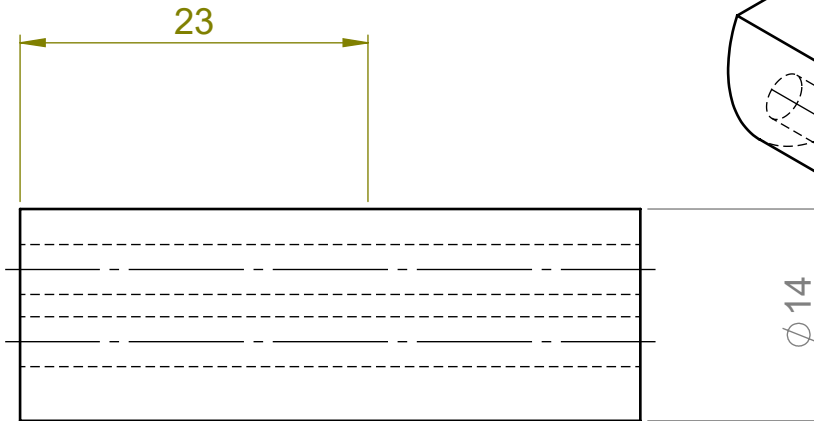
SECTION D-D

M4x0.7

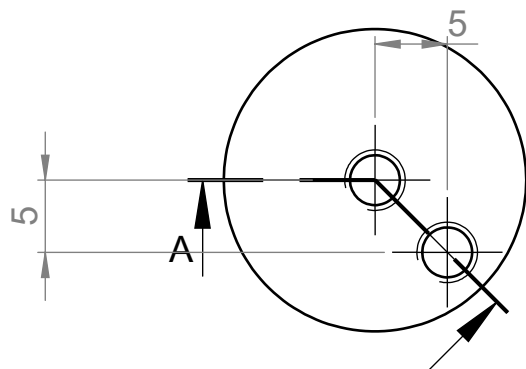
PROJECTION METHOD 	UNLESS STATED OTHERWISE: TOLERANCES ± 0.5 MM	DRAWN	Manuel	DATE	23-1-2025
		CHECKED	Thijm	SCALE	2:1
MATERIAL Free-machining steel --	SURFACE TREATMENT	TITLE Big assymetric part			REV. 01
		DRAWING NO. 01			A4
UNIVERSITY OF TWENTE.		FILE / PART NAME			
		Project Drawing Big assemetric release mechnsm part			
FACULTY OF ENGINEERING		DIMENSIONS IN MILLIMETERS		SHEET 1 OF 1	



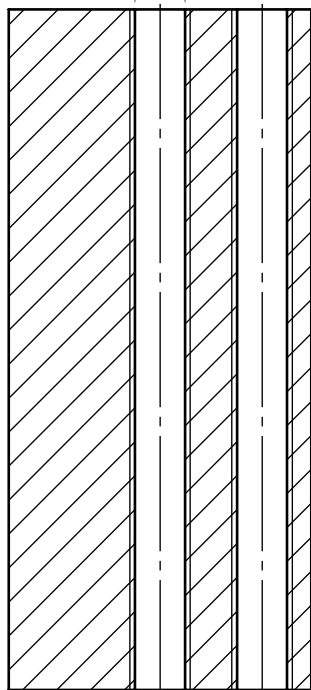
SECTION A-A



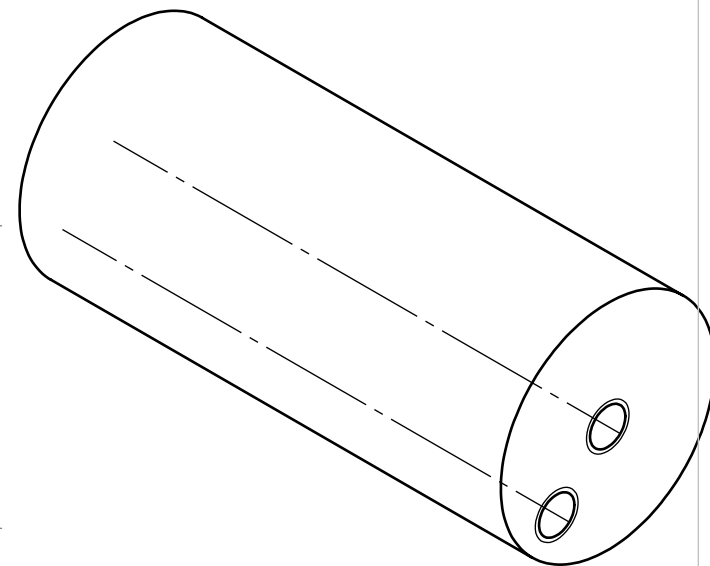
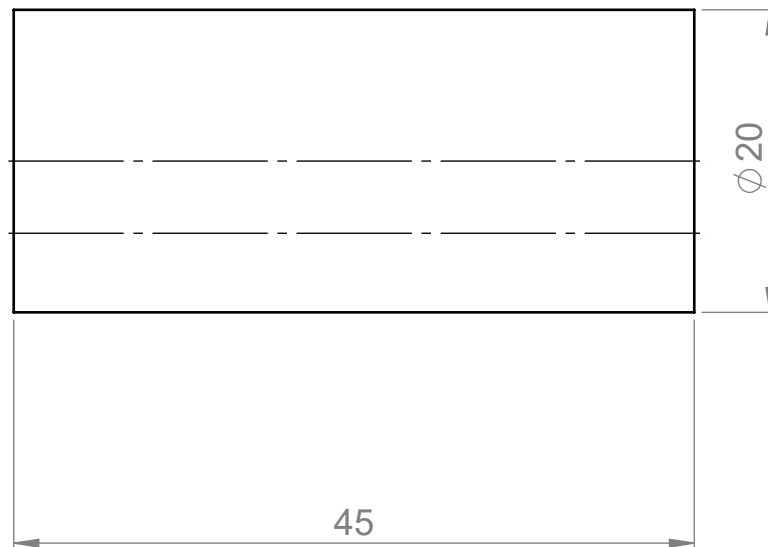
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			CHECKED	Thijm	SCALE	2:1
MATERIAL Free-machining steel --	SURFACE TREATMENT	TITLE Small assymetric bar 2	REV.		01	
			DRAWING NO.		01	
UNIVERSITY OF TWENTE.		FILE / PART NAME		A4		
		Project 2 Small assemetric realease mechnsm.				
FACULTY OF ENGINEERING		DIMENSIONS IN MILLIMETERS		SHEET 1 OF 1		



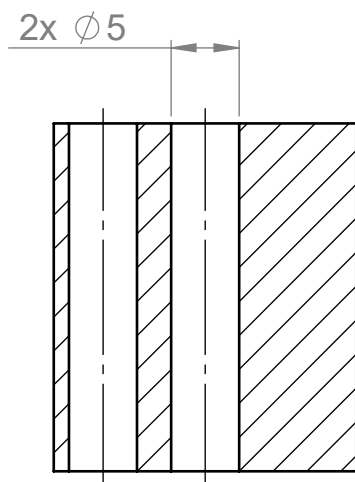
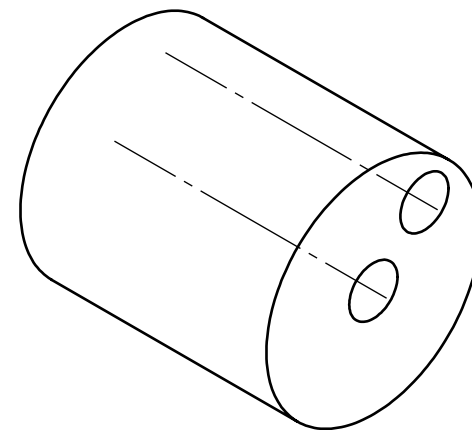
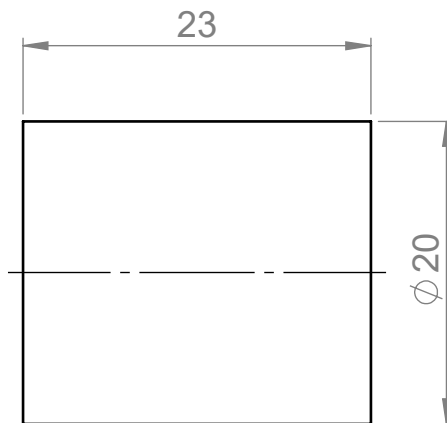
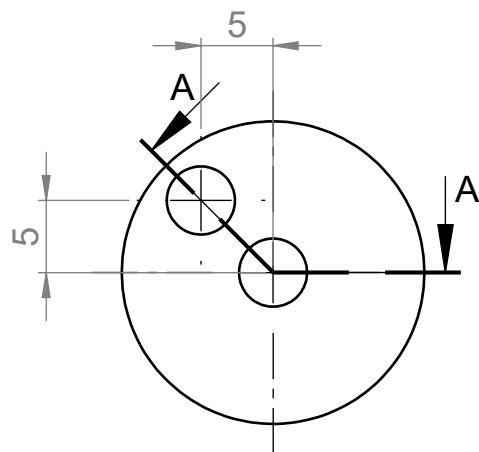
2x $\varnothing 3$



SECTION A-A

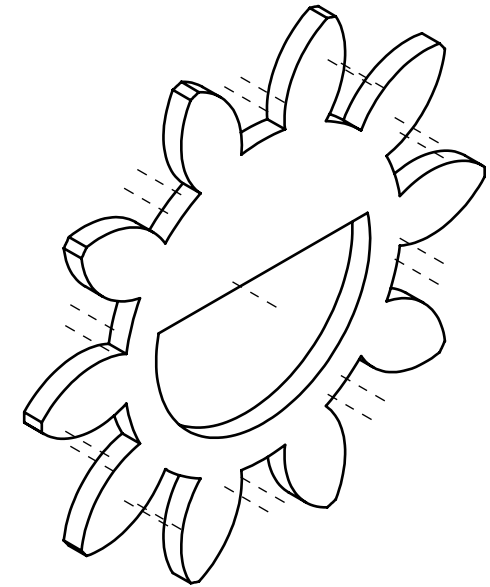
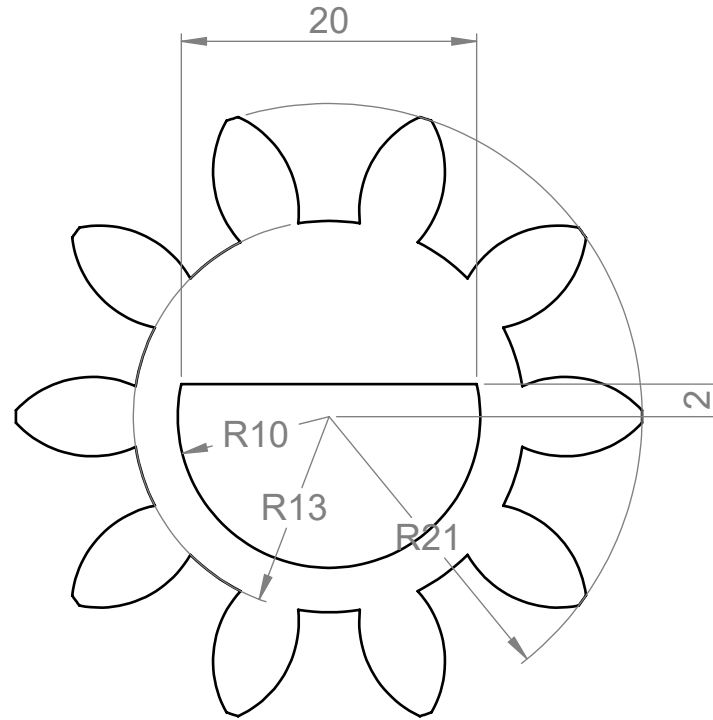
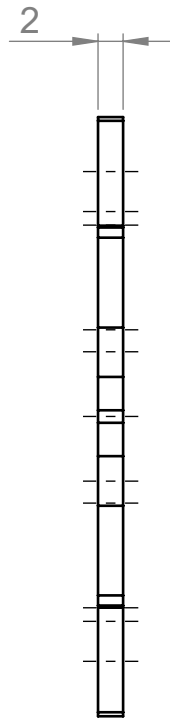


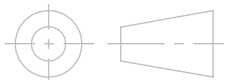
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			CHECKED	Thijm	SCALE	2:1	
MATERIAL		SURFACE TREATMENT		TITLE		REV.	
Free-machining steel				Small assymetric bar 1		01	
--				DRAWING NO.			
UNIVERSITY OF TWENTE.				01		A4	
				FILE / PART NAME			
		Project Drawing 1 Small assemetric realease mechnsm					
FACULTY OF ENGINEERING			DIMENSIONS IN MILLIMETERS			SHEET 1 OF 1	



SECTION A-A

PROJECTION METHOD		UNLESS STATED OTHERWISE: TOLERANCES ± 0,5 MM	DRAWN	Manuel	DATE	23-1-2025
			CHECKED	Thijm	SCALE	2:1
MATERIAL		SURFACE TREATMENT		TITLE	REV.	
Free-machining steel				Small assymetric bar 3	01	
--				DRAWING NO.		
				01		
UNIVERSITY OF TWENTE.		FILE / PART NAME			A4	
		Project Small assymetric bar 3				
FACULTY OF ENGINEERING			DIMENSIONS IN MILLIMETERS		SHEET 1 OF 1	



PROJECTION METHOD 	UNLESS STATED OTHERWISE: TOLERANCES ± 0.5 MM	DRAWN	Manuel	DATE	23-1-2025
		CHECKED	Thijm	SCALE	2:1
MATERIAL Sheet metal 2mm --	SURFACE TREATMENT	TITLE Gear Release mechanism			REV. 01
		DRAWING NO. 01			A4
UNIVERSITY OF TWENTE.					
FACULTY OF ENGINEERING		DIMENSIONS IN MILLIMETERS		SHEET 1 OF 1	

Appendix Six – Work Preparation Sheets

Part; T-Hinge/Baseplate Spacer

Work preparation sheet			
Part name: Part T-Hinge+Baseplate Spacer		Date: 06/12/2024	Sheet number: 1/2
Belonging to drawing: Technical Drawing - Part T-Hinge+Baseplate Spacer		Made by: Simon Hengeveld	
Part no: 04	Amount: 2	Material: Bright Drawn Steel	
Starting dimensions material: Ø10x161			
Machining sequence:			
Operation	Tool fixtures, tools and other aids	Settings speed (V), number of revolutions (N), feed (f), depth of cut (t ₀) etc.	Remarks
Cut stock to length	Saw		Cut to 161 mm length
Clamping	Three-jaw chuck		Have it stick out 90 mm
Face end	Toolholder for 80° insert	N=1000 rpm f=manual t=0.5mm	This is the first part you are working on!
Centerdrill	Centerdrill, cutting fluid	N= 1500 rpm	
Drill Ø3.2x10	Ø3.2 Drill, cutting fluid	N=2000 rpm	
Countersink	Countersink bit	N=50 rpm	
Tap M4x8	M4 Tap, tap chuck, cutting fluid	N=50 rpm	
Deburr edges	File	N=50 rpm	While spinning on the lathe
Cut length of 79mm off	Parting tool, cutting fluid	---	
Face the end of the material that still sticks out of the lathe	Toolholder for 80° insert	N=1000 rpm f=manual t=0.5mm	This is the second part you are now working on!
Centerdrill	Centerdrill, cutting fluid	N= 1500 rpm	

Drill Ø3.2x10	Ø3.2 Drill, cutting fluid	N=2000 rpm	
Countersink	Countersink bit	N=50 rpm	
Tap M4x8	M4 Tap, tap chuck, cutting fluid	N=50 rpm	
Deburr edges	File	N=50 rpm	While spinning on the lathe
Face the other end of the first part to a length of 77 mm	Toolholder for 80° insert	N=1000 rpm f=manual t=0.5mm	
Centerdrill	Centerdrill, cutting fluid	N= 1500 rpm	
Drill Ø3.2x10	Ø3.2 Drill, cutting fluid	N=2000 rpm	
Countersink	Countersink bit	N=50 rpm	
Tap M4x8	M4 Tap, tap chuck, cutting fluid	N=50 rpm	
Deburr edges	File	N=50 rpm	While spinning on the lathe
Face the other end of the second part to a length of 77 mm	Toolholder for 80° insert	N=1000 rpm f=manual t=0.5mm	
Centerdrill	Centerdrill, cutting fluid	N= 1500 rpm	
Drill Ø3.2x10	Ø3.2 Drill, cutting fluid	N=2000 rpm	
Countersink	Countersink bit	N=50 rpm	
Tap M4x8	M4 Tap, tap chuck, cutting fluid	N=50 rpm	
Deburr edges	File	N=50 rpm	While spinning on the lathe

Part; Pulley axle

Work preparation sheet		
Part name: Part Pulley Axle	Date: 06/12/2024	Sheet number: 1/2
Belonging to drawing: Part Pulley Axle	Made by: Simon Hengeveld	
Part no: 02	Amount: 1	Material: Bright Drawn Steel Roundbar
Starting dimensions material: Ø20x??		

Machining sequence:			
Operation	Tool fixtures, tools and other aids	Settings speed (V), number of revolutions (N), feed (f), depth of cut (t ₀) etc.	Remarks
Cut stock to length	Saw		<u>How long should the part be that sticks into the lathe?</u>
Face end	Toolholder for 80° insert	N=1000 rpm f>manual t=0.5mm	
Drilling centerhole	Centerdrill, cutting fluid	N= 1500 rpm	
Tailstock support	Tailstock		<u>Necessary?</u>
Bring OD to dimension Ø16h6	Toolholder for 80° insert	N=1000 rpm f>manual or 0.1mm/r t=0.5mm	
Remove Tailstock support			
Drill hole in centre Ø3.2x11mm	Ø3.2 Drill, cutting fluid	N=2000 rpm	
Tap hole in centre M4x9	M4 Tap, tap chuck, cutting fluid	N=50 rpm	
Deburr edges	File	N=50 rpm	While spinning on the lathe
Cut to length	Parting tool, cutting fluid	??	
Face other end	Toolholder for 80° insert	N=1000 rpm f>manual t=0.5mm	
Drilling a centerhole	Centerdrill, cutting fluid	N=1500 rpm	
Drill hole in centre Ø3.2x11mm	Ø3.2 Drill, cutting fluid	N=2000 rpm	
Tap hole in centre M4x9	M4 Tap, cutting fluid	N=50 rpm	
Deburr edges	File	N=50 rpm	While spinning on the lathe

Part; Pulley Axle Spacer

Work preparation sheet			
Part name: Part Pulley Axle Spacer		Date: 06/12/2024	Sheet number: 1/1
Belonging to drawing: Part Pulley Axle Spacer		Made by: Simon Hengeveld	
Part no: 01	Amount: 2	Material: Bright Welded Round Tube	
Starting dimensions material: Ø20x2 x 65			
Machining sequence:			
Operation	Tool fixtures, tools and other aids	Settings speed (V), number of revolutions (N), feed (f), depth of cut (t ₀) etc.	Remarks
Cut stock to length	Saw		Cut to 65 mm length
Clamping	Three-jaw chuck		Have it stick out 50 mm
Face end	Toolholder for 80° insert	N=1000 rpm f>manual t=0.5mm	This is the first part you are working on!
Deburr edges	File	N=50 rpm	While spinning on the lathe
Cut length of 30mm off	Parting tool, cutting fluid	??	
Face the end of the material that still sticks out of the lathe	Toolholder for 80° insert	N=1000 rpm f>manual t=0.5mm	This is the second part you are now working on!
Deburr edges	File	N=50 rpm	While spinning on the lathe
Face the other end of the first part to a length of 29 mm	Toolholder for 80° insert	N=1000 rpm f>manual t=0.5mm	
Deburr edges	File	N=50 rpm	While spinning on the lathe
Face the other end of the second part to a length of 29 mm	Toolholder for 80° insert	N=1000 rpm f>manual t=0.5mm	
Deburr edges	File	N=50 rpm	While spinning on the lathe

Part; Release Pin

Work preparation sheet			
Part name: Release pin		Date: 12-18-2024	Sheet number: 1 of 1
Belonging to drawing: Technical drawing-release pin		Made by: Naagie	
Part no: 1	Amount: 2	Material: Bright Dawn Steel	
Starting dimensions material: 60mm x Ø10mm			
Machining sequence: lathe -> milling -> lathe			
Operation	Tool	Settings	Remarks
Cut stock to 60mm length	Hacksaw	Manual	
Facing one end to the exact length of 60mm	Toolholder for 80° insert on lathe	N = 100rpm f = manual t ₀ = 0.5mm	
Drilling a centerhole	Centerdrill	N = 800 rpm	
Drilling a hole for M4 tap, depth 10mm	Drill Ø3.3mm, cutting fluid	N = 800 rpm	
Tapping M4 thread	M4 tap on the tapping chuck, cutting fluid	N = 50 rpm f = manual	

Subassy; Acylindrical Bars

Work preparation sheet		
Part name: Lever Release mechanism	Date: 19-12-2024	Sheet number:1
Belonging to drawing: Lever Release mechanism	Made by: Manuel	
Part no: 1	Amount:1	Material: Cold-formed square tube

Starting dimensions material: 25 mm x 25 mm x >300mm			
Machining sequence: Sawing → Milling			
Operation	Tool fixtures, tools and other aids	Settings speed (V), number of revolutions (N), feed (f), depth of cut (t ₀) etc.	Remarks
Cutting to length	Saw		Debur edges
Centerdrilling	Centerdrill	N=1500 rpm	
Drilling Hole	Drill ø 20mm with cutting fluid	N=318 rpm	
Debur and break all hole edges with countersink drill	Countersink with cutting fluid	N=200 rpm	

Work preparation sheet			
Part name: Small asymmetric bar 1		Date: 19-12-2024	Sheet number:1
Belonging to drawing: Small asymmetric bar 1		Made by: Manuel	
Part no: 2	Amount: 1	Material: Free-machining steel	
Starting dimensions material: >30mm x ø25mm			
Machining sequence: Sawing → Turning → Milling			
Operation	Tool fixtures, tools and other aids	Settings speed (V), number of revolutions (N), feed (f), depth of cut (t ₀) etc.	Remarks
Cutting to length	Saw		Deburr edges
Find middle of the circle to place origin	Probe	Feature circle centre	Use ground parallels to clamp the stock
Centerdrilling	Centerdrill	N=1500 rpm	
Drilling holes	Drill ø5mm with cutting fluid	N= 1274 rpm	

Debur and break all hole edges with countersink drill	Countersink with cutting fluid	N=200 rpm	
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Work preparation sheet			
Part name: Small asymmetric bar 3		Date: 19-12-2024	Sheet number:1
Belonging to drawing: Small asymmetric bar 3		Made by: Manuel	
Part no: 3	Amount: 1	Material: Free-machining steel	
Starting dimensions material: >30mm x ø25mm			
Machining sequence: Sawing → Turning → Milling			
Operation	Tool	Settings	Remarks
	fixtures, tools and other aids	speed (V), number of revolutions (N), feed (f), depth of cut (t ₀) etc.	
Cutting to length	Saw		Deburr edges
Turning diameter to 20 mm	Toolholder for 80° insert	N= 1000 rpm f= 0,1 mm/r t0= 1 mm	
Find middle of the circle to place origin	Probe	Feature circle centre	Use ground parallels to clamp the stock
Centerdrilling	Centerdrill	N=1500 rpm	
Drilling holes	Drill ø5mm with cutting fluid	N= 1274 rpm	
Debur and break all hole edges with countersink drill	Countersink with cutting fluid	N=200 rpm	

Work preparation sheet		
Part name: Small asymmetric bar 2	Date: 19-12-2024	Sheet number: 1
Belonging to drawing: Small asymmetric bar 2	Made by: Manuel	

Part no: 4	Amount: 2	Material: Free-machining steel	
Starting dimensions material: >30mm x ø25mm			
Machining sequence: Sawing → Milling			
Operation	Tool	Settings	Remarks
	fixtures, tools and other aids	speed (V), number of revolutions (N), feed (f), depth of cut (t ₀) etc.	
Cutting to length	Saw		Debur edges
Turning diameter to 20 mm	Toolholder for 80° insert	N= 1000 rpm f= 0,1 mm/r t0= 1 mm	
Find middle of the circle to place origin	Probe	Feature circle centre	Use ground parallels to clamp the stock
Centerdrilling	Centerdrill	N=1500 rpm	
Drilling Hole	Drill ø 20mm with cutting fluid	N=318 rpm	Debur edges
Debur and break all hole edges with countersink drill	Countersink with cutting fluid	N=200 rpm	
Turn part over			
Milling flat side	Mill ø 20mm	N=398 rpm	

Work preparation sheet			
Part name: Big asymmetric Bar		Date: 19-12-2024	Sheet number:1
Belonging to drawing: Big asymmetric Bar		Made by: Manuel	
Part no: 5	Amount: 1	Material: Free-machining steel	
Starting dimensions material: >400mm x ø25mm			
Machining sequence: Sawing → Turning → Milling			
Operation	Tool	Settings	Remarks

	fixtures, tools and other aids	speed (V), number of revolutions (N), feed (f), depth of cut (t_0) etc.	
Cutting to length	Saw		Debur edges
Turning diameter to 20 mm	Toolholder for 80° insert	N= 1000 rpm f= 0,1 mm/r t_0 = 1 mm	
Find middle of the circle to place origin	Probe	Feature circle centre	Use ground parallels to clamp the stock
Centerdrilling	Centerdrill	N=1500 rpm	
Drilling Holes	Drill $\varnothing$ 4mm with cutting fluid	N=1592 rpm	Debur edges
Debur and break all hole edges with countersink drill	Countersink with cutting fluid	N=200 rpm	
Tapping holes	Tap M4	N= 50 rpm	
Turn product 180 degrees.			
Centerdrilling	Centerdrill	N=1500 rpm	
Drilling Holes	Drill $\varnothing$ 4mm with cutting fluid	N=1592 rpm	Debur edges
Debur and break all hole edges with countersink drill	Countersink with cutting fluid	N=200 rpm	
Tapping holes	Tap M4	N= 50 rpm	

Appendix Seven - Measurement Report

Part	No.	Tolerance	Accuracy	Measuring tool	Conclusion
Winch Axle	1	20h6 20.000 19.989	20.041	Micrometer	Okay, it spins well.
Pulley Axle	1	16h6 16.000 15.989	16.009	Micrometer	Okay, it spins well.
Pulley Axle	1	70±0.50 70.50 69.50	69.75	Calliper	Okay, it fits well.
T-Hinge	1	10h6 10.000 9.989	9.986	Micrometer	Okay, it spins well.
T-Shaft	1	10h6 10.000 9.989	9.987	Micrometer	Okay, it spins well.
T-Shaft	2	10h6 10.000 9.989	9.982	Micrometer	Okay, it spins well.
Small Bar Gear	1	14±0.50 14.50 13.50	14.00	Calliper	Okay. gear fits.
Small Bar Gear	2	14±0.50 14.50 13.50	13.20	Calliper	Too small, gear needs to be remade.
T-Frame	1	70±0.50 70.50 69.50	69.85	Calliper	Okay, everything fits.
T-Frame	1	32±0.50 32.50 31.50	31.35	Calliper	Too small, nominal size on cart frame should now be this value.
Cart Frame	1	$32^{+0.00}_{-1.00}$ 32.00 31.00	30.50	Calliper	Too small, but still works.
Cart Frame	1	$32^{+0.00}_{-1.00}$ 32.00 31.00	31.20	Calliper	Okay, slides well.

Appendix Eight - Trajectory Model

```
In[*]:= Remove["Global`*"]
```

General equations (Newton)

```
In[*]:= DSolve[D[Vx[t], t] == 0, Vx[t], t]
DSolve[D[Vy[t], t] == -g, Vy[t], t]
```

```
Out[*]=
{ {Vx[t] -> c1} }
```

```
Out[*]=
{ {Vy[t] -> -g t + c1} }
```

```
In[*]:= Vx[t_] = c1 /. {c1 -> vbegin * Cos[α]};
Vy[t_] = -g * t + c2 /. {c2 -> vbegin * Sin[α]};
```

Solving for position equations

Integrating velocity equations to gain position equations
Position equations (where $x[0] = 0$ en $y[0] = 0$ are defined)

```
In[*]:= X[t_] = t vbegin Cos[α] + c1 /. {c1 -> 0}
Y[t_] = -\frac{g t^2}{2} + t vbegin Sin[α] + c2 /. {c2 -> 0}
```

```
Out[*]=
t vbegin Cos[α]
```

```
Out[*]=
-\frac{g t^2}{2} + t vbegin Sin[α]
```

Branch 1 - Peak parabola, {input: height and distance - output: shooting angle and start velocity}

Peak parabola

```
In[*]:= peak = Solve[0 == Vy[τ], τ][[1]]
```

```
Out[*]=
{ τ -> \frac{vbegin Sin[α]}{g} }
```

Height (at peak of parabola)

```
In[*]:= condΔh1 = Solve[Δh == Y[τ], Δh][[1]] /. peak // FullSimplify
```

```
Out[*]=
{ Δh -> \frac{vbegin^2 Sin[α]^2}{2 g} }
```

Distance (at peak of parabola)

```
In[*]:= condΔs1 = Solve[X[τ] == Δs, Δs][[1]] /. peak
```

```
Out[*]=
{ Δs -> \frac{vbegin^2 Cos[α] Sin[α]}{g} }
```

Branch 2 - Not peak parabola, {input: height, distance, start velocity - output: shooting angle}

Height (at a certain time)

In[*]:= `condΔh2 = Solve[Δh == Y[ττ], ττ] // FullSimplify`

Out[*]=

$$\left\{ \left\{ \tau\tau \rightarrow \frac{v_{\text{begin}} \sin[\alpha] - \sqrt{-2 g \Delta h + v_{\text{begin}}^2 \sin[\alpha]^2}}{g} \right\}, \right. \\ \left. \left\{ \tau\tau \rightarrow \frac{v_{\text{begin}} \sin[\alpha] + \sqrt{-2 g \Delta h + v_{\text{begin}}^2 \sin[\alpha]^2}}{g} \right\} \right\}$$

Distance (at the time corresponding to the height)

In[*]:= `condΔs2 = Solve[X[ττ] == Δs, Δs] /. condΔh2`

Out[*]=

$$\left\{ \left\{ \left\{ \Delta s \rightarrow \frac{v_{\text{begin}} \cos[\alpha] \left(v_{\text{begin}} \sin[\alpha] - \sqrt{-2 g \Delta h + v_{\text{begin}}^2 \sin[\alpha]^2} \right)}{g} \right\}, \right. \right. \\ \left. \left\{ \left\{ \Delta s \rightarrow \frac{v_{\text{begin}} \cos[\alpha] \left(v_{\text{begin}} \sin[\alpha] + \sqrt{-2 g \Delta h + v_{\text{begin}}^2 \sin[\alpha]^2} \right)}{g} \right\} \right\} \right\}$$

Determining trajectory

Parameters can be altered here

In[*]:= `cond1 = {g → 9.81, vbegin → 25.83, Δs → 6.7, Δh → 0.4};`

Branch 1 - output of shooting angle and start velocity

$$\frac{v_{\text{begin}}^2 \cos[\alpha] \sin[\alpha]}{g} == \Delta s \quad \&\& \quad \frac{v_{\text{begin}}^2 \sin[\alpha]^2}{2 g} == \Delta h$$

$$v_{\text{begin}}^2 \sin[\alpha]^2 = \Delta s g \frac{\sin[\alpha]}{\cos[\alpha]} \quad \&\& \quad v_{\text{begin}}^2 \sin[\alpha]^2 = 2 g \Delta h$$

$$\Delta s g \frac{\sin[\alpha]}{\cos[\alpha]} = 2 g \Delta h$$

$$\frac{\sin[\alpha]}{\cos[\alpha]} = \tan[\alpha] = \frac{2 \Delta h}{\Delta s}$$

```
In[*]:= alpha = ArcTan[ $\frac{2 \Delta h}{\Delta s}$ ] /. cond1;

Solve[ $\frac{vstart^2 \sin[\alpha]^2}{2 g} == \Delta h, vstart]$  /. cond1;

vstart /. %;
vstart = DeleteCases[%, x_ /; x ≤ 0][[1]];
```

```
alpha *  $\frac{180}{\pi}$  (*°*)
```

 **Solve:** Solve was unable to solve the system with inexact coefficients. The answer was obtained by solving a corresponding exact system and numericizing the result.

```
Out[*]=
6.80905
```

```
In[*]:= vstart (*m/s*)
Out[*]=
23.6286
```

Branch 2 - input of shooting velocity and output of shooting angle

```
In[*]:= systemofEq2 = Δs /. condΔs2
Out[*]=
```

$$\left\{ \left\{ \frac{vbegin \cos[\alpha] \left(vbegin \sin[\alpha] - \sqrt{-2 g \Delta h + vbegin^2 \sin[\alpha]^2} \right)}{g}, \right. \right.$$

$$\left. \left\{ \frac{vbegin \cos[\alpha] \left(vbegin \sin[\alpha] + \sqrt{-2 g \Delta h + vbegin^2 \sin[\alpha]^2} \right)}{g} \right\} \right\}$$

```
In[*]:= NSolve[systemofEq2[[1]] == Δs, α] /. cond1;
angles = DeleteCases[α /. %, x_ /; x ≤ 0 ∨ x ≥  $\frac{\pi}{2}$ ];
angles *  $\frac{180}{\pi}$ 
```

 **NSolve:** Inverse functions are being used by NSolve, so some solutions may not be found; use Reduce for complete solution information.

```
Out[*]=
{87.1648, 6.25178}
```

Solving equations:

```
In[*]:= cond11 = Append[Append[Delete[cond1, 2], α → alpha], vbegin → vstart]
cond21 = Append[cond1, α → angles[[1]]]
cond22 = Append[cond1, α → angles[[2]]]

Out[*]=
{g → 9.81, Δs → 6.7, Δh → 0.4, α → 0.11884, vbegin → 23.6286}

Out[*]=
{g → 9.81, vbegin → 25.83, Δs → 6.7, Δh → 0.4, α → 1.52131}

Out[*]=
{g → 9.81, vbegin → 25.83, Δs → 6.7, Δh → 0.4, α → 0.109114}
```

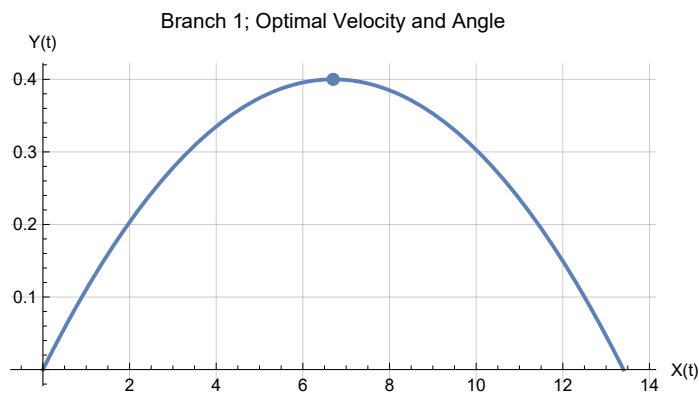
In[]:=

```
plt01 = ListPlot[{Δs, Δh} /. cond11, PlotMarkers → Automatic];
plt02 =
  ParametricPlot[{X[t] /. cond11, Y[t] /. cond11}, {t, 0,  $\frac{v_{\text{begin}} \sin[\alpha]}{g} * 2$  /. cond11},
    GridLines → Automatic, AxesLabel → {"X(t)", "Y(t)"}, PlotRange → Full,
    AspectRatio → 0.5, PlotLabel → "Branch 1; Optimal Velocity and Angle"];
Show[{plt02, plt01}]
```

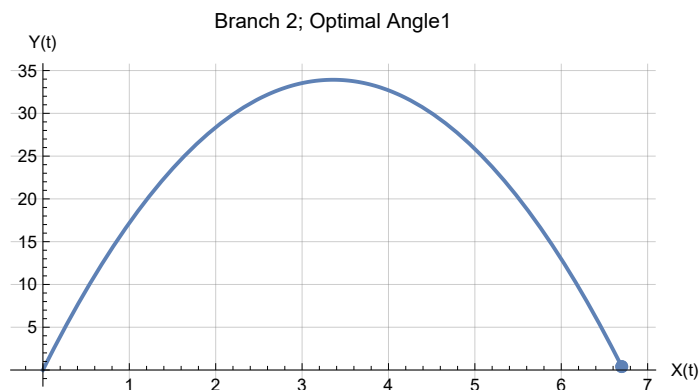
```
plt1 = ListPlot[{Δs, Δh} /. cond21, PlotMarkers → Automatic];
plt2 =
  ParametricPlot[{X[t] /. cond21, Y[t] /. cond21}, {t, 0,  $\frac{v_{\text{begin}} \sin[\alpha]}{g} * 2$  /. cond21},
    GridLines → Automatic, AxesLabel → {"X(t)", "Y(t)"}, PlotRange → Full,
    AspectRatio → 0.5, PlotLabel → "Branch 2; Optimal Angle1"];
Show[{plt2, plt1}]
```

```
plt3 = ListPlot[{Δs, Δh} /. cond22, PlotMarkers → Automatic];
plt4 =
  ParametricPlot[{X[t] /. cond22, Y[t] /. cond22}, {t, 0,  $\frac{v_{\text{begin}} \sin[\alpha]}{g} * 2$  /. cond22},
    GridLines → Automatic, AxesLabel → {"X(t)", "Y(t)"}, PlotRange → Full,
    AspectRatio → 0.5, PlotLabel → "Branch 2; Optimal Angle"];
Show[{plt4, plt3}]
```

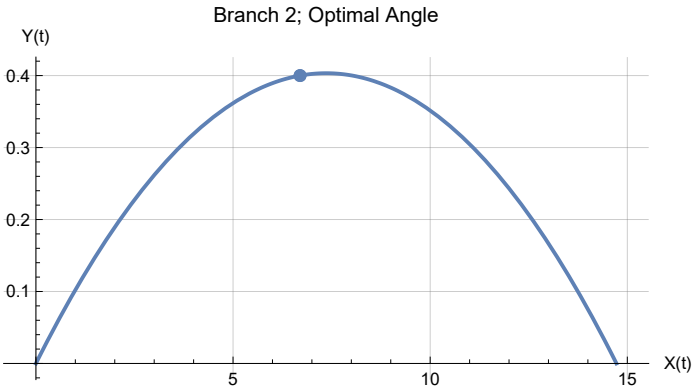
Out[]:=



Out[]:=



Out[]=



Angle check

```
In[ ]:=  $\beta = \text{ArcTan}\left[\frac{V_y[\tau\tau]}{V_x[\tau\tau]}\right] /. \text{cond}\Delta h2;$ 
```

```
Abs[ $\beta[[1]] /. \text{cond}21$ ] *  $\frac{180}{\pi}$  (*Plot Branch 2; Optimal Angle1*)
```

```
Abs[ $\beta[[1]] /. \text{cond}22$ ] *  $\frac{180}{\pi}$  (*Plot Branch 2; Optimal Angle2*)
```

Out[]=

87.148

Out[]=

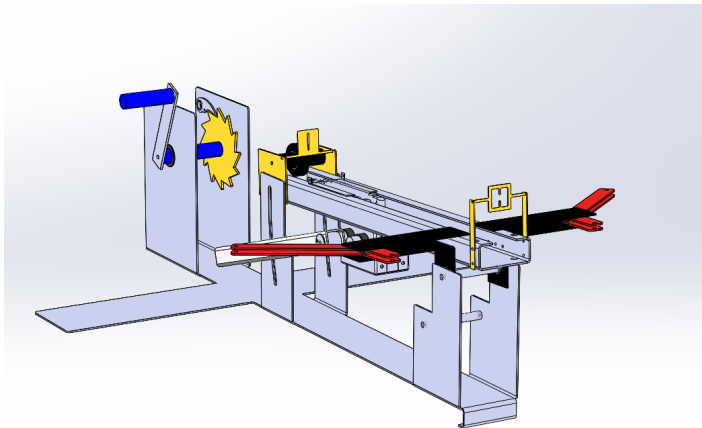
0.564557

Appendix Nine - Geometry Model

The geometry model is made in the online programme Desmos. It can be accessed through this link: <https://www.desmos.com/calculator/li9mlgdwmr?lang=nl>.

Appendix Ten - Test Protocol

Group Number	12	Tutor	Roy Damgrave	Date	22.1.2025
Group Members					
1. Manuel Meijnen	m.j.meijnen@student.utwente.nl	3464385			
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6. Miriam Kalds	mariamhanygeorgehemayakalds@student.utwente.nl	3276309			

Project name	Project prototyping
Product description (max 100 words)	The assignment is to develop a prototype for a Dart Shooter. This is supposed to be a completely mechanical device that allows you to play darts matches. The prototype (placed/fixed on a table) should be able to shoot darts to a blazon (target) located 4-8 meters from the table.
Image representing the product	

Scope of the testing
The product will be tested with a focus on the requirements, measuring if the design met the requirements. This means, measuring and seeing if the human interaction with the product is as we wanted it to be, for example the amount of force needed to pull. Our plan to do it is to let people use the Dart shooter, write down their experience and analyse at the shot and derive a conclusion. This testing method is chosen, because this way we mimic a real scenario, for which the requirements need to be met.

Expected duration of the testing	Starting: Ending: N* of sessions: 1 Duration of each session: 15 min.
Materials needed for the testing	- Dart shooter - Darts - Clamps - Target - Table
Describe the testing set-up addressing contextual information, participants, informed consent and possible risks.	The testing will be taking place in the hangar, this environment does not differ from that of the expected context of use. So, there will not be a measurable difference in product performance between those situations. For the test there are 2(people with different strengths) people needed. These participants will be from the study IDE, because these are the people that are at hand. We will also test different genders, because there is a difference in strength between genders. Before participating, participants are asked to sign under their answer a simple question. The risk of the testing is that someone can get hurt. There was tried to make the the testing setup very chill for the participants.
Others	-.

Execution of the testing
1. The participant will be explained how the machine works and which things to be conscious of, such as moving parts. 2. After that, the participant needs to pick up the prototype and walk a little distance and then secure in on a table (this will be timed). 3. When the participant will have secured the prototype to the table, the participant can operate the Dart shooter and shoot 3 darts (this will be timed). After these 3 shots, the results will be noted down, after which the participant can shoot 3 darts for another 3 times. 4. When this all is done, the participant needs to answer a small question and then it is done. The question is: "Is it hard to pull the cart back?"
Results
Give an objective overview of the results without analysing them (yet) (max 250 words) The dart shooter is carriable by 1 person. The dart shooter is operable by 2 persons, but not by 1. Because the ratchet needs to be unwinded, while the release lever is down. This is very hard, but doable with 1 person. The average time taken to secure the dart shooter to the table is 8 sec. The average time to shoot 3 darts is 1:35 min. The number of shots hit for points was 1 out of 5. The answer to the question was always "yes."

Analysis of results

Out of our results, we can draw some positive and some negative conclusions.

Negative:

The amount of people that is needed to operate the dart shooter is more than we wanted.

The distance the dart travelled is not enough to shoot every distance, it landed short a lot of times, because the springs are not strong enough.

Positive:

It is carriable by on person, so we met our requirement.

The participants did not need to put in a force that was very hard for them, so we also met this requirement.

The dart shooter can be installed, well in time, so we also met our requirement.

The dart shooter can shoot 3 darts within the wanted time frame, we are even left with some spare time.

Proposal for improvement

Improvements that can be made is changing the springs, to stronger ones, which will result in more power and a longer travel distance.

For the operability of the device something has to be made to hold the lever, while unwinding the winch.

Appendix Eleven - FMEA

Eleven A - FMEA

No	Function	Failure mode	Cause of failure	Failure Effect	Detection and prevention	Severity (S)	Occurrence (O)	Detection (D)	Action (Yes/No)	RPN	Actions recommend	Executed	Discussion
1 Baseplate	Stabilizing the dart shooter	Too little functioning: wrong measurements could result in instability.	Faulty measurements or connection	Decreased accuracy	Check measurements and calculations	2	2	1	No	4			To keep the dart shooter stable, it cannot have much moving space and what we filled in can be a cause.
2 Baseplates	Clamping dart shooter to the table	Too little functioning: the clamps might not be strong enough	Wrong calculations or faulty design: clamps in the wrong positions.	Decreased safety and accuracy	Bigger clamps, test without and with shooting to see if it can endure the high forces of the firing process.	4	1	1	No	4			For this to fail, it needs to break or just not clamp the dart shooter enough, since breaking does not feel realistic did we choose the insufficient clamping.
3 Winch: Crank	Convert manpower into usable energy	The user isn't able to supply enough force, resulting in no function. The lever can break off, resulting in unexpected functioning or no function.	The springs are too strong for the mechanical advantage used in the device. Excessive use of force.	The dart cannot be fired, the user might sustain an injury.	Bigger mechanical advantage. Test if the lever can endure the force necessary to pull back the dart.	6	2	3	No	36			Here we looked at what needs to happen with the crank and this thing is in direct contact with the user, this is where problems arise, that's why we have these failures.
4 Winch: Ratchet	Store rotational energy	The pawel can break off, resulting in unexpected functioning or no function at all.	The pawel can break off due to a weak/wrong joining method or excessive force.	The dart cannot be fired, the user might sustain an injury.	Test parts strength	6	1	5	No	30			We could only think of breaking when it came to the malfunction of the ratchet.
5 Winch: Bar	Hold the rope	No function, the part attached to the rope could break off, resulting in unexpected functioning or no function.	Fragile parts	The dart cannot be fired, the user might sustain an injury.	Test parts strength	6	1	3	No	18			For this one the same: We could only think of breaking when it came to the malfunction of the bar.
6 Cart: Backplate	Hold the dart in place	The backplate might not suffice in holding the dart in place,	Faulty measurements	Decreased accuracy	Check measurements,	4	2	1	No	8			The dart cannot have much moving space, this is a thing that can fail

		resulting in no function, since the dart won't be held down.											and results in less accurate shots or no shots at all.
7 Cart: Backplate	Hold the rope	Rope detaches from the backplate, resulting in unexpected functioning.	Fragile parts	The dart cannot be fired, the user might sustain an injury.	Test parts strength	6	1	3	No	18			For this to malfunction we could only think of breaking.
8 Cart: Wheels	Reduce friction	Doesn 't reduce enough friction. Can wear down the material due to excessive use.	Too many colliding surfaces	Decreased speed. Broken material can result in danger for the user.	Decrease colliding surfaces	7	8	2	Yes	112	Replace wheels by ridges in order to reduce the friction created by the cart.	Yes	The function of the wheels is to reduce friction and when it fails this then we need to adjust it, that is what we did.
9 Cart: Wheels	Accelerate Cart/dart	Axles can break, wheels cannot turn anymore.	Fragile parts	The dart cannot be fired. Greatly decreased speed.	Test parts strength	7	1	4	Yes	28	Replace wheels by ridges in order to eliminate the risk of breaking the axles.	Yes	The wheels add a lot of complexity and risk to the design. Replacing them with a simpler design, without losing any functionality is very useful.
10 L-shapes, springs and rope	Accelerate Cart/dart	No function: The rope releases from the cart or L-shapes and renders the device useless. The user could hurt themselves if the L-shapes spring back after releasing the cart.	Weak joining method used	The dart cannot be fired, the user gets hurt.	Test the joined parts.	8	4	5	Yes	160	Make sure to secure the rope very well, if the testing results say that a stronger rope needs to be used, use a stronger rope or elastic rope.	Yes	The L-shapes are the part of the design which accelerate the cart, allowing it to be shot at certain angles and distances. The only risk for this part is for the joining method to be too weak.
11 Release mechanism: Pinplate	Store linear energy	Unexpected functioning: The pins deform or break and release the cart unexpectedly.	Bendable or fragile parts.	The dart fires unexpectedly, potentially hurting someone. The rope hasn't been detached from the backplate yet and breaks the device.	Test parts in use.	7	1	6	No	42			The sole function of this part is to hold the cart in place until it is fired. If it breaks or is deformed, the entire device will not function.
12 Release mechanism: Lever	Release energy	No function: The lever breaks. User doesn't understand which way	Fragile parts, bad explanation.	The device is rendered useless. The user sustains an injury.	Test whether or not the lever can survive the necessary force to release the cart. Prevent this from	8	2	6	Yes	96	Clearly state how to use the dart shooter, also include arrows or any indication of direction on the side	Yes	The lever might not come across as a very clear part to some users, calling for the need to properly explain how

		to pull and hurts themselves.			happening by indication which way to pull the lever on the device.						of the device next to the lever. Use more material to ensure strength.		the lever works, intuitively.
13 T-shape: Hinge	Adjust angle of the dart shooter	No function due to deformation of the part. Too much/little functioning due to wrong measurements.	Material bends due to usage or thermal effects. Human error.	Decreased accuracy.	While testing, look for deformations of the part to see whether the part can still function properly.	2	2	3	No	12			This parts needs to be able to move properly.
14 T-shape: Hinge	Holding the dart shooter at the right angle	No function due to releasing unexpectedly. Not consistent(moves around).	Too much space between parts, fragile parts.	Decreased accuracy.	Running a lot of tests to see if the angle changes whilst firing the dart.	4	3	6	Yes	72	Use thicker metal, to have more strength and laser cut it, to ensure measurements.	Yes	The hinge needs to be able to sustain the force it is subjected to during the firing of the dart.
15 T-shape	Encase and/or attach to all other parts	Poorly attached to other parts, the connections could break, resulting in no function.	Wide gaps/wrong measurements and bent parts.	The device could stop functioning if the T-shape's attachment to another part breaks.	This is very noticeable during the production of the device. Easy to detect	2	2	2	No	8			Welding this part could obstruct the movement of the cart, and the movement of the cart could damage the connection.
16 T-shape	Guide the cart in a straight line	No function due to deformation	Use of the material, thermal effects or human error	Decreased speed and accuracy	Use laser cutting to minimize human error, preventing possible deformation during the machining process.	7	6	3	yes	126	Create this part with laser cutting instead of using metal bars and machining them, that way you reduce the risk of human error.	Yes	Keeping our inexperience in the workshop in mind, we took into account the serious possibility of failing to properly machine this part. To minimize the chance of that happening, we decided to laser cut this part. Allowing for more precision and more effective use of our time in the workshop.

Eleven B - Functional Model

Part	Functions
Baseplate	Stabilizing the dart shooter Clamping dart shooter to the table
Winch: Crank	Convert manpower into usable energy
Winch: Ratchet	Store rotational energy
Winch: Bar	Hold the rope
Cart: Backplate	Hold the dart in place Hold the rope
Cart: Wheels (Ridges)	Reduce friction Accelerate Cart/dart
L-shapes, springs and rope	Accelerate Cart/dart
Release mechanism: Pinplate	Store linear energy
Release mechanism: Lever	Release energy
T-Shape: Hinge	Adjust angle of the dart shooter Holding the dart shooter at the right angle
T-Shape	Encase and/or attach to all other parts Guide the cart in a straight line

